

Robust Reinforcement Learning under Stochastic Action Failure

Experiential Learning - Assignment 2 | Deep Reinforcement Learning (S2-25_AIMLCZG512)

Group 148

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Group contribution declaration

MANDATORY BEFORE SUBMISSION: These contribution percentages are declared but must be confirmed by all five group members.

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Submission field	Value
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Filename note: Check the exact final filename against the latest instructor guidance before upload. The build currently follows the requested pattern Group148_Q_learning_DQN_DDQN.pdf.

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Executive summary

This report implements the specified hidden 15% actuator-failure wrapper and compares DQN and Double DQN in a controlled 2 x 2 experiment. The four agents use the same seed, network, initialization procedure, optimizer, replay settings, exploration schedule, target-update cadence, duration, and fixed validation-state set. The wrapper is the only environment difference; the target calculation is the only DQN-versus-DDQN difference.

Across 250 random-policy verification episodes, 2,402 of 16,460 attempted thruster actions misfired (14.593%). The target 15% lies in the Wilson 95% interval [14.062% to 15.141%]. The internal fuel-penalty count equals the attempted-action count, and no returned info object was changed.

For the recorded single-seed study, DDQN on the modified environment achieved greedy mean reward -225.82 and safe-landing rate 3%. DQN achieved 48.02 and 44%. These values are evidence for this execution only, not a claim of statistical significance.

1. Introduction

LunarLander-v3 has an eight-dimensional continuous observation: horizontal and vertical position, horizontal and vertical velocity, angle, angular velocity, and two leg-contact indicators. Its four discrete actions are:

Action	Meaning
0	Do nothing
1	Fire left orientation engine
2	Fire main engine
3	Fire right orientation engine

Actuator failures are realistic in robotics because commands can be lost or partially ineffective through mechanical, electrical, communication, or control faults. Here, the agent selects a thruster action but may unknowingly execute no-op. This hidden replacement creates transition uncertainty: similar observed state-action pairs can lead to different physical next states.

The reward charges 0.3 for the **selected** thruster action even when it misfires. It can therefore encourage a conservative policy, but only if the learner discovers that avoiding unnecessary attempts improves long-run return. DQN can overestimate action values because a maximization over noisy estimates uses the same target network for selection and evaluation. DDQN reduces this bias by selecting the next action with the online network and evaluating that action with the target network.

2. Reproducibility and environment inspection

2.1 BITS virtual-lab installation

The environment is explicitly LunarLander-v3; no fallback version is used. On a Debian/Ubuntu-style virtual lab, install Box2D build support and this repository as follows:

```
sudo apt-get update
sudo apt-get install -y swig build-essential python3-dev
python -m pip install --upgrade pip setuptools wheel
python -m pip install -e ".[dev]"
python -c "import gymnasium as gym; env = gym.make('LunarLander-v3'); print(env.observation_space, env.action_space);
    ↪ env.close()"
```

If sudo is unavailable, ask the lab administrator to install swig and the compiler packages; do not silently switch LunarLander versions.

2.2 Seeding and provenance

One function seeds Python random, NumPy, PyTorch CPU, all CUDA devices, Gymnasium reset, and the action space. It also requests deterministic PyTorch algorithms and disables cuDNN benchmarking where available. Box2D, floating-point kernels, drivers, and hardware can still produce cross-platform nondeterminism, so versions and hardware are recorded.

Parameter	Shared value
Global seed	148
Training episodes per agent	800
Maximum steps per episode	1000
Evaluation episodes per agent	100
Network	8-128-128-4, ReLU
Trainable parameters	18,180
Initialization	PyTorch Linear default; same seed before every agent
Optimizer / loss	Adam / Smooth L1
Learning rate	0.0005
Discount factor	0.99
Replay capacity / batch	100,000 / 64
Learning warm-up	1,000 steps
Target update	Hard copy every 500 updates
Epsilon schedule	1.00 to 0.01 over 100,000 steps
Fixed validation states	512
Failure probability	0.15
Attempted-thruster penalty	0.3
Safe-landing bonus	50.0
Device	cpu

Recorded provenance:

- Python: 3.13.12
- Gymnasium: 1.2.3
- PyTorch: 2.13.0
- NumPy / pandas / Seaborn: 2.4.4 / 3.0.2 / 0.13.2
- Platform / logical CPUs: macOS-26.5.2-arm64-arm-64bit-Mach-O / 14
- CUDA available / device: False / None
- Device used: cpu
- Global random seed: 148
- Git commit at execution: 3eb6b9bf0e382fc67c02976cdb55ed8dde1dfffa7
- Git worktree dirty at execution: True

3. Custom environment and verification - 2.5 marks

For selected action a , the wrapper samples a private RNG only when $a \in \{1, 2, 3\}$. With probability 0.15 it executes action 0; otherwise it executes a . A private RNG avoids consuming the base LunarLander RNG and hence keeps ordinary transition randomness isolated. Observation/action spaces, termination, and truncation are delegated unchanged.

The modified reward is

$$R = R_{base} - 0.3 \mathbf{1}(a \neq 0) + B,$$

where $B = 50$ only for a terminated, non-truncated transition with both legs in contact and absolute horizontal velocity, vertical velocity, and angle each strictly below 0.10. Private counters are externally readable for verification but are never inserted into the observation or returned info.

3.1 Deterministic mock-environment tests

Case	Selected	Executed	Bonus	Penalty	Info unchanged	Passed
Safe terminal	0	0	50.0	0.0	True	True
Firing + safe	2	2	50.0	0.3	True	True
Misfire + safe	2	0	50.0	0.3	True	True
Not terminated	0	0	0.0	0.0	True	True

Case	Selected	Executed	Bonus	Penalty	Info unchanged	Passed
Truncated	0	0	0.0	0.0	True	True
Left leg absent	0	0	0.0	0.0	True	True
Right leg absent	0	0	0.0	0.0	True	True
Excess x velocity	0	0	0.0	0.0	True	True
Excess y velocity	0	0	0.0	0.0	True	True
Excess angle	0	0	0.0	0.0	True	True

The automated unit suite additionally checks action 0, successful firing, misfiring, strict threshold equality, both leg contacts, truncation, unchanged spaces, invalid actions, and exact info identity.

3.2 Random-policy evidence

Measure	Recorded result
Episodes	250
Environment steps	21,912
Attempted thruster actions	16,460
Misfires	2,402
Observed misfire rate	14.593%
Absolute difference from 0.15	0.004070
Wilson 95% interval	14.062% to 15.141%
Fuel-penalty count	16,460
Fuel-penalty count equals attempts	True
Fuel-penalty mismatches	0
Returned-info identity mismatches	0
Random-policy safe bonuses observed	12

Successful-fire and forced-misfire mock cases both show the 0.3 selected-action penalty. Deterministic landing-boundary cases are authoritative for the +50 bonus because random policies do not guarantee useful landing coverage.

4. Replay, network, DQN, and DDQN - 8 marks

4.1 Replay and terminal masking

The replay buffer stores state, action, reward, next state, terminated, and truncated separately. The target masks only true terminal states:

$$y = r + \gamma(1 - \text{terminated}) Q_{\text{bootstrap}}.$$

A time-limit truncation still bootstraps because the underlying MDP state is not terminal; the episode ended due to an external horizon. This choice is explicit and unit-tested.

4.2 Shared Q-network

All experiments use the same 8-128-128-4 multilayer perceptron, ReLU activations, PyTorch Linear initialization after the same global seed, Adam, and Smooth L1 loss. It has **18,180 trainable parameters**.

4.3 The only algorithmic branch

For DQN:

$$y_{DQN} = r + \gamma(1 - t) \max_a Q_{target}(s', a).$$

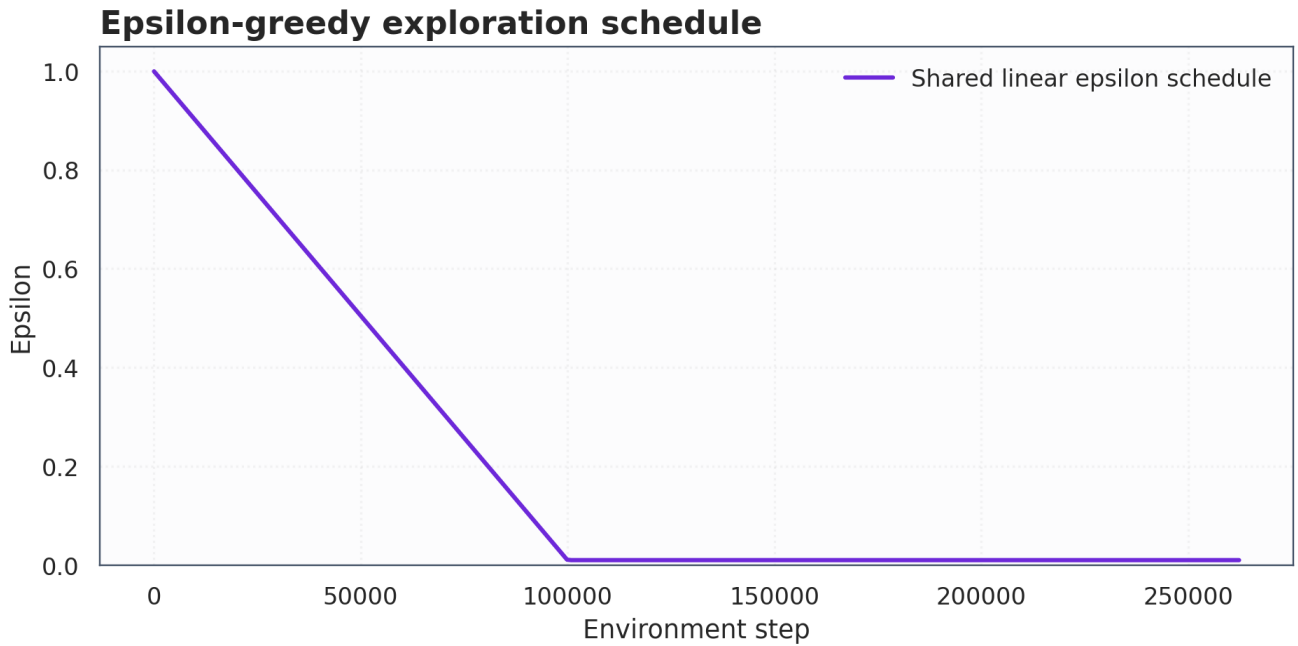
For DDQN:

$$a^* = \arg \max_a Q_{online}(s', a), \quad y_{DDQN} = r + \gamma(1 - t) Q_{target}(s', a^*).$$

The online/target architectures, optimizer, replay, epsilon policy, discount, warm-up, batch size, learning rate, target updates, seed, and training duration are otherwise identical.

4.4 Shared epsilon-greedy schedule

Checkpoint	Environment step	Epsilon
Start	0	1.000
Half decay	50000	0.505
End of decay	100000	0.010
After decay	200000	0.010



The same start, end, and step-based decay are applied to all four agents.

Figure 1: Shared epsilon-greedy schedule

5. Fixed validation states, metrics, and controlled runs

The validation states were collected once from the original environment with a reproducible random policy and never resampled during training. Reusing an identical set prevents state-distribution drift from being confused with algorithmic differences in predicted Q-values.

- Shape: (512, 8)
- SHA-256: b765a8c76d9ffa0e93207cc98d3bda2bf16adca80d38c8871380e832e25c7f8a

Every episode records reward, fixed-set average maximum Q-value, strict safe landing, moving safe-landing rate over up to 100 episodes, attempted and executed thruster actions, cumulative average attempts per episode, steps, epsilon, mean training loss, environment type, algorithm, global seed, episode seed, global step, and measured duration.

Four fresh agents and replay buffers are trained:

1. DQN - Original Environment
2. DDQN - Original Environment
3. DQN - Modified Environment
4. DDQN - Modified Environment

Each episode prints one compact progress line and writes one log record. The full study is parameterized, so a short smoke run and the full run use the same code path.

6. Performance evaluation - 2 marks

DQN vs. DDQN under stochastic actuator failure

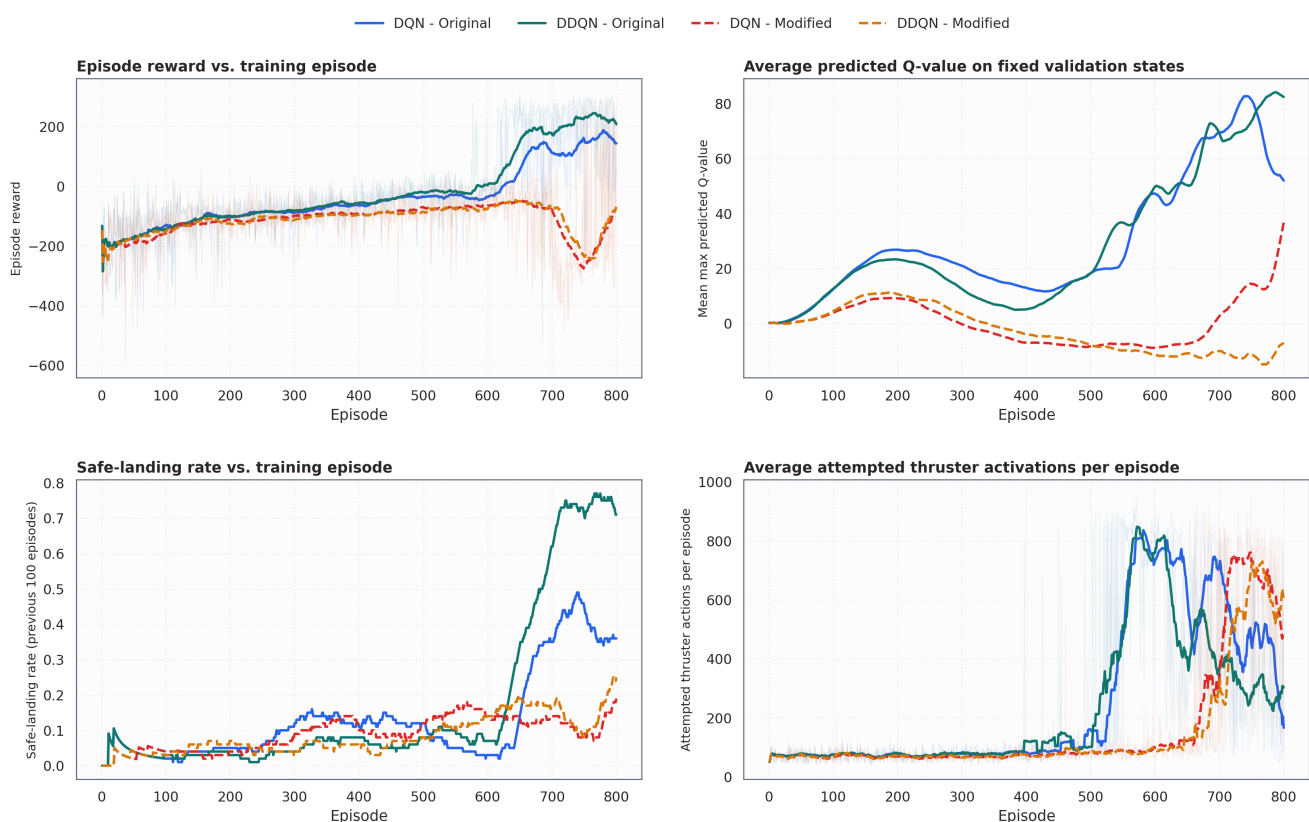


Figure 2: All four required training metrics

The figure contains the four required comparisons: episode reward, fixed-set average predicted Q-value, 100-episode moving safe-landing rate, and attempted thruster activations per episode. Raw reward and activation traces are light; moderate moving means make trends legible without concealing instability. Colours and line styles are consistent across panels.

6.1 Required final comparison table

The two panels below form one comparison table while remaining legible on A4.

Panel A - performance and value estimates

Algorithm	Environment	Mean reward final 100	Reward SD final 100	Best MA(100) reward	Final fixed-set Q	Safe rate final 100
DQN	Original	150.52	113.67	154.40	49.12	36%

Algorithm	Environment	Mean reward final 100	Reward SD final 100	Best MA(100) reward	Final fixed-set Q	Safe rate final 100
DDQN	Original	218.24	98.01	226.34	80.51	71%
DQN	Modified	-172.73	192.67	-56.87	48.56	19%
DDQN	Modified	-150.42	200.69	-54.05	-6.72	25%

Panel B - action use, successes, and duration

Algorithm	Environment	Mean attempted final 100	Mean executed final 100	Safe landings total	Training seconds
DQN	Original	409.80	409.80	116	166.63
DDQN	Original	304.11	304.11	166	137.97
DQN	Modified	655.06	557.94	86	89.09
DDQN	Modified	595.74	507.37	83	98.72

Mean reward, reward SD, safe rate, and both action-use means use the final 100 training episodes. The best moving average, final Q-value, total successes, and duration use the scopes stated in their labels. Executed actions equal attempts in the original environment; modified runs report actual non-misfired executions.

6.2 Greedy evaluation on shared seeds

Panel A - reward and strict landing performance

Algorithm	Environment	Mean reward	Reward SD	Safe landing rate
DDQN	Modified	-225.82	102.17	3%
DDQN	Original	169.05	136.75	61%
DQN	Modified	48.02	186.73	44%
DQN	Original	67.15	95.71	24%

Panel B - action use and episode length

Algorithm	Environment	Mean attempted	Mean executed	Mean steps
DDQN	Modified	730.92	622.07	972.34
DDQN	Original	250.72	250.72	360.51
DQN	Modified	423.85	359.97	615.37
DQN	Original	138.07	138.07	188.34

This evaluation uses 100 shared episode seeds per experiment and epsilon 0. It is supplementary to the assignment's final-100 training summary.

7. Discussion - 2.5 marks

7.1 Does failure increase the DQN-DDQN Q-value difference?

For this execution, yes. The final-100 mean absolute Q-gap is 16.23 in the original environment and 28.68 in the modified environment, an increase of 12.46. The fixed-state Q-value panel provides the trajectory; these numerical values provide the final-window comparison. A single seed cannot establish a population-level effect.

7.2 Why is temporal credit assignment harder?

The same selected thruster action can yield an impulse or a no-op while the agent receives no replacement indicator. Replay therefore contains higher-variance outcomes for apparently similar state-action inputs. The

attempted-action cost is charged in either case, separating a certain immediate cost from an uncertain physical effect and making delayed crash or landing credit harder to assign.

7.3 Does the 0.3 penalty encourage a conservative strategy?

Not in the attempted-action metric for this execution. Greedy DDQN attempts 730.92 actions in the modified environment versus 250.72 in the original. Greedy DQN attempts 423.85 versus 138.07. Both modified policies therefore attempt more, not fewer, thruster actions. The longer modified episodes and need to compensate for no-op commands may outweigh the 0.3 cost. The penalty alone did not produce a demonstrably conservative strategy.

7.4 Which algorithm performs better under failure?

The answer is metric-dependent. In the final 100 training episodes, DDQN has the higher mean reward (-150.42 versus -172.73) and safe-landing rate (25% versus 19%). However, frozen-checkpoint greedy evaluation strongly favors DQN: mean reward 48.02 and safe-landing rate 44% versus DDQN's -225.82 and 3%. Therefore this run does not give an unambiguous DDQN advantage and is not cleanly consistent with the theoretical expectation.

7.5 Limitation and improvement

The executed study uses one training seed, so between-run variance is not estimated. The strongest improvement is a preregistered paired multi-seed study using the same four-run design, followed by confidence intervals for final reward, safe-landing rate, Q-gap, and thruster use. A later failure-probability sweep could test whether any DDQN advantage scales with uncertainty.

8. Virtual-lab evidence

The following evidence must be genuine. Placeholders are intentional and are never substituted with fabricated screenshots.

8.0 Beginning-of-execution evidence

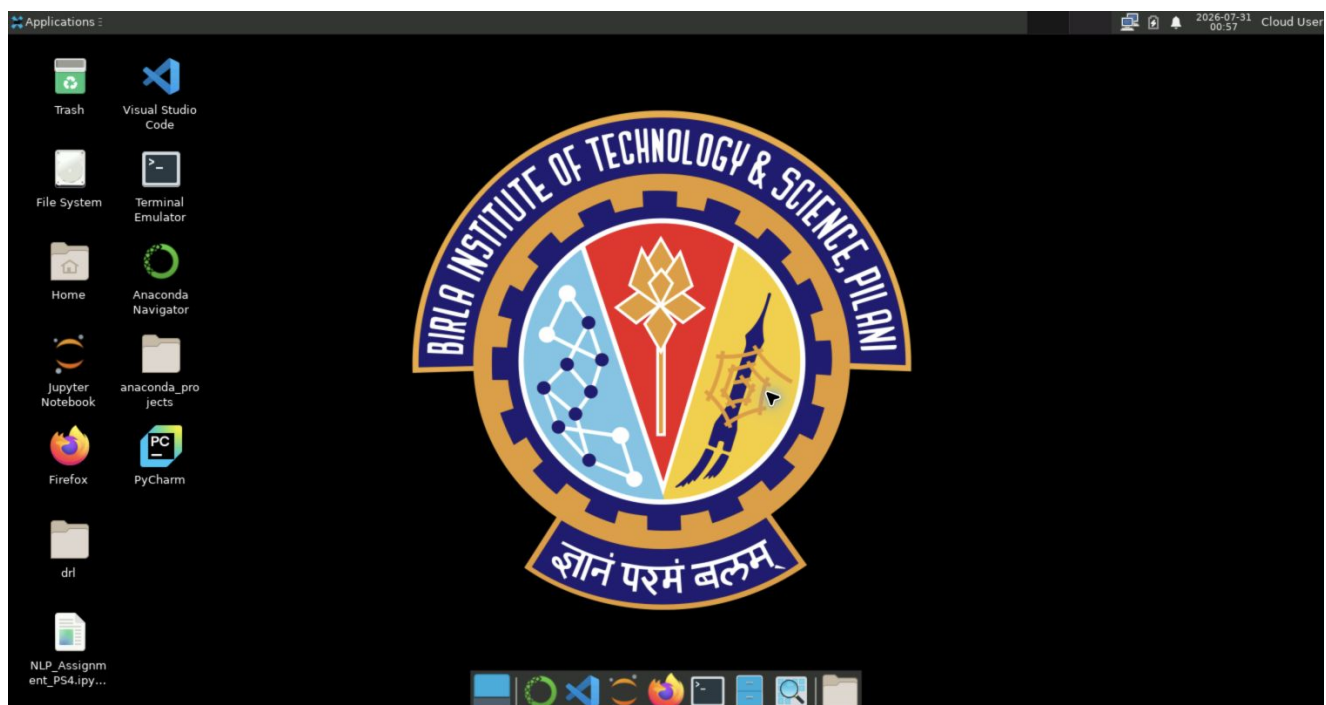


Figure 3: Beginning-of-execution timestamp and institutional virtual-lab identity

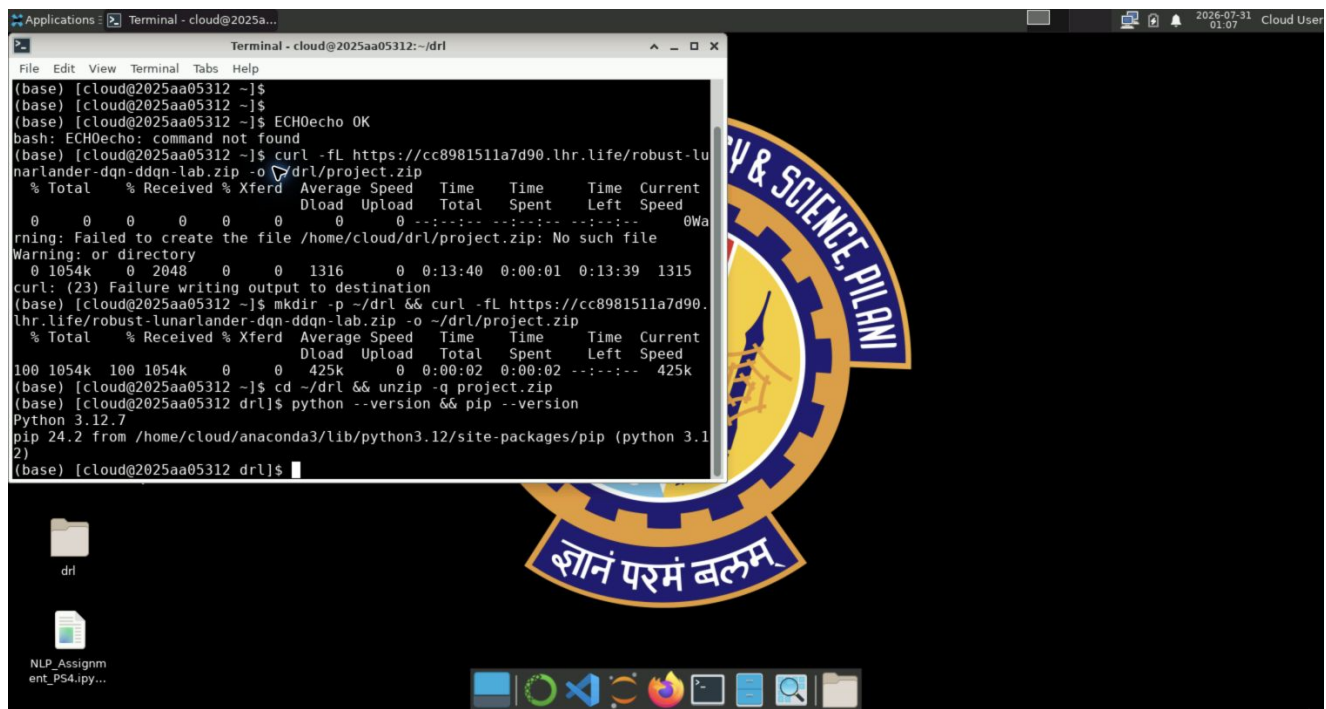


Figure 4: Environment and package-version evidence

8.1 Package and environment evidence

8.2 Timestamped training evidence

8.3 Timestamped final outputs and plots

8.4 Timestamped saved-file and checkpoint evidence

The saved study artifacts are included with this submission and are listed below. The final-output screenshot above provides the timestamped virtual-lab completion evidence; this inventory avoids inserting a non-genuine replacement image.

Artifact class	Saved location
Per-episode ledgers and progress logs	artifacts/metrics/, artifacts/logs/
Model checkpoints	artifacts/checkpoints/
Fixed validation states	artifacts/validation_states.npy
Evaluation data and summary	artifacts/evaluation_episodes.csv, artifacts/evaluation_summary.csv
Plots and final comparison	artifacts/plots/, artifacts/final_comparison.csv

9. Conclusion

In the recorded single-seed experiment, hidden actuator failure increased the DQN-DDQN fixed-state Q-value gap from 16.23 to 28.68. The selected-action fuel penalty did not reduce attempted activations in the modified runs. DDQN was slightly stronger in the final training window, while DQN was substantially stronger in greedy evaluation, so no algorithm wins every metric. The main limitation is the single seed; paired multi-seed replication is the next improvement. This conclusion must be regenerated if the official virtual-lab run changes the recorded outputs.

```

Applications: Terminal - cloud@2025a...
Terminal - cloud@2025aa05312:~/drl

[dqn_original] episode 69/800 reward= -113.00 q= 8.29 safe100= 4.3% attempted= 62 executed= 62 steps= 88 epsilon=0.9337 loss=1.316969
[dqn_original] episode 70/800 reward= -147.69 q= 8.45 safe100= 4.3% attempted= 81 executed= 81 steps= 108 epsilon=0.9326 loss=1.433465
[dqn_original] episode 71/800 reward= -109.82 q= 8.48 safe100= 4.2% attempted= 92 executed= 92 steps= 120 epsilon=0.9314 loss=1.442143
[dqn_original] episode 72/800 reward= -90.23 q= 9.18 safe100= 4.2% attempted= 90 executed= 90 steps= 116 epsilon=0.9303 loss=1.539167
[dqn_original] episode 73/800 reward= -127.92 q= 8.78 safe100= 4.1% attempted= 51 executed= 51 steps= 80 epsilon=0.9295 loss=1.479289
[dqn_original] episode 74/800 reward= -83.21 q= 9.14 safe100= 4.1% attempted= 61 executed= 61 steps= 80 epsilon=0.9287 loss=1.427909
[dqn_original] episode 75/800 reward= -54.93 q= 9.33 safe100= 4.0% attempted= 57 executed= 57 steps= 74 epsilon=0.9280 loss=1.253281
[dqn_original] episode 76/800 reward= -191.63 q= 8.99 safe100= 3.9% attempted= 80 executed= 80 steps= 94 epsilon=0.9270 loss=1.341112
[dqn_original] episode 77/800 reward= -108.70 q= 9.11 safe100= 3.9% attempted= 65 executed= 65 steps= 84 epsilon=0.9262 loss=1.608913
[dqn_original] episode 78/800 reward= -132.54 q= 11.04 safe100= 3.8% attempted= 65 executed= 65 steps= 87 epsilon=0.9253 loss=1.550955
[dqn_original] episode 79/800 reward= -102.55 q= 10.47 safe100= 3.8% attempted= 54 executed= 54 steps= 70 epsilon=0.9246 loss=1.558437
[dqn_original] episode 80/800 reward= -44.38 q= 10.31 safe100= 5.0% attempted= 73 executed= 73 steps= 94 epsilon=0.9237 loss=1.408165
[dqn_original] episode 81/800 reward= -58.02 q= 10.25 safe100= 4.9% attempted= 55 executed= 55 steps= 68 epsilon=0.9230 loss=1.257996
[dqn_original] episode 82/800 reward= -135.37 q= 10.61 safe100= 4.9% attempted= 104 executed= 104 steps= 126 epsilon=0.9218 loss=1.455430
[dqn_original] episode 83/800 reward= -96.54 q= 10.33 safe100= 4.8% attempted= 79 executed= 79 steps= 93 epsilon=0.9209 loss=1.362653
[dqn_original] episode 84/800 reward= -15.45 q= 11.78 safe100= 4.8% attempted= 57 executed= 57 steps= 75 epsilon=0.9201 loss=1.639377
[dqn_original] episode 85/800 reward= -108.43 q= 11.85 safe100= 4.7% attempted= 79 executed= 79 steps= 98 epsilon=0.9192 loss=1.258529
[dqn_original] episode 86/800 reward= -112.49 q= 11.82 safe100= 4.7% attempted= 72 executed= 72 steps= 95 epsilon=0.9182 loss=1.47778
[dqn_original] episode 87/800 reward= -240.74 q= 11.50 safe100= 4.6% attempted= 81 executed= 81 steps= 98 epsilon=0.9172 loss=1.572468
[dqn_original] episode 88/800 reward= -241.40 q= 11.79 safe100= 4.5% attempted= 47 executed= 47 steps= 65 epsilon=0.9166 loss=1.400864
[dqn_original] episode 89/800 reward= -98.49 q= 11.84 safe100= 4.5% attempted= 58 executed= 58 steps= 78 epsilon=0.9158 loss=1.371317
[dqn_original] episode 90/800 reward= -151.74 q= 12.53 safe100= 4.4% attempted= 53 executed= 53 steps= 73 epsilon=0.9151 loss=1.557905
[dqn_original] episode 91/800 reward= -202.71 q= 12.88 safe100= 4.4% attempted= 75 executed= 75 steps= 103 epsilon=0.9141 loss=1.150353
[dqn_original] episode 92/800 reward= -213.81 q= 12.81 safe100= 4.3% attempted= 75 executed= 75 steps= 84 epsilon=0.9133 loss=1.654998
[dqn_original] episode 93/800 reward= -228.81 q= 12.64 safe100= 4.3% attempted= 99 executed= 99 steps= 129 epsilon=0.9120 loss=1.411842
[dqn_original] episode 94/800 reward= -74.44 q= 14.80 safe100= 4.3% attempted= 83 executed= 83 steps= 117 epsilon=0.9108 loss=1.629821
[dqn_original] episode 95/800 reward= -102.52 q= 13.91 safe100= 4.2% attempted= 95 executed= 95 steps= 115 epsilon=0.9097 loss=1.560421
[dqn_original] episode 96/800 reward= -237.89 q= 13.31 safe100= 4.2% attempted= 102 executed= 102 steps= 133 epsilon=0.9084 loss=1.411588
[dqn_original] episode 97/800 reward= -47.49 q= 13.79 safe100= 4.1% attempted= 99 executed= 99 steps= 126 epsilon=0.9071 loss=1.524896
[dqn_original] episode 98/800 reward= -76.46 q= 13.60 safe100= 4.1% attempted= 60 executed= 60 steps= 79 epsilon=0.9063 loss=1.427872
[dqn_original] episode 99/800 reward= -74.18 q= 14.90 safe100= 4.0% attempted= 52 executed= 52 steps= 64 epsilon=0.9057 loss=1.754195
[dqn_original] episode 100/800 reward= -115.63 q= 15.17 safe100= 4.0% attempted= 92 executed= 92 steps= 119 epsilon=0.9045 loss=1.435516
[dqn_original] episode 101/800 reward= -64.01 q= 14.73 safe100= 4.0% attempted= 47 executed= 47 steps= 62 epsilon=0.9039 loss=1.396633
[dqn_original] episode 102/800 reward= -72.94 q= 14.99 safe100= 4.0% attempted= 88 executed= 88 steps= 107 epsilon=0.9029 loss=1.548436
[dqn_original] episode 103/800 reward= -140.69 q= 14.95 safe100= 4.0% attempted= 91 executed= 91 steps= 111 epsilon=0.9018 loss=1.416810
[dqn_original] episode 104/800 reward= -32.60 q= 15.05 safe100= 4.0% attempted= 55 executed= 55 steps= 71 epsilon=0.9010 loss=1.644511
[dqn_original] episode 105/800 reward= -168.49 q= 16.24 safe100= 4.0% attempted= 91 executed= 91 steps= 115 epsilon=0.8999 loss=1.644961

```

Figure 5: Timestamped training progress showing compact per-episode output

```

← → ↻ G |
Applications: Terminal - cloud@2025a...
Terminal - cloud@2025aa05312:~/drl

{
  "mean_episode_steps": 415.28
},
{
  "experiment": "ddqn_original",
  "algorithm": "DDQN",
  "environment_type": "Original",
  "mean_reward": 122.48670707316191,
  "reward_std": 126.00028449129385,
  "safe_landing_rate": 0.51,
  "mean_attempted_thruster_activations": 539.08,
  "mean_executed_thruster_activations": 539.08,
  "mean_episode_steps": 711.47
},
{
  "experiment": "dqn_modified",
  "algorithm": "DQN",
  "environment_type": "Modified",
  "mean_reward": 21.58272936370111,
  "reward_std": 165.21994489604148,
  "safe_landing_rate": 0.6,
  "mean_attempted_thruster_activations": 479.02,
  "mean_executed_thruster_activations": 407.84,
  "mean_episode_steps": 687.26
},
{
  "experiment": "dqn_original",
  "algorithm": "DQN",
  "environment_type": "Original",
  "mean_reward": 83.71733004326639,
  "reward_std": 179.90001122055736,
  "safe_landing_rate": 0.42,
  "mean_attempted_thruster_activations": 517.99,
  "mean_executed_thruster_activations": 517.99,
  "mean_episode_steps": 586.95
}
},
"best_algorithm_under_modified_environment": "DDQN"
}

(base) [cloud@2025aa05312 drl]$
(base) [cloud@2025aa05312 drl]$
(base) [cloud@2025aa05312 drl]$

```

Figure 6: Timestamped final outputs and required plots

References

1. Mnih, V. et al. (2015). Human-level control through deep reinforcement learning. *Nature*, 518, 529-533. [DOI record](#).
2. van Hasselt, H., Guez, A., and Silver, D. (2016). Deep Reinforcement Learning with Double Q-Learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 30(1). [DOI record](#).
3. Farama Foundation. [Gymnasium LunarLander-v3 documentation](#).

Appendix A - Complete commented source

src/robust_lunarlander/config.py

```

"""Central configuration for fair and reproducible DQN/DDQN experiments."""

from __future__ import annotations

from dataclasses import asdict, dataclass
from pathlib import Path
from typing import Any

@dataclass(frozen=True)
class TrainingConfig:
    """Store every experimental choice shared by all four training runs."""

    seed: int = 148
    episodes: int = 800
    max_steps_per_episode: int = 1_000
    gamma: float = 0.99
    learning_rate: float = 5e-4
    batch_size: int = 64
    replay_capacity: int = 100_000
    learning_starts: int = 1_000
    train_frequency: int = 1
    target_update_interval: int = 500
    epsilon_start: float = 1.0
    epsilon_end: float = 0.01
    epsilon_decay_steps: int = 100_000
    hidden_sizes: tuple[int, int] = (128, 128)
    gradient_clip_norm: float = 10.0
    validation_state_count: int = 512
    evaluation_episodes: int = 100
    failure_probability: float = 0.15
    attempted_thruster_penalty: float = 0.3
    safe_landing_bonus: float = 50.0
    device: str = "cpu"
    output_dir: Path = Path("artifacts")

    def as_serializable_dict(self) -> dict[str, Any]:
        """Return a JSON-safe representation suitable for provenance files."""

        values = asdict(self)
        values["output_dir"] = str(self.output_dir)
        values["hidden_sizes"] = list(self.hidden_sizes)
        return values

def linear_epsilon(step: int, config: TrainingConfig) -> float:
    """Linearly anneal epsilon from its start value to its final value."""

    progress = min(max(step, 0) / config.epsilon_decay_steps, 1.0)
    return config.epsilon_start + progress * (config.epsilon_end - config.epsilon_start)

```

src/robust_lunarlander/envs.py

```

"""Environment construction and the assignment-specified action-failure wrapper."""

from __future__ import annotations

from copy import deepcopy
from typing import Any

import gymnasium as gym
import numpy as np

def is_safe_landing(
    observation: np.ndarray,
    terminated: bool,
    truncated: bool,
) -> bool:
    """Return True only when every assignment safe-landing condition is met."""

    observation = np.asarray(observation)
    return bool(
        terminated
        and not truncated
        and observation[6] == 1
        and observation[7] == 1
        and abs(float(observation[2])) < 0.10
        and abs(float(observation[3])) < 0.10
        and abs(float(observation[4])) < 0.10
    )

```

```

class StochasticActionFailureWrapper(gym.Wrapper):
    """Apply hidden thruster failures, attempted-fuel cost, and safe-landing bonus.

    The selected and executed actions are retained only in private verification
    counters. The returned observation, termination flags, truncation flag, and
    info object come directly from the base environment. In particular, no failure
    indicator is leaked to the agent.

    A private RNG is used for actuator failures so drawing the failure event does
    not consume random numbers from LunarLander's own transition generator.
    """

    def __init__(
        self,
        env: gym.Env,
        failure_probability: float = 0.15,
        attempted_thruster_penalty: float = 0.3,
        safe_landing_bonus: float = 50.0,
    ) -> None:
        """Initialize the wrapper while preserving the base action and observation spaces."""

        super().__init__(env)
        if not 0.0 <= failure_probability <= 1.0:
            raise ValueError("failure_probability must lie in [0, 1].")
        if attempted_thruster_penalty < 0.0:
            raise ValueError("attempted_thruster_penalty must be non-negative.")
        self.failure_probability = float(failure_probability)
        self.attempted_thruster_penalty = float(attempted_thruster_penalty)
        self.safe_landing_bonus = float(safe_landing_bonus)
        self._failure_rng = np.random.default_rng()
        self._verification_counters = {
            "attempted_thruster_actions": 0,
            "misfired_thruster_actions": 0,
            "executed_thruster_actions": 0,
            "applied_fuel_penalties": 0,
            "safe_landing_bonuses": 0,
        }

    @property
    def verification_counters(self) -> dict[str, int]:
        """Return a defensive copy of private counters for external verification."""

        return deepcopy(self._verification_counters)

    def clear_verification_counters(self) -> None:
        """Reset private counters without changing any agent-visible state."""

        for key in self._verification_counters:
            self._verification_counters[key] = 0

    def reset(
        self,
        *,
        seed: int | None = None,
        options: dict[str, Any] | None = None,
    ) -> tuple[np.ndarray, dict[str, Any]]:
        """Reset LunarLander and independently seed the hidden failure process."""

        if seed is not None:
            # A distinct SeedSequence component avoids coupling failure draws to
            # the base environment even though both originate from one run seed.
            failure_seed = np.random.SeedSequence([int(seed), 0xFA11])
            self._failure_rng = np.random.default_rng(failure_seed)
        return self.env.reset(seed=seed, options=options)

    def step(
        self,
        action: int,
    ) -> tuple[np.ndarray, float, bool, bool, dict[str, Any]]:
        """Execute one assignment-compliant transition in the specified order."""

        selected_action = int(action)
        if not self.action_space.contains(selected_action):
            raise ValueError(f"Invalid action {selected_action!r} for {self.action_space}.")

        # Only a selected thruster command can fail. Action zero is never replaced.
        executed_action = selected_action
        if selected_action in (1, 2, 3):
            self._verification_counters["attempted_thruster_actions"] += 1
            if self._failure_rng.random() < self.failure_probability:
                executed_action = 0
                self._verification_counters["misfired_thruster_actions"] += 1
            else:
                self._verification_counters["executed_thruster_actions"] += 1

        # Delegate the final action to the unchanged LunarLander transition logic.
        observation, base_reward, terminated, truncated, info = self.env.step(executed_action)

        fuel_cost = self.attempted_thruster_penalty if selected_action != 0 else 0.0
        self._verification_counters["applied_fuel_penalties"] += int(selected_action != 0)
        landing_bonus = (
            self.safe_landing_bonus if is_safe_landing(observation, terminated, truncated) else 0.0
        )

```



```

        self._verification_counters["safe_landing_bonuses"] += int(landing_bonus != 0.0)
        modified_reward = float(base_reward) - fuel_cost + landing_bonus

        # Return the base info object exactly; diagnostics remain private.
        return observation, modified_reward, terminated, truncated, info

def make_environment(
    *,
    modified: bool,
    failure_probability: float = 0.15,
    attempted_thruster_penalty: float = 0.3,
    safe_landing_bonus: float = 50.0,
    render_mode: str | None = None,
) -> gym.Env:
    """Create either original LunarLander-v3 or the specified modified variant."""

    base_environment = gym.make("LunarLander-v3", render_mode=render_mode)
    if not modified:
        return base_environment
    return StochasticActionFailureWrapper(
        base_environment,
        failure_probability=failure_probability,
        attempted_thruster_penalty=attempted_thruster_penalty,
        safe_landing_bonus=safe_landing_bonus,
    )

```

src/robust_lunarlander/network.py

```

"""Neural-network definition shared without alteration by DQN and DDQN."""

from __future__ import annotations
from collections.abc import Sequence

import torch
from torch import nn

class QNetwork(nn.Module):
    """Map an eight-dimensional LunarLander state to four action values."""

    def __init__(
        self,
        observation_size: int,
        action_count: int,
        hidden_sizes: Sequence[int] = (128, 128),
    ) -> None:
        """Build a multilayer perceptron with ReLU hidden activations."""

        super().__init__()
        dimensions = [observation_size, *hidden_sizes, action_count]
        layers: list[nn.Module] = []
        for input_size, output_size in zip(dimensions[:-2], dimensions[1:-1], strict=True):
            layers.extend((nn.Linear(input_size, output_size), nn.ReLU()))
        layers.append(nn.Linear(dimensions[-2], dimensions[-1]))
        self.model = nn.Sequential(*layers)

    def forward(self, observations: torch.Tensor) -> torch.Tensor:
        """Return one predicted Q-value per discrete action."""

        return self.model(observations)

```

src/robust_lunarlander/replay.py

```

"""Preallocated experience replay used identically by DQN and DDQN."""

from __future__ import annotations
from typing import NamedTuple

import numpy as np
import torch

class ReplayBatch(NamedTuple):
    """Hold one tensor batch sampled uniformly from replay memory."""

    observations: torch.Tensor
    actions: torch.Tensor
    rewards: torch.Tensor
    next_observations: torch.Tensor
    terminated: torch.Tensor
    truncated: torch.Tensor

```

```

class ReplayBuffer:
    """Store transitions in a fixed-size circular buffer."""

    def __init__(
        self,
        capacity: int,
        observation_size: int,
        seed: int,
    ) -> None:
        """Allocate storage once and initialize an isolated sampling generator."""

        self.capacity = int(capacity)
        self.observations = np.empty((capacity, observation_size), dtype=np.float32)
        self.actions = np.empty(capacity, dtype=np.int64)
        self.rewards = np.empty(capacity, dtype=np.float32)
        self.next_observations = np.empty((capacity, observation_size), dtype=np.float32)
        self.terminated = np.empty(capacity, dtype=np.float32)
        self.truncated = np.empty(capacity, dtype=np.float32)
        self.position = 0
        self.size = 0
        self.rng = np.random.default_rng(seed)

    def add(
        self,
        observation: np.ndarray,
        action: int,
        reward: float,
        next_observation: np.ndarray,
        terminated: bool,
        truncated: bool,
    ) -> None:
        """Insert one transition with both Gymnasium ending flags preserved."""

        index = self.position
        self.observations[index] = observation
        self.actions[index] = action
        self.rewards[index] = reward
        self.next_observations[index] = next_observation
        self.terminated[index] = float(terminated)
        self.truncated[index] = float(truncated)
        self.position = (self.position + 1) % self.capacity
        self.size = min(self.size + 1, self.capacity)

    def sample(self, batch_size: int, device: torch.device) -> ReplayBatch:
        """Sample transitions uniformly and move the resulting tensors to a device."""

        if self.size < batch_size:
            raise ValueError("Cannot sample more transitions than the buffer currently stores.")
        indices = self.rng.integers(0, self.size, size=batch_size)
        return ReplayBatch(
            torch.as_tensor(self.observations[indices], device=device),
            torch.as_tensor(self.actions[indices], device=device).unsqueeze(1),
            torch.as_tensor(self.rewards[indices], device=device).unsqueeze(1),
            torch.as_tensor(self.next_observations[indices], device=device),
            torch.as_tensor(self.terminated[indices], device=device).unsqueeze(1),
            torch.as_tensor(self.truncated[indices], device=device).unsqueeze(1),
        )

    def __len__(self) -> int:
        """Return the number of valid transitions currently stored."""

        return self.size

```

src/robust_lunarlander/agent.py

```

"""A single agent implementation whose target calculation selects DQN or DDQN."""

from __future__ import annotations
from typing import Literal

import numpy as np
import torch
from torch import nn

from .config import TrainingConfig
from .network import QNetwork
from .replay import ReplayBatch, ReplayBuffer

Algorithm = Literal["dqn", "ddqn"]

class ValueAgent:
    """Train DQN or DDQN with an otherwise identical architecture and optimizer."""

    def __init__(
        self,
        observation_size: int,
        action_count: int,

```



```

    algorithm: Algorithm,
    config: TrainingConfig,
) -> None:
    """Create online/target networks, Adam optimizer, replay memory, and RNG."""

    if algorithm not in ("dqn", "ddqn"):
        raise ValueError("algorithm must be either 'dqn' or 'ddqn'.")
    self.algorithm = algorithm
    self.action_count = action_count
    self.config = config
    self.device = torch.device(config.device)
    self.rng = np.random.default_rng(config.seed)

    self.online_network = QNetwork(
        observation_size,
        action_count,
        config.hidden_sizes,
    ).to(self.device)
    self.target_network = QNetwork(
        observation_size,
        action_count,
        config.hidden_sizes,
    ).to(self.device)
    self.target_network.load_state_dict(self.online_network.state_dict())
    self.target_network.eval()

    self.optimizer = torch.optim.Adam(
        self.online_network.parameters(),
        lr=config.learning_rate,
    )
    self.loss_function = nn.SmoothL1Loss()
    self.replay = ReplayBuffer(
        config.replay_capacity,
        observation_size,
        config.seed + 1,
    )
    self.update_count = 0

def select_action(self, observation: np.ndarray, epsilon: float) -> int:
    """Choose a random action with probability epsilon, otherwise act greedily."""

    if self.rng.random() < epsilon:
        return int(self.rng.integers(self.action_count))
    with torch.no_grad():
        state = torch.as_tensor(
            observation,
            dtype=torch.float32,
            device=self.device,
        ).unsqueeze(0)
        return int(self.online_network(state).argmax(dim=1).item())

def _bootstrap_values(self, batch: ReplayBatch) -> torch.Tensor:
    """Compute the only algorithm-specific term: the next-state target value."""

    with torch.no_grad():
        if self.algorithm == "dqn":
            return self.target_network(batch.next_observations).max(dim=1, keepdim=True).values

        online_actions = self.online_network(batch.next_observations).argmax(
            dim=1,
            keepdim=True,
        )
        return self.target_network(batch.next_observations).gather(1, online_actions)

def _target_values(self, batch: ReplayBatch) -> torch.Tensor:
    """Build TD targets, masking terminals but bootstrapping truncations."""

    bootstrap_values = self._bootstrap_values(batch)
    # A true MDP terminal state has no future return. Time-limit truncation is
    # stored separately and still bootstraps because the underlying state is
    # not terminal; the episode ended only because of an external step limit.
    return batch.rewards + self.config.gamma * (1.0 - batch.terminated) * bootstrap_values

def update(self) -> float:
    """Perform one replay-based temporal-difference optimization step."""

    batch = self.replay.sample(self.config.batch_size, self.device)
    predicted_values = self.online_network(batch.observations).gather(1, batch.actions)
    target_values = self._target_values(batch)

    loss = self.loss_function(predicted_values, target_values)
    self.optimizer.zero_grad(set_to_none=True)
    loss.backward()
    nn.utils.clip_grad_norm_(
        self.online_network.parameters(),
        self.config.gradient_clip_norm,
    )
    self.optimizer.step()

    self.update_count += 1
    if self.update_count % self.config.target_update_interval == 0:
        self.target_network.load_state_dict(self.online_network.state_dict())
    return float(loss.item())

```

```

def average_max_q(self, validation_states: np.ndarray) -> float:
    """Measure mean max action value on one unchanged validation-state set."""
    with torch.no_grad():
        states = torch.as_tensor(
            validation_states,
            dtype=torch.float32,
            device=self.device,
        )
        values = self.online_network(states).max(dim=1).values
    return float(values.mean().item())

def save(self, path: str) -> None:
    """Persist trained weights and the experimental configuration."""
    torch.save(
        {
            "algorithm": self.algorithm,
            "online_network": self.online_network.state_dict(),
            "target_network": self.target_network.state_dict(),
            "optimizer": self.optimizer.state_dict(),
            "config": self.config.as_serializable_dict(),
        },
        path,
    )

def load(self, path: str) -> None:
    """Restore online and target weights from a saved checkpoint."""
    checkpoint = torch.load(path, map_location=self.device, weights_only=False)
    if checkpoint["algorithm"] != self.algorithm:
        raise ValueError("Checkpoint algorithm does not match this agent.")
    self.online_network.load_state_dict(checkpoint["online_network"])
    self.target_network.load_state_dict(checkpoint["target_network"])
    self.optimizer.load_state_dict(checkpoint["optimizer"])

```

src/robust_lunarlander/verification.py

```

"""Experimental and boundary verification of the modified environment."""

from __future__ import annotations

import argparse
import json
import math
import platform
import sys
from dataclasses import dataclass
from datetime import datetime
from pathlib import Path
from typing import Any

import gymnasium as gym
import numpy as np
import pandas as pd
from gymnasium import spaces

from .envs import StochasticActionFailureWrapper, is_safe_landing

class ActionRewardSpy(gym.Wrapper):
    """Record base-environment inputs and outputs outside the agent-facing wrapper."""

    def __init__(self, env: gym.Env) -> None:
        """Initialize an empty last-transition record."""
        super().__init__(env)
        self.last: dict[str, Any] | None = None

    def step(
        self,
        action: int,
    ) -> tuple[np.ndarray, float, bool, bool, dict[str, Any]]:
        """Forward the action and retain the unmodified response for verification."""
        observation, reward, terminated, truncated, info = self.env.step(action)
        self.last = {
            "executed_action": int(action),
            "observation": np.asarray(observation).copy(),
            "base_reward": float(reward),
            "terminated": bool(terminated),
            "truncated": bool(truncated),
            "info": info,
        }
        return observation, reward, terminated, truncated, info

class ScriptedTerminalEnvironment(gym.Env):

```

```

"""Return one specified terminal transition for controlled reward tests."""

metadata: dict[str, Any] = {}

def __init__(
    self,
    observation: np.ndarray,
    *,
    base_reward: float = 7.0,
    terminated: bool = True,
    truncated: bool = False,
) -> None:
    """Store the exact response that step must return."""

    super().__init__()
    self.action_space = spaces.Discrete(4)
    self.observation_space = spaces.Box(-np.inf, np.inf, shape=(8,), dtype=np.float32)
    self.response_observation = np.asarray(observation, dtype=np.float32)
    self.base_reward = float(base_reward)
    self.terminated = bool(terminated)
    self.truncated = bool(truncated)
    self.received_action: int | None = None
    self.info_object = {"source": "scripted-base"}

def reset(
    self,
    *,
    seed: int | None = None,
    options: dict[str, Any] | None = None,
) -> tuple[np.ndarray, dict[str, Any]]:
    """Reset Gymnasium's random generator and return a neutral observation."""

    super().reset(seed=seed)
    del options
    return np.zeros(8, dtype=np.float32), {}

def step(
    self,
    action: int,
) -> tuple[np.ndarray, float, bool, bool, dict[str, Any]]:
    """Return the configured response while recording the executed action."""

    self.received_action = int(action)
    return (
        self.response_observation.copy(),
        self.base_reward,
        self.terminated,
        self.truncated,
        self.info_object,
    )

@dataclass(frozen=True)
class BoundaryCase:
    """Describe one controlled safe-landing or fuel-penalty check."""

    name: str
    observation: list[float]
    terminated: bool
    truncated: bool
    selected_action: int
    failure_probability: float
    expected_bonus: float
    expected_executed_action: int

def _safe_observation() -> np.ndarray:
    """Return an observation satisfying all five state thresholds."""

    return np.array([0.0, 0.0, 0.02, -0.03, 0.04, 0.0, 1.0, 1.0], dtype=np.float32)

def controlled_boundary_cases() -> list[BoundaryCase]:
    """Enumerate positive, negative, boundary, truncation, and misfire cases."""

    safe = _safe_observation()
    cases: list[BoundaryCase] = [
        BoundaryCase("all_conditions_true", safe.tolist(), True, False, 0, 0.0, 50.0, 0),
        BoundaryCase("attempted_thruster_success", safe.tolist(), True, False, 2, 0.0, 50.0, 2),
        BoundaryCase("attempted_thruster_misfire", safe.tolist(), True, False, 2, 1.0, 50.0, 0),
    ]
    mutations = [
        ("not_terminated", 2, 0.02, False, False),
        ("episode_truncated", 2, 0.02, True, True),
        ("left_leg_absent", 6, 0.0, True, False),
        ("right_leg_absent", 7, 0.0, True, False),
        ("excess_horizontal_velocity", 2, 0.11, True, False),
        ("excess_vertical_velocity", 3, -0.11, True, False),
        ("excess_orientation_angle", 4, 0.11, True, False),
    ]
    for name, index, value, terminated, truncated in mutations:
        observation = safe.copy()
        observation[index] = value

```

```

        cases.append(
            BoundaryCase(
                name,
                observation.tolist(),
                terminated,
                truncated,
                0,
                0.0,
                0.0,
                0,
            )
        )
    return cases

def run_controlled_boundary_verification() -> pd.DataFrame:
    """Execute every controlled case and prove exact reward/action semantics."""
    rows: list[dict[str, Any]] = []
    for case in controlled_boundary_cases():
        base = ScriptedTerminalEnvironment(
            np.asarray(case.observation, dtype=np.float32),
            terminated=case.terminated,
            truncated=case.truncated,
        )
        wrapped = StochasticActionFailureWrapper(
            base,
            failure_probability=case.failure_probability,
        )
        wrapped.reset(seed=148)
        observation, reward, terminated, truncated, returned_info = wrapped.step(
            case.selected_action
        )
        counters = wrapped.verification_counters
        expected_penalty = 0.3 if case.selected_action != 0 else 0.0
        expected_reward = base.base_reward - expected_penalty + case.expected_bonus
        rows.append(
            {
                "case": case.name,
                "selected_action": case.selected_action,
                "executed_action": base.received_action,
                "expected_executed_action": case.expected_executed_action,
                "safe_landing_detected": is_safe_landing(observation, terminated, truncated),
                "expected_bonus": case.expected_bonus,
                "observed_reward": float(reward),
                "expected_reward": expected_reward,
                "fuel_penalty": expected_penalty,
                "counter_attempted_actions": counters["attempted_thruster_actions"],
                "counter_executed_actions": counters["executed_thruster_actions"],
                "counter_misfires": counters["misfired_thruster_actions"],
                "info_identity_preserved": returned_info is base.info_object,
                "passed": bool(
                    base.received_action == case.expected_executed_action
                    and math.isclose(float(reward), expected_reward, abs_tol=1e-7)
                    and returned_info is base.info_object
                    and counters["applied_fuel_penalties"] == int(case.selected_action != 0)
                    and counters["safe_landing_bonuses"] == int(case.expected_bonus != 0.0)
                ),
            }
        )
    wrapped.close()
    return pd.DataFrame(rows)

def _wilson_interval(successes: int, trials: int) -> tuple[float, float]:
    """Return a 95 percent Wilson score interval for a binomial proportion."""
    if trials == 0:
        return float("nan"), float("nan")
    z = 1.959963984540054
    proportion = successes / trials
    denominator = 1.0 + z * z / trials
    centre = (proportion + z * z / (2.0 * trials)) / denominator
    radius = (
        z
        * math.sqrt(proportion * (1.0 - proportion) / trials + z * z / (4.0 * trials * trials))
        / denominator
    )
    return centre - radius, centre + radius

def run_random_policy_verification(
    *,
    episodes: int,
    seed: int,
) -> tuple[dict[str, Any], pd.DataFrame]:
    """Run many random LunarLander episodes and compare selected/executed actions."""
    spy = ActionRewardSpy(gym.make("LunarLander-v3"))
    environment = StochasticActionFailureWrapper(spy)
    action_rng = np.random.default_rng(seed + 10_000)
    records: list[dict[str, Any]] = []
    total_steps = 0

```

```

attempted_thrusters = 0
misfires = 0
penalty_checks = 0
penalty_mismatches = 0
info_identity_mismatches = 0
unexpected_action_replacements = 0
observed_safe_landings = 0

for episode in range(1, episodes + 1):
    observation, _ = environment.reset(seed=seed + episode - 1)
    del observation
    terminated = truncated = False
    episode_step = 0
    while not (terminated or truncated):
        selected_action = int(action_rng.integers(environment.action_space.n))
        observation, reward, terminated, truncated, info = environment.step(selected_action)
        if spy.last is None:
            raise RuntimeError("The verification spy did not record a base transition.")

        executed_action = int(spy.last["executed_action"])
        safe = is_safe_landing(observation, terminated, truncated)
        expected_penalty = 0.3 if selected_action != 0 else 0.0
        expected_bonus = 50.0 if safe else 0.0
        expected_reward = float(spy.last["base_reward"]) - expected_penalty + expected_bonus
        penalty_ok = math.isclose(float(reward), expected_reward, abs_tol=1e-6)

        if selected_action != 0:
            attempted_thrusters += 1
            penalty_checks += 1
            misfires += int(executed_action == 0)
            penalty_mismatches += int(not penalty_ok)
            unexpected_action_replacements += int(executed_action not in (0, selected_action))
        else:
            unexpected_action_replacements += int(executed_action != 0)
            info_identity_mismatches += int(info is not spy.last["info"])
            observed_safe_landings += int(safe)
            total_steps += 1
            episode_step += 1

        if len(records) < 500:
            records.append(
                {
                    "episode": episode,
                    "step": episode_step,
                    "selected_action": selected_action,
                    "executed_action": executed_action,
                    "misfire": selected_action != 0 and executed_action == 0,
                    "base_reward": float(spy.last["base_reward"]),
                    "modified_reward": float(reward),
                    "expected_fuel_penalty": expected_penalty,
                    "safe_landing_bonus": expected_bonus,
                    "reward_check_passed": penalty_ok,
                }
            )

    internal_counters = environment.verification_counters
    environment.close()
    interval_low, interval_high = _wilson_interval(misfires, attempted_thrusters)
    observed_rate = misfires / attempted_thrusters
    summary = {
        "episodes": episodes,
        "total_steps": total_steps,
        "attempted_thruster_actions": attempted_thrusters,
        "misfired_thruster_actions": misfires,
        "observed_misfire_rate": observed_rate,
        "target_misfire_rate": 0.15,
        "misfire_rate_absolute_error": abs(observed_rate - 0.15),
        "misfire_rate_wilson_95_low": interval_low,
        "misfire_rate_wilson_95_high": interval_high,
        "target_inside_wilson_interval": interval_low <= 0.15 <= interval_high,
        "fuel_penalty_checks": penalty_checks,
        "fuel_penalty_count": internal_counters["applied_fuel_penalties"],
        "fuel_penalty_count_matches": (
            internal_counters["applied_fuel_penalties"] == attempted_thrusters
        ),
        "fuel_penalty_mismatches": penalty_mismatches,
        "internal_attempt_count_matches": (
            internal_counters["attempted_thruster_actions"] == attempted_thrusters
        ),
        "internal_misfire_count_matches": (
            internal_counters["misfired_thruster_actions"] == misfires
        ),
        "executed_thruster_actions": internal_counters["executed_thruster_actions"],
        "safe_landing_bonus_count": internal_counters["safe_landing_bonuses"],
        "safe_landing_bonus_count_matches": (
            internal_counters["safe_landing_bonuses"] == observed_safe_landings
        ),
        "info_identity_mismatches": info_identity_mismatches,
        "unexpected_action_replacements": unexpected_action_replacements,
        "safe_landings_observed_under_random_policy": observed_safe_landings,
        "passed": bool(
            interval_low <= 0.15 <= interval_high
            and penalty_mismatches == 0
        )
    }

```

```

        and internal_counters["applied_fuel_penalties"] == attempted_thrusters
        and internal_counters["attempted_thruster_actions"] == attempted_thrusters
        and internal_counters["misfired_thruster_actions"] == misfires
        and internal_counters["safe_landing_bonuses"] == observed_safe_landings
        and info_identity_mismatches == 0
        and unexpected_action_replacements == 0
    ),
}
return summary, pd.DataFrame(records)

def save_verification_artifacts(
    *,
    output_dir: Path,
    episodes: int,
    seed: int,
) -> dict[str, Any]:
    """Run all checks and write machine-readable evidence files."""

    output_dir.mkdir(parents=True, exist_ok=True)
    boundary_results = run_controlled_boundary_verification()
    random_summary, random_samples = run_random_policy_verification(
        episodes=episodes,
        seed=seed,
    )
    summary = {
        "generated_at": datetime.now().astimezone().isoformat(timespec="seconds"),
        "seed": seed,
        "python_version": sys.version.split()[0],
        "platform": platform.platform(),
        "gymnasium_version": gym.__version__,
        "random_policy": random_summary,
        "controlled_boundary_cases": {
            "count": int(len(boundary_results)),
            "passed": int(boundary_results["passed"].sum()),
            "all_passed": bool(boundary_results["passed"].all()),
        },
        "overall_passed": bool(random_summary["passed"] and boundary_results["passed"].all()),
    }
    (output_dir / "wrapper_verification.json").write_text(
        json.dumps(summary, indent=2) + "\n",
        encoding="utf-8",
    )
    boundary_results.to_csv(output_dir / "controlled_boundary_cases.csv", index=False)
    random_samples.to_csv(output_dir / "random_policy_transition_sample.csv", index=False)
    return summary

def main() -> None:
    """Parse command-line options and produce the complete verification bundle."""

    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument("--episodes", type=int, default=250)
    parser.add_argument("--seed", type=int, default=148)
    parser.add_argument(
        "--output-dir",
        type=Path,
        default=Path("artifacts/verification"),
    )
    args = parser.parse_args()
    summary = save_verification_artifacts(
        output_dir=args.output_dir,
        episodes=args.episodes,
        seed=args.seed,
    )
    print(json.dumps(summary, indent=2))
    if not summary["overall_passed"]:
        raise SystemExit("Wrapper verification failed.")

if __name__ == "__main__":
    main()

```

src/robust_lunarlander/experiment.py

```

"""Reproducible training, evaluation, and evidence generation for all four agents."""

from __future__ import annotations

import argparse
import hashlib
import json
import os
import platform
import random
import socket
import subprocess
import sys
import time

```

```

from dataclasses import replace
from datetime import datetime
from pathlib import Path
from typing import Any

import gymnasium as gym
import numpy as np
import pandas as pd
import seaborn as sns
import torch

from .agent import Algorithm, ValueAgent
from .config import TrainingConfig, linear_epsilon
from .envs import StochasticActionFailureWrapper, is_safe_landing, make_environment
from .plotting import create_all_plots, summarize_experiments

EXPERIMENTS: tuple[tuple[Algorithm, bool], ...] = (
    ("dqn", False),
    ("ddqn", False),
    ("dqn", True),
    ("ddqn", True),
)

def experiment_name(algorithm: Algorithm, modified: bool) -> str:
    """Return a stable artifact name for one algorithm/environment pairing."""

    environment_name = "modified" if modified else "original"
    return f"{algorithm}_{environment_name}"

def set_reproducible_seeds(seed: int, environment: gym.Env | None = None) -> None:
    """Seed all supported RNGs and request deterministic PyTorch behavior.

    Box2D and some GPU kernels can still vary across platforms, so provenance is
    recorded and exact cross-machine bitwise equality is not promised.
    """

    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    if torch.cuda.is_available():
        torch.cuda.manual_seed_all(seed)
    torch.use_deterministic_algorithms(True, warn_only=True)
    if torch.backends.cudnn.is_available():
        torch.backends.cudnn.benchmark = False
        torch.backends.cudnn.deterministic = True
    if environment is not None:
        environment.action_space.seed(seed)
        environment.reset(seed=seed)

def collect_fixed_validation_states(
    config: TrainingConfig,
    destination: Path,
) -> np.ndarray:
    """Collect one deterministic random-policy state set shared by all agents."""

    if destination.exists():
        states = np.load(destination)["states"]
        expected_shape = (config.validation_state_count, 8)
        if states.shape != expected_shape:
            raise ValueError(
                f"Existing validation set has shape {states.shape}, expected {expected_shape}."
            )
        return states.astype(np.float32, copy=False)

    environment = make_environment(modified=False)
    set_reproducible_seeds(config.seed, environment)
    action_rng = np.random.default_rng(config.seed + 20_000)
    states: list[np.ndarray] = []
    episode = 0
    while len(states) < config.validation_state_count:
        observation, _ = environment.reset(seed=config.seed + episode)
        terminated = truncated = False
        while not (terminated or truncated) and len(states) < config.validation_state_count:
            states.append(np.asarray(observation, dtype=np.float32).copy())
            action = int(action_rng.integers(environment.action_space.n))
            observation, _, terminated, truncated, _ = environment.step(action)
        episode += 1
    environment.close()

    validation_states = np.stack(states)
    destination.parent.mkdir(parents=True, exist_ok=True)
    np.savez_compressed(destination, states=validation_states)
    return validation_states

def validation_set_sha256(validation_states: np.ndarray) -> str:
    """Hash the exact validation states to prove that all runs use one fixed set."""

    return hashlib.sha256(validation_states.tobytes()).hexdigest()

```



```

def train_agent(
    *,
    algorithm: Algorithm,
    modified: bool,
    config: TrainingConfig,
    validation_states: np.ndarray,
    force: bool = False,
) -> pd.DataFrame:
    """Train one agent and write per-episode metrics plus its final checkpoint."""

    name = experiment_name(algorithm, modified)
    metrics_dir = config.output_dir / "metrics"
    checkpoints_dir = config.output_dir / "checkpoints"
    metrics_dir.mkdir(parents=True, exist_ok=True)
    checkpoints_dir.mkdir(parents=True, exist_ok=True)
    metrics_path = metrics_dir / f"{name}.csv"
    checkpoint_path = checkpoints_dir / f"{name}.pt"

    required_columns = {
        "episode",
        "episode_reward",
        "average_predicted_q",
        "successful_safe_landing",
        "moving_safe_landing_rate_100",
        "attempted_thruster_activations",
        "executed_thruster_activations",
        "average_attempted_thruster_activations_per_episode",
        "episode_steps",
        "epsilon",
        "training_loss",
        "environment_type",
        "algorithm",
        "random_seed",
    }

    if metrics_path.exists() and checkpoint_path.exists() and not force:
        metrics = pd.read_csv(metrics_path)
        if len(metrics) == config.episodes and required_columns.issubset(metrics.columns):
            print(f"[{name}] Reusing complete artifacts ({config.episodes} episodes).", flush=True)
            return metrics
        print(f"[{name}] Existing artifacts use an incomplete schema; retraining.", flush=True)

    environment = make_environment(
        modified=modified,
        failure_probability=config.failure_probability,
        attempted_thruster_penalty=config.attempted_thruster_penalty,
        safe_landing_bonus=config.safe_landing_bonus,
    )

    set_reproducible_seeds(config.seed, environment)
    observation_size = int(np.prod(environment.observation_space.shape))
    action_count = int(environment.action_space.n)
    agent = ValueAgent(observation_size, action_count, algorithm, config)

    rows: list[dict[str, Any]] = []
    global_step = 0
    run_started = time.perf_counter()
    progress_dir = config.output_dir / "logs"
    progress_dir.mkdir(parents=True, exist_ok=True)
    progress_path = progress_dir / f"{name}.log"
    progress_path.write_text(
        "episode,total_reward,average_q,safe_rate_100,attempted,executed,"
        "steps,epsilon,training_loss\n",
        encoding="utf-8",
    )

    for episode in range(1, config.episodes + 1):
        episode_started = time.perf_counter()
        observation, _ = environment.reset(seed=config.seed + episode - 1)
        counters_before = (
            environment.verification_counters
            if isinstance(environment, StochasticActionFailureWrapper)
            else None
        )
        terminated = truncated = False
        episode_reward = 0.0
        attempted_thrusters = 0
        episode_losses: list[float] = []
        last_observation = np.asarray(observation, dtype=np.float32)
        episode_steps = 0

        for _ in range(config.max_steps_per_episode):
            episode_steps += 1
            epsilon = linear_epsilon(global_step, config)
            action = agent.select_action(observation, epsilon)
            attempted_thrusters += int(action != 0)
            next_observation, reward, terminated, truncated, _ = environment.step(action)
            episode_done = terminated or truncated

            agent.replay.add(
                np.asarray(observation, dtype=np.float32),
                action,
                float(reward),
                np.asarray(next_observation, dtype=np.float32),
            )

```

```

        bool(terminated),
        bool(truncated),
    )
    if (
        global_step >= config.learning_starts
        and global_step % config.train_frequency == 0
        and len(agent.replay) >= config.batch_size
    ):
        episode_losses.append(agent.update())

        episode_reward += float(reward)
        observation = next_observation
        last_observation = np.asarray(next_observation, dtype=np.float32)
        global_step += 1
        if episode_done:
            break

    successful_landing = is_safe_landing(last_observation, terminated, truncated)
    previous_successes = [int(row["successful_safe_landing"]) for row in rows[-99:]]
    moving_success_rate = float(np.mean([*previous_successes, int(successful_landing)]))
    average_q = agent.average_max_q(validation_states)
    if isinstance(environment, StochasticActionFailureWrapper):
        counters_after = environment.verification_counters
        assert counters_before is not None
        executed_thrusters = (
            counters_after["executed_thruster_actions"]
            - counters_before["executed_thruster_actions"]
        )
    else:
        executed_thrusters = attempted_thrusters
    average_attempts = (
        sum(float(row["attempted_thruster_activations"]) for row in rows) + attempted_thrusters
    ) / episode
    training_loss = float(np.mean(episode_losses)) if episode_losses else np.nan
    episode_seconds = time.perf_counter() - episode_started
    rows.append(
        {
            "experiment": name,
            "algorithm": algorithm.upper(),
            "environment_type": "Modified" if modified else "Original",
            "episode": episode,
            "episode_reward": episode_reward,
            "average_predicted_q": average_q,
            "successful_safe_landing": int(successful_landing),
            "moving_safe_landing_rate_100": moving_success_rate,
            "attempted_thruster_activations": attempted_thrusters,
            "executed_thruster_activations": executed_thrusters,
            "average_attempted_thruster_activations_per_episode": average_attempts,
            "episode_steps": episode_steps,
            "epsilon": linear_epsilon(global_step, config),
            "training_loss": training_loss,
            "global_step": global_step,
            "episode_seconds": episode_seconds,
            "random_seed": config.seed,
            "episode_seed": config.seed + episode - 1,
        }
    )

    loss_text = "nan" if np.isnan(training_loss) else f"{training_loss:.6f}"
    progress_line = (
        f"[{name}] episode {episode:4d}/{config.episodes} "
        f"reward={episode_reward:9.2f} q={average_q:8.2f} "
        f"safe100={moving_success_rate:6.1%} "
        f"attempted={attempted_thrusters:4d} executed={executed_thrusters:4d} "
        f"steps={episode_steps:4d} epsilon={linear_epsilon(global_step, config):.4f} "
        f"loss={loss_text}"
    )
    print(progress_line, flush=True)
    with progress_path.open("a", encoding="utf-8") as progress_file:
        progress_file.write(
            f"{episode},{episode_reward:.10f},{average_q:.10f}, "
            f"{moving_success_rate:.10f},{attempted_thrusters}, "
            f"{executed_thrusters},{episode_steps}, "
            f"{linear_epsilon(global_step, config):.10f},{loss_text}\n"
        )

    environment.close()
    agent.save(str(checkpoint_path))
    metrics = pd.DataFrame(rows)
    metrics.to_csv(metrics_path, index=False)
    elapsed = time.perf_counter() - run_started
    print(f"[{name}] completed in {elapsed / 60.0:.2f} minutes.", flush=True)
    return metrics

def evaluate_agent(
    *,
    algorithm: Algorithm,
    modified: bool,
    config: TrainingConfig,
) -> pd.DataFrame:
    """Evaluate one trained agent greedily over a shared deterministic seed list."""

```

```

name = experiment_name(algorithm, modified)
checkpoint_path = config.output_dir / "checkpoints" / f"{name}.pt"
if not checkpoint_path.exists():
    raise FileNotFoundError(f"Missing checkpoint: {checkpoint_path}")

environment = make_environment(
    modified=modified,
    failure_probability=config.failure_probability,
    attempted_thruster_penalty=config.attempted_thruster_penalty,
    safe_landing_bonus=config.safe_landing_bonus,
)
set_reproducible_seeds(config.seed, environment)
observation_size = int(np.prod(environment.observation_space.shape))
action_count = int(environment.action_space.n)
agent = ValueAgent(observation_size, action_count, algorithm, config)
agent.load(str(checkpoint_path))
rows: list[dict[str, Any]] = []

for episode in range(1, config.evaluation_episodes + 1):
    observation, _ = environment.reset(seed=config.seed + 100_000 + episode - 1)
    counters_before = (
        environment.verification_counters
        if isinstance(environment, StochasticActionFailureWrapper)
        else None
    )
    terminated = truncated = False
    episode_reward = 0.0
    attempted_thrusters = 0
    last_observation = np.asarray(observation, dtype=np.float32)
    episode_steps = 0

    for _ in range(config.max_steps_per_episode):
        episode_steps += 1
        action = agent.select_action(observation, epsilon=0.0)
        attempted_thrusters += int(action != 0)
        observation, reward, terminated, truncated, _ = environment.step(action)
        episode_reward += float(reward)
        last_observation = np.asarray(observation, dtype=np.float32)
        if terminated or truncated:
            break

    if isinstance(environment, StochasticActionFailureWrapper):
        counters_after = environment.verification_counters
        assert counters_before is not None
        executed_thrusters = (
            counters_after["executed_thruster_actions"]
            - counters_before["executed_thruster_actions"]
        )
    else:
        executed_thrusters = attempted_thrusters
    rows.append(
        {
            "experiment": name,
            "algorithm": algorithm.upper(),
            "environment_type": "Modified" if modified else "Original",
            "evaluation_episode": episode,
            "reward": episode_reward,
            "successful_safe_landing": int(
                is_safe_landing(last_observation, terminated, truncated)
            ),
            "attempted_thruster_activations": attempted_thrusters,
            "executed_thruster_activations": executed_thrusters,
            "episode_steps": episode_steps,
            "random_seed": config.seed,
            "episode_seed": config.seed + 100_000 + episode - 1,
        }
    )

environment.close()
return pd.DataFrame(rows)

def write_provenance(
    config: TrainingConfig,
    validation_states: np.ndarray,
) -> None:
    """Record exact configuration, library versions, hardware, and wall-clock time."""
    try:
        git_status = subprocess.run(
            ["git", "status", "--porcelain"],
            check=True,
            capture_output=True,
            text=True,
        ).stdout
        git_worktree_dirty = bool(git_status.strip())
    except (OSError, subprocess.CalledProcessError):
        git_worktree_dirty = None

    config.output_dir.mkdir(parents=True, exist_ok=True)
    (config.output_dir / "training_config.json").write_text(
        json.dumps(config.as_serializable_dict(), indent=2) + "\n",
        encoding="utf-8",
    )

```

```

)
try:
    git_commit = subprocess.run(
        ["git", "rev-parse", "HEAD"],
        check=True,
        capture_output=True,
        text=True,
    ).stdout.strip()
except (OSError, subprocess.CalledProcessError):
    git_commit = "unavailable"
cuda_available = torch.cuda.is_available()
provenance = {
    "generated_at": datetime.now().astimezone().isoformat(timespec="seconds"),
    "hostname": socket.gethostname(),
    "platform": platform.platform(),
    "processor": platform.processor(),
    "logical_cpu_count": os.cpu_count(),
    "python": sys.version,
    "gymnasium": gym.__version__,
    "numpy": np.__version__,
    "pandas": pd.__version__,
    "seaborn": sns.__version__,
    "torch": torch.__version__,
    "device": config.device,
    "cuda_available": cuda_available,
    "cuda_device_name": torch.cuda.get_device_name(0) if cuda_available else None,
    "deterministic_algorithms_requested": torch.are_deterministic_algorithms_enabled(),
    "git_commit": git_commit,
    "git_worktree_dirty_at_execution": git_worktree_dirty,
    "environment_id": "LunarLander-v3",
    "validation_set_shape": list(validation_states.shape),
    "validation_set_sha256": validation_set_sha256(validation_states),
}
(config.output_dir / "system_provenance.json").write_text(
    json.dumps(provenance, indent=2) + "\n",
    encoding="utf-8",
)

def run_complete_study(config: TrainingConfig, *, force: bool = False) -> dict[str, Any]:
    """Run the four fair experiments, greedy evaluations, plots, and summaries."""

    validation_path = config.output_dir / "validation" / "fixed_validation_states.npz"
    validation_states = collect_fixed_validation_states(config, validation_path)
    set_reproducible_seeds(config.seed)
    write_provenance(config, validation_states)

    metrics: dict[str, pd.DataFrame] = {}
    for algorithm, modified in EXPERIMENTS:
        name = experiment_name(algorithm, modified)
        metrics[name] = train_agent(
            algorithm=algorithm,
            modified=modified,
            config=config,
            validation_states=validation_states,
            force=force,
        )

    evaluations = pd.concat(
        [
            evaluate_agent(algorithm=algorithm, modified=modified, config=config)
            for algorithm, modified in EXPERIMENTS
        ],
        ignore_index=True,
    )
    evaluations.to_csv(config.output_dir / "evaluation_episodes.csv", index=False)
    evaluation_summary = (
        evaluations.groupby(
            ["experiment", "algorithm", "environment_type"],
            as_index=False,
        )
        .agg(
            mean_reward=("reward", "mean"),
            reward_std=("reward", "std"),
            safe_landing_rate=("successful_safe_landing", "mean"),
            mean_attempted_thruster_activations=("attempted_thruster_activations", "mean"),
            mean_executed_thruster_activations=("executed_thruster_activations", "mean"),
            mean_episode_steps=("episode_steps", "mean"),
        )
        .sort_values("experiment")
    )
    evaluation_summary.to_csv(config.output_dir / "evaluation_summary.csv", index=False)

    create_all_plots(metrics, config.output_dir / "plots")
    summary = summarize_experiments(metrics, evaluation_summary)
    final_comparison = pd.DataFrame(summary["final_comparison"])
    final_comparison.to_csv(config.output_dir / "final_comparison.csv", index=False)
    (config.output_dir / "study_summary.json").write_text(
        json.dumps(summary, indent=2) + "\n",
        encoding="utf-8",
    )
    print(json.dumps(summary, indent=2), flush=True)
    return summary

```

```
def main() -> None:
    """Parse command-line options and launch the complete assignment experiment."""

    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument("--episodes", type=int, default=800)
    parser.add_argument("--evaluation-episodes", type=int, default=100)
    parser.add_argument("--seed", type=int, default=148)
    parser.add_argument("--device", default="cpu")
    parser.add_argument("--output-dir", type=Path, default=Path("artifacts"))
    parser.add_argument("--force", action="store_true")
    args = parser.parse_args()
    config = replace(
        TrainingConfig(),
        episodes=args.episodes,
        evaluation_episodes=args.evaluation_episodes,
        seed=args.seed,
        device=args.device,
        output_dir=args.output_dir,
    )
    run_complete_study(config, force=args.force)

if __name__ == "__main__":
    main()
```

src/robust_lunarlander/plotting.py

```
"""Publication-quality plots and evidence summaries for the four experiments."""

from __future__ import annotations

from pathlib import Path
from typing import Any

import matplotlib

matplotlib.use("Agg")
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

DISPLAY_ORDER = [
    "dqn_original",
    "ddqn_original",
    "dqn_modified",
    "ddqn_modified",
]
DISPLAY_LABELS = {
    "dqn_original": "DQN - Original",
    "ddqn_original": "DDQN - Original",
    "dqn_modified": "DQN - Modified",
    "ddqn_modified": "DDQN - Modified",
}
COLORS = {
    "dqn_original": "#2563EB",
    "ddqn_original": "#0F766E",
    "dqn_modified": "#DC2626",
    "ddqn_modified": "#D97706",
}
LINESTYLES = {
    "dqn_original": "-",
    "ddqn_original": "_",
    "dqn_modified": "--",
    "ddqn_modified": "-.-",
}

def _configure_style() -> None:
    """Apply one accessible visual system to all report figures."""

    sns.set_theme(style="whitegrid", context="notebook")
    plt.rcParams.update(
        {
            "font.family": "DejaVu Sans",
            "font.size": 11,
            "axes.titlesize": 15,
            "axes.labelsize": 12,
            "axes.edgecolor": "#475569",
            "axes.linewidth": 0.8,
            "axes.grid": True,
            "grid.alpha": 0.22,
            "grid.linestyle": ":",
            "legend.frameon": False,
            "figure.facecolor": "white",
            "axes.facecolor": "#FCFCFD",
        }
    )
```

```

def _save_figure(figure: plt.Figure, output_dir: Path, stem: str) -> None:
    """Write sharp PNG and editable SVG copies of one figure."""

    output_dir.mkdir(parents=True, exist_ok=True)
    figure.savefig(output_dir / f"{stem}.png", dpi=220, bbox_inches="tight")
    figure.savefig(output_dir / f"{stem}.svg", bbox_inches="tight")
    plt.close(figure)

def _plot_series(
    axis: plt.Axes,
    metrics: dict[str, pd.DataFrame],
    *,
    column: str,
    rolling_window: int | None,
    raw_alpha: float = 0.0,
) -> None:
    """Draw all four experiments with optional raw and smoothed traces."""

    for name in DISPLAY_ORDER:
        frame = metrics[name]
        color = COLORS[name]
        if raw_alpha > 0.0:
            axis.plot(
                frame["episode"],
                frame[column],
                color=color,
                alpha=raw_alpha,
                linewidth=0.55,
            )
        values = (
            frame[column].rolling(rolling_window, min_periods=1).mean()
            if rolling_window
            else frame[column]
        )
        axis.plot(
            frame["episode"],
            values,
            label=DISPLAY_LABELS[name],
            color=color,
            linestyle=LINESTYLES[name],
            linewidth=2.0,
        )

def create_all_plots(metrics: dict[str, pd.DataFrame], output_dir: Path) -> None:
    """Create each required plot and a compact four-panel overview."""

    _configure_style()
    specifications = [
        (
            "episode_reward",
            "episode_reward",
            "Episode reward vs. training episode",
            "Episode reward",
            50,
            0.15,
            "Thin lines show raw rewards; strong lines show a 50-episode moving mean.",
        ),
        (
            "average_predicted_q",
            "average_predicted_q",
            "Average predicted Q-value on fixed validation states",
            "Mean max predicted Q-value",
            20,
            0.0,
            "The same 512 validation states are used at every episode for all four agents.",
        ),
        (
            "moving_safe_landing_rate_100",
            "success_rate_100",
            "Safe-landing rate vs. training episode",
            "Safe-landing rate (previous 100 episodes)",
            None,
            0.0,
            "Success uses the assignment's terminal leg-contact, velocity, and angle criterion.",
        ),
        (
            "attempted_thruster_activations",
            "thruster_activations",
            "Average attempted thruster activations per episode",
            "Attempted thruster actions per episode",
            20,
            0.08,
            "Strong lines show a 20-episode moving mean of selected actions 1, 2, and 3.",
        ),
    ],
]

for column, stem, title, ylabel, window, raw_alpha, caption in specifications:
    figure, axis = plt.subplots(figsize=(11.2, 6.2))
    _plot_series(

```

```

        axis,
        metrics,
        column=column,
        rolling_window=window,
        raw_alpha=raw_alpha,
    )
    axis.set_title(title, loc="left", fontweight="bold")
    axis.set_xlabel("Training episode")
    axis.set_ylabel(ylabel)
    axis.legend(ncol=2, loc="best")
    figure.text(0.125, 0.015, caption, color="#475569", fontsize=9)
    figure.subplots_adjust(bottom=0.14)
    _save_figure(figure, output_dir, stem)

overview, axes = plt.subplots(2, 2, figsize=(15.2, 10.0))
for axis, (column, _, title, ylabel, window, raw_alpha, _) in zip(
    axes.flat,
    specifications,
    strict=True,
):
    _plot_series(
        axis,
        metrics,
        column=column,
        rolling_window=window,
        raw_alpha=min(raw_alpha, 0.10),
    )
    axis.set_title(title, loc="left", fontweight="bold", fontsize=12)
    axis.set_xlabel("Episode")
    axis.set_ylabel(ylabel, fontsize=10)
handles, labels = axes.flat[0].get_legend_handles_labels()
overview.legend(
    handles,
    labels,
    loc="upper center",
    bbox_to_anchor=(0.52, 0.943),
    ncol=4,
    frameon=False,
)
overview.suptitle(
    "DQN vs. DDQN under stochastic actuator failure",
    x=0.07,
    y=0.985,
    ha="left",
    fontsize=17,
    fontweight="bold",
)
overview.subplots_adjust(
    left=0.07,
    right=0.985,
    bottom=0.07,
    top=0.86,
    hspace=0.34,
    wspace=0.18,
)
_save_figure(overview, output_dir, "four_metric_overview")

epsilon_figure, epsilon_axis = plt.subplots(figsize=(10.0, 4.8))
reference = metrics[DISPLAY_ORDER[0]]
epsilon_axis.plot(
    reference["global_step"],
    reference["epsilon"],
    color="#6D28D9",
    linewidth=2.0,
    label="Shared linear epsilon schedule",
)
epsilon_axis.set_title("Epsilon-greedy exploration schedule", loc="left", fontweight="bold")
epsilon_axis.set_xlabel("Environment step")
epsilon_axis.set_ylabel("Epsilon")
epsilon_axis.legend(loc="best")
epsilon_axis.set_ylim(0.0, 1.05)
epsilon_figure.text(
    0.125,
    0.015,
    "The same start, end, and step-based decay are applied to all four agents.",
    color="#475569",
    fontsize=9,
)
epsilon_figure.subplots_adjust(bottom=0.18)
_save_figure(epsilon_figure, output_dir, "epsilon_schedule")

def summarize_experiments(
    metrics: dict[str, pd.DataFrame],
    evaluation_summary: pd.DataFrame,
) -> dict[str, Any]:
    """Compute report-ready comparisons without inventing unsupported conclusions."""

    final_window: dict[str, dict[str, float]] = {}
    final_comparison: list[dict[str, Any]] = []
    for name in DISPLAY_ORDER:
        tail = metrics[name].tail(100)
        moving_window = min(100, len(metrics[name]))

```

```

final_window[name] = {
    "mean_training_reward_last_100": float(tail["episode_reward"].mean()),
    "reward_std_last_100": float(tail["episode_reward"].std()),
    "best_100_episode_moving_average_reward": float(
        metrics[name]["episode_reward"]
        .rolling(moving_window, min_periods=moving_window)
        .mean()
        .max()
    ),
    "final_fixed_set_average_q": float(metrics[name]["average_predicted_q"].iloc[-1]),
    "safe_landing_rate_last_100": float(tail["successful_safe_landing"].mean()),
    "mean_predicted_q_last_100": float(tail["average_predicted_q"].mean()),
    "mean_attempted_thruster_activations_last_100": float(
        tail["attempted_thruster_activations"].mean()
    ),
    "mean_executed_thruster_activations_last_100": float(
        tail["executed_thruster_activations"].mean()
    ),
    "successful_safe_landings_total": int(metrics[name]["successful_safe_landing"].sum()),
    "training_duration_seconds": float(metrics[name]["episode_seconds"].sum()),
}
final_comparison.append(
    {
        "experiment": name,
        "algorithm": str(metrics[name]["algorithm"].iloc[0]),
        "environment": str(metrics[name]["environment_type"].iloc[0]),
        "mean_reward_final_100": final_window[name]["mean_training_reward_last_100"],
        "reward_std_final_100": final_window[name]["reward_std_last_100"],
        "best_moving_average_reward_100": final_window[name][
            "best_100_episode_moving_average_reward"
        ],
        "final_fixed_set_average_q": final_window[name]["final_fixed_set_average_q"],
        "safe_landing_rate_final_100": final_window[name]["safe_landing_rate_last_100"],
        "mean_attempted_thrusters_final_100": final_window[name][
            "mean_attempted_thruster_activations_last_100"
        ],
        "mean_executed_thrusters_final_100": final_window[name][
            "mean_executed_thruster_activations_last_100"
        ],
        "successful_safe_landings_total": final_window[name][
            "successful_safe_landings_total"
        ],
        "training_duration_seconds": final_window[name]["training_duration_seconds"],
    }
)

original_gap = float(
    (
        metrics["dqn_original"]["average_predicted_q"].tail(100).reset_index(drop=True)
        - metrics["ddqn_original"]["average_predicted_q"].tail(100).reset_index(drop=True)
    )
    .abs()
    .mean()
)
modified_gap = float(
    (
        metrics["dqn_modified"]["average_predicted_q"].tail(100).reset_index(drop=True)
        - metrics["ddqn_modified"]["average_predicted_q"].tail(100).reset_index(drop=True)
    )
    .abs()
    .mean()
)
evaluation_records = evaluation_summary.to_dict(orient="records")
modified_evaluation = evaluation_summary[
    evaluation_summary["environment_type"] == "Modified"
].sort_values(["safe_landing_rate", "mean_reward"], ascending=False)
best_modified = str(modified_evaluation.iloc[0]["algorithm"])

return {
    "final_100_training_episodes": final_window,
    "q_value_gap": {
        "original_environment_mean_absolute_gap": original_gap,
        "modified_environment_mean_absolute_gap": modified_gap,
        "modified_minus_original": modified_gap - original_gap,
        "failure_increased_gap": modified_gap > original_gap,
    },
    "attempted_thruster_change_modified_minus_original": {
        "DQN": final_window["dqn_modified"]["mean_attempted_thruster_activations_last_100"]
        - final_window["dqn_original"]["mean_attempted_thruster_activations_last_100"],
        "DDQN": final_window["ddqn_modified"]["mean_attempted_thruster_activations_last_100"]
        - final_window["ddqn_original"]["mean_attempted_thruster_activations_last_100"],
    },
    "final_comparison": final_comparison,
    "greedy_evaluation": evaluation_records,
    "best_algorithm_under_modified_environment": best_modified,
}

```


Appendix B - Per-iteration output evidence

Every training episode emits a compact console line and is persisted in both a CSV ledger and a progress log. This appendix reproduces all 3,200 per-iteration records: 800 rows for each of the four controlled experiments. Values follow the compact progress output convention: reward, fixed-set average Q, moving safe-landing rate, attempted/executed thrusters, steps, epsilon, and training loss. nan denotes the pre-warm-up episodes where no gradient update was yet performed.

Experiment	CSV rows	Log records	Episode range	Complete
dqn_original	800	800	1-800	True
ddqn_original	800	800	1-800	True
dqn_modified	800	800	1-800	True
ddqn_modified	800	800	1-800	True

Table 1: Complete per-iteration training output - Dqn Original (800 episodes).

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
1	-132.79	0.24	0.0%	51/51	71	0.9993	nan
2	-436.57	0.24	0.0%	66/66	93	0.9984	nan
3	-112.58	0.24	0.0%	104/104	124	0.9971	nan
4	-74.45	0.24	0.0%	67/67	86	0.9963	nan
5	-141.56	0.24	0.0%	104/104	139	0.9949	nan
6	-168.37	0.24	0.0%	82/82	113	0.9938	nan
7	-170.35	0.24	0.0%	69/69	88	0.9929	nan
8	-264.71	0.24	0.0%	43/43	63	0.9923	nan
9	-262.58	0.24	0.0%	88/88	114	0.9912	nan
10	-256.16	-0.01	0.0%	94/94	126	0.9899	2.111354
11	-57.57	-0.18	9.1%	51/51	65	0.9893	2.197836
12	-174.41	0.17	8.3%	67/67	86	0.9884	1.900430
13	-304.37	0.38	7.7%	68/68	92	0.9875	2.101976
14	-285.82	0.69	7.1%	74/74	102	0.9865	1.573831
15	-245.45	0.75	6.7%	86/86	122	0.9853	1.507664
16	-108.64	0.92	6.2%	65/65	84	0.9845	1.839810
17	-214.15	0.90	5.9%	86/86	104	0.9834	1.404884
18	-182.99	0.97	5.6%	62/62	85	0.9826	1.330281
19	-78.73	0.82	10.5%	81/81	104	0.9816	1.395608
20	-76.50	1.19	10.0%	57/57	74	0.9808	1.516414
21	-187.82	1.13	9.5%	51/51	71	0.9801	1.515786
22	-378.98	1.45	9.1%	80/80	104	0.9791	1.434591
23	-350.93	1.19	8.7%	69/69	87	0.9782	1.317473
24	-26.12	1.51	8.3%	75/75	97	0.9773	1.059367
25	-79.35	1.37	8.0%	76/76	99	0.9763	1.418375
26	-174.05	1.71	7.7%	50/50	61	0.9757	1.130719
27	-95.26	1.75	7.4%	58/58	83	0.9749	1.399712
28	-173.95	1.61	7.1%	92/92	120	0.9737	1.178418
29	-102.62	1.75	6.9%	47/47	67	0.9730	0.985096
30	-240.45	1.83	6.7%	67/67	87	0.9722	1.205984
31	-164.48	1.86	6.5%	118/118	146	0.9707	1.207793
32	-89.86	2.06	6.2%	52/52	74	0.9700	1.239478
33	-200.89	2.37	6.1%	84/84	113	0.9689	1.086603
34	-20.86	2.36	5.9%	98/98	134	0.9675	1.228904
35	-251.55	2.49	5.7%	86/86	110	0.9665	1.167238
36	-335.83	2.66	5.6%	84/84	106	0.9654	1.139542
37	-106.60	2.87	5.4%	64/64	74	0.9647	1.179119
38	-112.47	3.09	5.3%	59/59	71	0.9640	1.299323
39	-88.76	2.89	5.1%	65/65	89	0.9631	1.138951
40	-161.58	2.79	5.0%	88/88	113	0.9620	1.123193
41	-195.98	2.85	4.9%	88/88	120	0.9608	1.115919
42	-187.76	3.72	4.8%	51/51	71	0.9601	1.253932
43	-239.04	3.67	4.7%	93/93	121	0.9589	1.294743
44	-103.19	3.33	4.5%	95/95	123	0.9577	1.101671
45	-92.13	4.00	4.4%	47/47	62	0.9571	1.253441
46	-234.35	3.67	4.3%	93/93	109	0.9560	1.171772
47	-117.76	4.71	4.3%	83/83	102	0.9550	1.327842
48	-208.50	4.37	4.2%	72/72	93	0.9540	1.232274
49	-133.23	4.49	4.1%	84/84	113	0.9529	1.210450
50	-287.21	4.50	4.0%	53/53	67	0.9523	1.417502
51	-16.45	4.31	3.9%	84/84	114	0.9511	1.245028
52	-71.91	5.40	3.8%	107/107	143	0.9497	1.329888
53	-163.40	5.03	3.8%	91/91	118	0.9485	1.228288

Continued: Dqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
54	-105.90	5.17	3.7%	45/45	63	0.9479	1.079115
55	-81.13	5.31	3.6%	63/63	84	0.9471	1.186178
56	-483.11	5.22	3.6%	95/95	128	0.9458	1.180866
57	11.33	6.19	3.5%	62/62	83	0.9450	1.452056
58	-98.10	6.02	3.4%	95/95	133	0.9437	1.216991
59	-66.84	6.01	3.4%	41/41	59	0.9431	1.334488
60	-108.26	5.94	3.3%	85/85	110	0.9420	1.317638
61	-212.80	6.19	3.3%	54/54	73	0.9413	1.312369
62	-124.19	7.90	3.2%	56/56	77	0.9405	1.325598
63	-245.88	7.00	3.2%	105/105	138	0.9392	1.350404
64	-204.32	6.85	3.1%	85/85	109	0.9381	1.379324
65	-208.55	7.07	3.1%	69/69	99	0.9371	1.309410
66	-122.98	6.97	3.0%	56/56	78	0.9363	1.123987
67	-111.92	8.43	3.0%	65/65	87	0.9355	1.279522
68	-215.86	7.95	2.9%	82/82	102	0.9345	1.667938
69	-108.49	7.98	2.9%	64/64	90	0.9336	1.294002
70	-140.15	8.30	2.9%	78/78	106	0.9325	1.422064
71	-98.17	7.97	2.8%	101/101	132	0.9312	1.435579
72	20.05	9.07	2.8%	104/104	145	0.9298	1.454593
73	-88.93	9.31	2.7%	51/51	72	0.9291	1.419533
74	-138.76	9.05	2.7%	55/55	69	0.9284	1.328961
75	-70.51	9.08	2.7%	62/62	78	0.9276	1.352209
76	-215.37	8.86	2.6%	70/70	87	0.9267	1.441207
77	-83.07	9.00	2.6%	69/69	89	0.9259	1.502981
78	-107.71	10.47	2.6%	66/66	86	0.9250	1.541481
79	-105.43	10.15	2.5%	43/43	60	0.9244	1.642682
80	-93.15	10.16	2.5%	66/66	82	0.9236	1.241167
81	-53.54	10.37	2.5%	56/56	69	0.9229	1.419300
82	-119.15	10.36	2.4%	108/108	132	0.9216	1.413143
83	-119.06	11.74	2.4%	85/85	101	0.9206	1.517266
84	-244.76	11.25	2.4%	54/54	71	0.9199	1.601497
85	-114.49	11.42	2.4%	73/73	93	0.9190	1.296057
86	-73.93	11.38	2.3%	86/86	105	0.9180	1.425455
87	-122.68	11.23	2.3%	79/79	107	0.9169	1.469567
88	-366.05	11.69	2.3%	80/80	102	0.9159	1.549518
89	-79.22	12.47	2.2%	49/49	65	0.9152	1.514544
90	-137.95	12.31	2.2%	50/50	75	0.9145	1.328946
91	-111.61	12.27	2.2%	87/87	103	0.9135	1.298701
92	-288.01	12.43	2.2%	67/67	86	0.9126	1.636529
93	-115.04	12.44	2.2%	78/78	98	0.9117	1.522967
94	-77.74	13.92	2.1%	100/100	134	0.9103	1.486537
95	-71.35	13.73	2.1%	110/110	139	0.9090	1.618968
96	-81.41	13.70	2.1%	89/89	115	0.9078	1.404128
97	-66.37	13.92	2.1%	85/85	114	0.9067	1.474638
98	-108.43	13.63	2.0%	56/56	68	0.9060	1.487921
99	-135.54	14.99	2.0%	48/48	63	0.9054	1.619277
100	-116.09	14.80	2.0%	96/96	123	0.9042	1.370284
101	-47.07	15.07	2.0%	54/54	69	0.9035	1.577216
102	-97.14	14.50	2.0%	66/66	79	0.9027	1.288804
103	-359.71	14.72	2.0%	98/98	118	0.9015	1.437954
104	-59.01	16.21	2.0%	58/58	75	0.9008	1.700163
105	-201.64	16.22	2.0%	89/89	116	0.8997	1.490932
106	-232.70	16.16	2.0%	98/98	119	0.8985	1.324214
107	-51.81	16.38	2.0%	58/58	67	0.8978	1.428422
108	-224.17	16.16	2.0%	64/64	73	0.8971	1.473464
109	-192.16	16.82	2.0%	97/97	122	0.8959	1.578725
110	-340.38	16.85	2.0%	73/73	88	0.8950	1.880671
111	-148.51	16.85	1.0%	55/55	71	0.8943	1.578203
112	-61.08	16.93	2.0%	91/91	121	0.8931	1.489684
113	-265.68	16.76	2.0%	78/78	97	0.8921	1.575589
114	-144.69	18.06	2.0%	131/131	155	0.8906	1.627576
115	-153.86	17.74	2.0%	83/83	103	0.8896	1.487052
116	-59.56	18.40	2.0%	79/79	114	0.8885	1.758846
117	-85.36	17.77	2.0%	74/74	93	0.8875	1.488655
118	-52.79	18.42	2.0%	67/67	82	0.8867	1.398832
119	-99.25	19.10	1.0%	72/72	93	0.8858	1.440740
120	-117.89	19.56	1.0%	75/75	102	0.8848	1.521538
121	-75.03	19.38	1.0%	57/57	67	0.8841	1.195997
122	-13.20	19.69	1.0%	116/116	147	0.8827	1.575629
123	-100.36	19.69	1.0%	47/47	64	0.8821	1.611541
124	-98.80	19.63	1.0%	51/51	67	0.8814	1.786981
125	-81.80	20.35	1.0%	81/81	95	0.8804	1.530408
126	-75.22	20.09	2.0%	92/92	124	0.8792	1.613012
127	-70.44	20.30	2.0%	102/102	119	0.8780	1.569762

Continued: Dqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
128	-89.57	19.91	2.0%	81/81	108	0.8770	1.465846
129	-85.75	20.69	2.0%	84/84	111	0.8759	1.558024
130	-72.32	20.87	3.0%	78/78	97	0.8749	1.826849
131	-146.71	20.61	3.0%	71/71	87	0.8741	1.722103
132	-124.50	20.80	3.0%	69/69	83	0.8732	1.524684
133	-117.62	20.99	3.0%	60/60	81	0.8724	1.516726
134	-69.80	20.80	4.0%	67/67	88	0.8716	1.400355
135	-73.01	21.62	4.0%	79/79	95	0.8706	1.694743
136	-57.95	21.61	4.0%	56/56	75	0.8699	1.833061
137	-64.27	21.93	4.0%	94/94	124	0.8686	1.502878
138	-85.23	21.79	4.0%	101/101	139	0.8673	1.478509
139	-61.53	21.58	4.0%	54/54	69	0.8666	1.267713
140	-208.89	22.61	4.0%	90/90	116	0.8654	1.626684
141	-53.10	22.71	4.0%	62/62	79	0.8647	1.584719
142	-154.71	22.62	4.0%	106/106	141	0.8633	1.491198
143	-106.15	22.89	4.0%	66/66	87	0.8624	1.371870
144	-67.55	22.66	4.0%	62/62	79	0.8616	1.619559
145	-61.37	23.79	4.0%	88/88	110	0.8605	1.450032
146	-84.68	23.60	4.0%	84/84	111	0.8594	1.635570
147	-273.23	23.42	4.0%	60/60	83	0.8586	1.267417
148	-53.19	23.66	4.0%	93/93	123	0.8574	1.750101
149	-61.36	23.39	4.0%	56/56	73	0.8567	1.442780
150	-64.59	24.09	4.0%	52/52	68	0.8560	1.293135
151	-101.49	24.24	4.0%	67/67	85	0.8552	1.323254
152	-89.19	24.47	4.0%	70/70	89	0.8543	1.591605
153	-84.92	24.32	4.0%	88/88	114	0.8531	1.457696
154	-79.96	24.31	4.0%	78/78	103	0.8521	1.524177
155	-129.26	25.11	4.0%	51/51	66	0.8515	1.656500
156	-120.89	25.09	4.0%	76/76	103	0.8505	1.548660
157	-40.73	25.38	4.0%	71/71	83	0.8496	1.442775
158	-57.12	25.09	4.0%	51/51	67	0.8490	1.550241
159	-58.15	25.14	4.0%	53/53	68	0.8483	1.584052
160	-64.84	25.33	4.0%	55/55	66	0.8476	1.096623
161	-95.97	25.06	4.0%	52/52	74	0.8469	1.311186
162	-123.13	26.29	4.0%	79/79	99	0.8459	1.487261
163	-81.78	26.53	4.0%	74/74	94	0.8450	1.628722
164	-68.70	26.17	4.0%	56/56	68	0.8443	1.708853
165	-118.86	26.42	4.0%	54/54	73	0.8436	1.485840
166	-85.77	26.24	4.0%	95/95	128	0.8423	1.315695
167	-86.04	26.25	4.0%	80/80	107	0.8413	1.854084
168	-143.64	26.35	4.0%	57/57	73	0.8406	1.381531
169	-90.46	26.81	4.0%	95/95	123	0.8393	1.628735
170	-100.95	26.31	4.0%	73/73	90	0.8384	1.255750
171	-237.26	26.78	4.0%	96/96	119	0.8373	1.478187
172	-78.27	26.72	4.0%	55/55	66	0.8366	1.343799
173	-89.39	26.90	4.0%	52/52	64	0.8360	1.377220
174	-71.24	26.73	5.0%	60/60	82	0.8352	1.483567
175	-123.16	26.76	5.0%	86/86	107	0.8341	1.891692
176	-113.16	26.92	5.0%	101/101	125	0.8329	1.489870
177	-92.12	26.53	5.0%	67/67	91	0.8320	1.175172
178	-85.90	26.29	5.0%	54/54	76	0.8312	1.958608
179	-89.97	26.53	5.0%	105/105	125	0.8300	1.342431
180	-349.17	26.70	5.0%	86/86	110	0.8289	1.586825
181	-85.20	26.71	5.0%	60/60	79	0.8281	1.707335
182	-85.18	26.39	5.0%	103/103	130	0.8268	1.544973
183	-61.03	26.76	5.0%	84/84	106	0.8258	1.973738
184	-87.31	27.16	5.0%	57/57	74	0.8250	1.638357
185	-106.14	26.47	5.0%	75/75	94	0.8241	1.564892
186	-90.28	27.58	5.0%	80/80	112	0.8230	1.636100
187	-138.45	27.07	5.0%	82/82	105	0.8220	1.774899
188	-128.95	27.11	5.0%	75/75	93	0.8210	1.820779
189	-22.25	26.98	5.0%	120/120	149	0.8196	1.833248
190	-105.43	26.88	5.0%	61/61	75	0.8188	1.361761
191	-107.00	26.55	5.0%	78/78	94	0.8179	1.471876
192	-90.21	27.45	5.0%	89/89	111	0.8168	1.780228
193	-59.57	27.37	5.0%	77/77	89	0.8159	1.653543
194	-121.87	26.84	5.0%	102/102	129	0.8146	1.655400
195	-82.33	27.27	5.0%	88/88	116	0.8135	1.572435
196	-105.63	26.55	5.0%	51/51	65	0.8128	1.487846
197	-100.08	25.93	5.0%	86/86	112	0.8117	1.613053
198	-243.21	26.68	5.0%	79/79	95	0.8108	1.649063
199	-67.37	27.04	5.0%	69/69	87	0.8099	1.890146
200	-112.97	26.55	5.0%	83/83	103	0.8089	1.764121
201	-117.23	26.09	5.0%	74/74	111	0.8078	1.965949

Continued: Dqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
202	-99.90	26.17	5.0%	88/88	109	0.8067	1.608318
203	-80.65	26.30	5.0%	62/62	76	0.8060	1.439143
204	-86.44	26.62	5.0%	61/61	77	0.8052	1.302299
205	-129.39	26.29	5.0%	89/89	106	0.8042	1.368889
206	-42.54	26.45	5.0%	53/53	69	0.8035	1.671000
207	-68.61	26.19	5.0%	62/62	79	0.8027	1.401159
208	-48.26	26.28	5.0%	65/65	78	0.8019	1.657288
209	-63.76	26.25	5.0%	50/50	63	0.8013	1.516010
210	-60.19	26.30	5.0%	92/92	118	0.8001	1.479153
211	-53.20	26.61	5.0%	81/81	98	0.7992	1.931113
212	-55.43	26.83	4.0%	46/46	64	0.7985	1.668519
213	-107.57	26.38	4.0%	71/71	94	0.7976	1.464684
214	-105.66	26.72	4.0%	66/66	87	0.7967	1.659493
215	-36.75	26.77	5.0%	56/56	72	0.7960	1.719652
216	-46.76	26.55	5.0%	80/80	102	0.7950	1.523213
217	-142.11	27.21	5.0%	81/81	101	0.7940	1.818099
218	-80.40	26.57	5.0%	78/78	101	0.7930	1.528571
219	-99.02	27.77	5.0%	81/81	97	0.7921	1.644388
220	-122.06	26.21	5.0%	70/70	94	0.7911	1.672793
221	-62.69	26.50	5.0%	68/68	90	0.7902	1.575978
222	-68.41	26.89	5.0%	54/54	65	0.7896	1.603405
223	-53.54	26.81	5.0%	54/54	66	0.7889	1.638188
224	-112.81	26.54	5.0%	75/75	91	0.7880	1.470540
225	-81.33	25.12	5.0%	83/83	104	0.7870	1.600476
226	-74.59	26.24	4.0%	50/50	65	0.7864	1.469472
227	-70.70	25.83	4.0%	70/70	86	0.7855	1.570140
228	-19.11	25.88	5.0%	93/93	119	0.7843	1.594381
229	-52.25	26.48	6.0%	79/79	97	0.7834	1.633891
230	-140.85	26.41	5.0%	70/70	101	0.7824	1.589230
231	-78.44	25.49	5.0%	57/57	68	0.7817	1.695189
232	-75.26	25.12	5.0%	65/65	78	0.7809	1.739530
233	-58.07	25.39	5.0%	72/72	91	0.7800	1.908603
234	-96.96	25.46	4.0%	51/51	68	0.7794	1.421748
235	-119.88	25.40	4.0%	75/75	93	0.7784	1.367479
236	-72.52	25.36	4.0%	78/78	95	0.7775	1.617569
237	-79.88	25.01	4.0%	60/60	79	0.7767	1.515743
238	-103.16	24.80	4.0%	68/68	90	0.7758	1.565967
239	-174.37	25.05	4.0%	110/110	131	0.7745	1.745135
240	-83.51	24.68	4.0%	88/88	106	0.7735	1.512653
241	-85.07	24.59	4.0%	49/49	63	0.7729	1.448766
242	-113.21	24.76	4.0%	75/75	91	0.7720	1.718916
243	-61.35	24.78	4.0%	77/77	97	0.7710	1.691580
244	-102.81	24.70	4.0%	93/93	112	0.7699	1.699395
245	-49.67	24.75	5.0%	57/57	74	0.7692	1.664775
246	-171.00	24.83	5.0%	60/60	77	0.7684	1.416763
247	-78.11	24.50	5.0%	51/51	60	0.7678	1.671559
248	-97.86	24.62	5.0%	73/73	95	0.7669	1.820242
249	-163.88	24.03	5.0%	71/71	94	0.7659	1.586567
250	-62.89	24.44	6.0%	81/81	103	0.7649	1.501691
251	-77.98	25.02	6.0%	72/72	84	0.7641	1.897511
252	-109.88	24.10	6.0%	60/60	84	0.7632	1.469516
253	-59.02	24.25	6.0%	49/49	58	0.7627	1.446086
254	-20.95	23.97	6.0%	107/107	128	0.7614	1.804672
255	-83.36	23.91	6.0%	93/93	114	0.7603	1.786525
256	-32.87	24.19	7.0%	99/99	126	0.7590	1.809394
257	-162.55	24.22	7.0%	83/83	107	0.7580	1.610611
258	-100.35	23.76	7.0%	67/67	85	0.7571	1.629845
259	-66.24	24.03	7.0%	78/78	95	0.7562	1.776336
260	-82.45	23.58	7.0%	69/69	93	0.7553	1.506323
261	-64.78	23.69	7.0%	53/53	69	0.7546	1.383717
262	-46.31	23.72	7.0%	76/76	95	0.7536	1.513800
263	-40.74	23.71	7.0%	56/56	68	0.7530	1.520614
264	-111.66	23.58	7.0%	92/92	111	0.7519	1.705278
265	-90.61	23.46	7.0%	109/109	132	0.7506	1.613424
266	-76.08	23.02	7.0%	103/103	123	0.7493	1.514840
267	-99.76	22.84	7.0%	81/81	99	0.7484	1.698353
268	-36.80	23.48	8.0%	50/50	62	0.7477	1.926447
269	-115.30	22.34	8.0%	71/71	87	0.7469	1.682132
270	-205.81	22.62	8.0%	95/95	126	0.7456	1.358899
271	-75.95	22.94	9.0%	72/72	94	0.7447	1.550614
272	-44.38	22.35	10.0%	68/68	86	0.7439	1.583684
273	-49.87	22.63	10.0%	82/82	98	0.7429	1.378509
274	-79.99	22.42	9.0%	67/67	88	0.7420	1.840617
275	-85.38	22.49	9.0%	75/75	92	0.7411	1.769359

Continued: Dqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
276	-50.50	22.20	9.0%	63/63	79	0.7403	1.159865
277	-26.16	21.96	10.0%	91/91	113	0.7392	1.496208
278	-87.29	22.52	10.0%	60/60	79	0.7384	1.763550
279	-116.31	21.72	10.0%	95/95	114	0.7373	1.702121
280	-50.01	21.68	10.0%	57/57	75	0.7366	1.669839
281	-77.81	21.72	10.0%	67/67	82	0.7357	1.885429
282	-240.73	21.69	10.0%	73/73	89	0.7349	1.551954
283	-125.46	22.14	10.0%	87/87	108	0.7338	1.378341
284	-44.15	21.07	10.0%	99/99	130	0.7325	1.753301
285	-89.33	21.53	10.0%	77/77	92	0.7316	1.560487
286	-81.95	21.58	10.0%	78/78	94	0.7307	1.980043
287	-33.56	21.75	10.0%	98/98	121	0.7295	1.645130
288	-82.51	21.70	10.0%	69/69	84	0.7286	1.686377
289	-105.22	21.94	10.0%	63/63	84	0.7278	1.840046
290	-40.99	20.33	10.0%	80/80	98	0.7268	1.785865
291	-19.14	20.96	11.0%	102/102	125	0.7256	1.576077
292	-222.88	20.33	11.0%	118/118	143	0.7242	1.512681
293	-64.72	20.62	11.0%	86/86	102	0.7232	1.608906
294	-108.12	20.76	11.0%	72/72	97	0.7222	1.628580
295	-134.69	20.59	11.0%	75/75	102	0.7212	1.738389
296	-64.54	20.49	11.0%	75/75	106	0.7201	1.780805
297	-124.59	20.16	11.0%	66/66	77	0.7194	1.993632
298	-281.73	20.99	11.0%	88/88	105	0.7183	1.648067
299	-80.71	19.93	11.0%	75/75	82	0.7175	2.074172
300	-35.27	19.43	12.0%	96/96	125	0.7163	1.417082
301	-32.82	20.17	12.0%	101/101	121	0.7151	2.076162
302	-55.22	20.18	12.0%	63/63	77	0.7143	1.831566
303	-50.48	20.06	13.0%	84/84	108	0.7133	1.952185
304	-49.00	19.45	13.0%	44/44	61	0.7127	2.066209
305	-143.29	19.10	13.0%	94/94	119	0.7115	1.735314
306	-68.87	19.14	13.0%	78/78	89	0.7106	1.659685
307	-28.66	19.19	13.0%	117/117	141	0.7092	1.707224
308	-119.57	19.22	13.0%	74/74	88	0.7083	1.851744
309	-44.52	18.02	14.0%	64/64	80	0.7075	1.936012
310	-122.64	18.77	14.0%	57/57	66	0.7069	1.756571
311	-62.81	18.36	14.0%	90/90	116	0.7057	1.791892
312	-84.96	17.93	14.0%	51/51	62	0.7051	1.770406
313	-57.51	18.11	14.0%	92/92	111	0.7040	1.537989
314	-57.37	18.08	14.0%	51/51	65	0.7034	1.687235
315	-78.11	18.00	13.0%	78/78	101	0.7024	1.836397
316	-69.08	18.18	13.0%	85/85	102	0.7014	1.861847
317	-61.93	17.53	13.0%	53/53	74	0.7006	1.800928
318	-50.22	17.85	14.0%	68/68	88	0.6998	1.455161
319	-126.45	18.24	14.0%	59/59	76	0.6990	1.668143
320	-59.63	18.04	15.0%	76/76	91	0.6981	1.678668
321	-53.40	17.32	15.0%	97/97	116	0.6970	1.746313
322	-46.70	17.01	15.0%	106/106	125	0.6957	1.801074
323	-58.41	17.50	15.0%	64/64	85	0.6949	1.871027
324	-24.62	16.58	15.0%	80/80	103	0.6939	1.579588
325	-11.01	16.78	15.0%	74/74	97	0.6929	1.591653
326	-72.28	16.75	15.0%	42/42	59	0.6923	1.531634
327	-48.81	17.12	16.0%	56/56	76	0.6916	1.387910
328	-116.32	17.44	15.0%	66/66	80	0.6908	1.773861
329	-18.86	17.19	14.0%	79/79	91	0.6899	1.872267
330	-41.07	17.17	14.0%	88/88	114	0.6888	1.865366
331	-42.60	16.93	14.0%	70/70	86	0.6879	1.754085
332	-82.95	16.28	14.0%	83/83	107	0.6868	1.603006
333	-16.44	16.69	14.0%	92/92	119	0.6857	1.799694
334	-46.29	16.81	14.0%	86/86	99	0.6847	1.799144
335	-201.85	16.88	14.0%	93/93	124	0.6835	1.972734
336	-22.46	15.80	14.0%	64/64	76	0.6827	1.535983
337	-110.99	16.66	14.0%	59/59	76	0.6820	1.759243
338	-61.40	15.89	14.0%	60/60	71	0.6812	1.931025
339	-122.31	16.34	14.0%	99/99	124	0.6800	1.741466
340	-98.47	15.84	14.0%	68/68	87	0.6792	1.690666
341	-62.02	16.15	14.0%	101/101	118	0.6780	1.672343
342	-97.70	16.08	14.0%	92/92	110	0.6769	1.869687
343	-25.27	15.61	15.0%	50/50	67	0.6762	1.834508
344	-98.17	15.58	15.0%	90/90	108	0.6752	1.687799
345	-83.80	16.12	14.0%	86/86	98	0.6742	2.044104
346	-106.66	15.25	14.0%	99/99	121	0.6730	1.888783
347	-73.39	15.82	14.0%	68/68	84	0.6722	1.757684
348	-9.87	15.22	14.0%	59/59	74	0.6714	1.783733
349	-54.74	14.97	14.0%	69/69	85	0.6706	1.955536

Continued: Dqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
350	-47.46	14.87	13.0%	94/94	121	0.6694	1.989640
351	-86.46	15.16	13.0%	72/72	85	0.6686	1.896518
352	-43.45	15.14	13.0%	103/103	126	0.6673	1.891866
353	-62.31	15.14	14.0%	72/72	90	0.6664	1.926734
354	-41.41	14.94	14.0%	75/75	92	0.6655	1.972740
355	-56.19	14.65	14.0%	69/69	86	0.6647	1.859285
356	-29.58	14.60	13.0%	61/61	69	0.6640	1.935376
357	-86.69	14.42	13.0%	83/83	100	0.6630	1.754790
358	16.64	14.16	14.0%	81/81	102	0.6620	1.905739
359	-72.64	14.47	14.0%	71/71	82	0.6612	1.889100
360	-47.72	15.00	14.0%	76/76	95	0.6602	1.564476
361	-30.67	14.67	14.0%	75/75	90	0.6593	1.759931
362	-32.85	15.33	15.0%	95/95	116	0.6582	1.738984
363	-109.04	14.54	15.0%	78/78	101	0.6572	1.747104
364	-60.96	14.67	15.0%	61/61	73	0.6565	1.769159
365	-36.60	14.91	15.0%	114/114	133	0.6551	1.621841
366	-50.91	15.00	15.0%	119/119	144	0.6537	1.537255
367	-98.73	14.60	15.0%	66/66	84	0.6529	1.754956
368	-174.26	15.07	14.0%	123/123	143	0.6515	1.600036
369	-125.69	14.20	14.0%	52/52	61	0.6509	1.744739
370	-58.38	14.04	14.0%	69/69	88	0.6500	1.956068
371	-89.24	14.40	13.0%	83/83	104	0.6490	1.681788
372	-9.69	14.58	12.0%	111/111	129	0.6477	1.884943
373	-51.27	14.10	12.0%	61/61	75	0.6469	1.660531
374	-39.50	14.37	13.0%	91/91	111	0.6458	1.971175
375	-38.58	14.41	13.0%	125/125	148	0.6444	1.926273
376	-63.76	13.40	13.0%	85/85	103	0.6434	1.853180
377	-52.76	13.43	12.0%	67/67	83	0.6425	1.974796
378	-66.53	12.23	12.0%	74/74	86	0.6417	1.657912
379	-112.87	13.47	12.0%	98/98	126	0.6404	1.860765
380	-66.60	13.38	12.0%	68/68	87	0.6396	1.601088
381	-62.88	13.46	12.0%	82/82	104	0.6386	1.715886
382	-43.05	13.49	12.0%	59/59	77	0.6378	2.087928
383	-58.24	13.76	12.0%	106/106	125	0.6366	1.657168
384	-110.16	13.41	12.0%	102/102	123	0.6353	1.602685
385	-48.22	13.57	12.0%	65/65	82	0.6345	1.992476
386	-86.67	13.52	12.0%	82/82	107	0.6335	1.923120
387	-70.20	12.60	12.0%	86/86	110	0.6324	1.752192
388	-84.18	13.42	12.0%	68/68	86	0.6315	1.909667
389	-61.04	13.77	12.0%	114/114	137	0.6302	1.505875
390	-63.23	13.20	12.0%	88/88	107	0.6291	1.565003
391	-78.04	12.72	11.0%	94/94	108	0.6280	1.953839
392	-13.37	12.80	11.0%	78/78	93	0.6271	1.757620
393	-54.91	12.44	11.0%	107/107	129	0.6258	2.053123
394	-47.02	13.15	11.0%	89/89	108	0.6248	1.960711
395	-77.94	13.20	11.0%	73/73	91	0.6239	1.822483
396	11.38	12.52	12.0%	73/73	90	0.6230	2.152686
397	-204.73	12.54	12.0%	99/99	123	0.6218	1.961305
398	-43.94	12.43	12.0%	59/59	70	0.6211	1.716983
399	-75.56	11.86	12.0%	60/60	76	0.6203	1.984464
400	-73.87	12.57	11.0%	80/80	100	0.6193	1.786896
401	-53.73	11.33	11.0%	52/52	62	0.6187	1.920979
402	-89.37	11.89	11.0%	71/71	99	0.6177	1.943319
403	-55.76	12.51	10.0%	58/58	67	0.6171	1.704164
404	-28.99	11.65	11.0%	52/52	67	0.6164	1.675194
405	-17.39	12.27	12.0%	82/82	97	0.6154	1.991226
406	-46.81	11.98	12.0%	60/60	77	0.6147	1.472614
407	8.53	12.36	13.0%	97/97	116	0.6135	1.723015
408	-167.15	12.75	13.0%	108/108	140	0.6121	2.057488
409	-135.32	12.03	12.0%	116/116	140	0.6108	1.896270
410	-42.92	12.49	12.0%	85/85	115	0.6096	1.895292
411	-14.33	11.21	13.0%	108/108	126	0.6084	1.891892
412	-21.99	11.50	13.0%	102/102	120	0.6072	1.749937
413	-24.66	12.10	14.0%	105/105	121	0.6060	1.985496
414	-50.44	11.71	14.0%	91/91	108	0.6049	2.170080
415	-22.54	11.82	14.0%	85/85	106	0.6039	1.908250
416	-20.56	11.58	14.0%	75/75	90	0.6030	1.747738
417	-13.57	11.77	14.0%	78/78	93	0.6021	2.044258
418	-62.50	12.42	13.0%	60/60	71	0.6014	1.760826
419	-40.11	12.06	13.0%	87/87	105	0.6003	1.740884
420	-52.61	11.36	12.0%	94/94	133	0.5990	2.121130
421	-60.55	11.18	12.0%	95/95	119	0.5978	1.796276
422	-24.93	11.32	12.0%	63/63	79	0.5970	1.776442
423	-51.15	11.32	12.0%	66/66	74	0.5963	1.855590

Continued: Dqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
424	-49.65	11.21	12.0%	62/62	70	0.5956	1.642970
425	-33.89	10.91	12.0%	124/124	138	0.5942	1.988169
426	-74.03	11.43	12.0%	87/87	97	0.5933	1.879806
427	-40.83	11.70	11.0%	102/102	120	0.5921	1.832551
428	3.22	12.53	11.0%	109/109	131	0.5908	1.968376
429	-7.45	12.05	11.0%	106/106	128	0.5895	1.903419
430	-41.22	12.23	12.0%	71/71	88	0.5887	1.692673
431	-27.13	12.34	12.0%	72/72	89	0.5878	2.015875
432	-33.22	11.59	12.0%	85/85	112	0.5867	1.738739
433	-1.32	12.24	13.0%	67/67	82	0.5859	1.749491
434	-63.08	11.73	13.0%	94/94	119	0.5847	1.767782
435	-28.35	12.47	13.0%	95/95	127	0.5834	1.717847
436	-37.12	12.88	13.0%	106/106	121	0.5822	1.806583
437	-5.40	12.30	14.0%	100/100	128	0.5810	1.738230
438	-23.57	12.74	15.0%	69/69	83	0.5801	2.036648
439	-67.88	12.97	15.0%	86/86	99	0.5792	1.909593
440	-82.79	13.00	15.0%	71/71	87	0.5783	1.883315
441	-58.96	12.65	15.0%	126/126	148	0.5768	2.060362
442	-96.91	12.09	15.0%	82/82	96	0.5759	1.958850
443	-33.46	12.31	14.0%	78/78	100	0.5749	1.747816
444	-27.79	14.60	14.0%	94/94	114	0.5738	1.743751
445	-56.47	13.96	14.0%	76/76	92	0.5729	1.867676
446	-21.51	14.32	14.0%	108/108	132	0.5715	1.577361
447	-50.34	13.12	14.0%	69/69	82	0.5707	1.837183
448	-95.64	13.37	14.0%	97/97	112	0.5696	1.896712
449	15.58	13.90	14.0%	84/84	93	0.5687	1.688727
450	-59.74	14.68	14.0%	93/93	115	0.5676	1.852338
451	-16.20	14.61	14.0%	75/75	86	0.5667	1.832542
452	-24.66	13.92	14.0%	60/60	82	0.5659	1.458904
453	-53.72	14.30	13.0%	60/60	69	0.5652	1.911603
454	-40.48	14.18	13.0%	686/686	820	0.5571	1.765085
455	-11.46	14.35	13.0%	94/94	104	0.5561	1.925525
456	-30.66	14.47	13.0%	103/103	132	0.5548	1.682301
457	-46.26	14.52	13.0%	105/105	128	0.5535	1.651991
458	-29.42	14.91	12.0%	107/107	128	0.5522	2.076428
459	-4.96	15.37	12.0%	97/97	112	0.5511	1.889755
460	-40.23	14.91	12.0%	82/82	91	0.5502	1.543212
461	-43.10	15.74	12.0%	105/105	131	0.5489	1.822628
462	-0.72	15.77	12.0%	78/78	93	0.5480	1.697952
463	-45.55	14.72	12.0%	95/95	117	0.5468	1.966412
464	-3.59	15.70	12.0%	116/116	131	0.5456	1.839905
465	-53.20	15.65	12.0%	73/73	81	0.5447	1.730525
466	-18.94	16.18	12.0%	73/73	84	0.5439	1.848235
467	-27.30	16.32	12.0%	130/130	155	0.5424	2.062738
468	-42.78	15.68	12.0%	77/77	86	0.5415	1.766384
469	-16.10	16.09	12.0%	71/71	89	0.5406	2.058623
470	-4.16	16.71	12.0%	85/85	95	0.5397	1.724385
471	-74.00	15.95	12.0%	73/73	84	0.5389	1.812072
472	-19.45	15.20	12.0%	91/91	107	0.5378	1.964569
473	-2.17	16.25	12.0%	153/153	174	0.5361	1.687279
474	-24.43	15.91	11.0%	111/111	139	0.5347	1.689843
475	-46.93	15.51	11.0%	102/102	122	0.5335	1.960945
476	-21.84	16.23	11.0%	61/61	65	0.5329	1.738283
477	-39.74	16.27	11.0%	63/63	71	0.5322	2.141520
478	-43.63	16.47	11.0%	89/89	108	0.5311	1.820170
479	-11.34	16.34	11.0%	139/139	156	0.5296	1.700067
480	27.12	16.30	11.0%	113/113	132	0.5282	1.595318
481	-87.07	17.40	11.0%	171/171	195	0.5263	2.034232
482	11.78	17.06	12.0%	141/141	168	0.5247	1.964190
483	-30.21	16.55	12.0%	75/75	85	0.5238	1.859590
484	-57.24	16.94	12.0%	85/85	103	0.5228	1.948736
485	-97.87	16.94	12.0%	84/84	98	0.5218	2.120703
486	-70.79	17.20	12.0%	117/117	136	0.5205	1.666331
487	-12.58	17.38	12.0%	77/77	97	0.5195	2.035150
488	-14.42	18.53	12.0%	108/108	119	0.5183	2.015028
489	-127.19	18.13	12.0%	97/97	115	0.5172	1.751508
490	-198.64	18.20	12.0%	110/110	125	0.5160	1.770783
491	-80.30	18.29	12.0%	857/857	1000	0.5061	1.934576
492	0.07	18.91	12.0%	117/117	144	0.5046	1.899997
493	-43.82	19.74	12.0%	92/92	101	0.5036	1.577335
494	-30.98	19.64	12.0%	89/89	102	0.5026	1.756455
495	-23.34	18.91	12.0%	113/113	133	0.5013	1.620740
496	-31.42	19.13	11.0%	428/428	497	0.4964	1.843104
497	4.25	19.64	12.0%	92/92	116	0.4952	2.150203

Continued: Dqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
498	6.91	19.19	12.0%	125/125	136	0.4939	1.850569
499	-43.01	19.43	12.0%	125/125	158	0.4923	1.662099
500	-35.87	19.28	12.0%	116/116	136	0.4910	1.829379
501	-17.86	20.06	12.0%	78/78	85	0.4901	1.945464
502	-47.86	20.43	12.0%	99/99	119	0.4890	1.726128
503	-5.19	19.19	12.0%	107/107	120	0.4878	1.985880
504	-25.15	19.41	11.0%	122/122	156	0.4862	1.911523
505	-30.27	19.67	10.0%	91/91	110	0.4851	1.929749
506	-27.79	19.58	10.0%	85/85	96	0.4842	1.851226
507	19.62	19.99	9.0%	96/96	116	0.4830	1.657479
508	-36.41	19.98	9.0%	83/83	92	0.4821	1.733831
509	-49.37	19.89	9.0%	101/101	130	0.4808	1.879646
510	18.57	19.89	9.0%	139/139	154	0.4793	1.791346
511	-3.74	19.92	8.0%	81/81	88	0.4784	1.842928
512	-70.47	20.10	8.0%	101/101	122	0.4772	1.852956
513	-45.22	19.36	7.0%	78/78	89	0.4764	1.731670
514	40.42	20.42	7.0%	159/159	183	0.4745	1.864583
515	-43.15	19.89	7.0%	129/129	147	0.4731	1.535419
516	-26.45	19.75	7.0%	115/115	132	0.4718	1.845210
517	36.17	20.52	8.0%	95/95	109	0.4707	1.850332
518	-112.17	19.88	8.0%	320/320	369	0.4671	1.748151
519	-3.01	19.92	8.0%	241/241	279	0.4643	1.696921
520	1.12	20.91	8.0%	90/90	107	0.4632	1.793097
521	-60.04	19.94	8.0%	79/79	89	0.4624	1.630981
522	-27.51	19.33	8.0%	112/112	123	0.4611	1.755271
523	-40.98	21.00	8.0%	77/77	88	0.4603	1.493924
524	-192.08	20.11	8.0%	874/874	1000	0.4504	1.738391
525	-9.49	19.74	8.0%	294/294	339	0.4470	1.818639
526	-68.02	19.01	8.0%	879/879	1000	0.4371	1.851008
527	17.19	18.80	8.0%	146/146	171	0.4354	1.723157
528	-29.28	20.01	8.0%	100/100	114	0.4343	2.212268
529	-26.64	19.80	8.0%	115/115	133	0.4330	1.684892
530	-43.36	20.33	7.0%	665/665	742	0.4256	1.807052
531	-241.18	20.46	7.0%	439/439	506	0.4206	1.838210
532	-44.26	20.20	7.0%	858/858	1000	0.4107	1.873838
533	30.60	20.33	6.0%	135/135	156	0.4092	1.795188
534	31.00	20.62	7.0%	135/135	163	0.4076	1.873580
535	37.61	20.63	7.0%	109/109	132	0.4062	1.756573
536	23.94	21.65	7.0%	111/111	122	0.4050	2.112042
537	-4.18	20.65	6.0%	96/96	109	0.4040	1.791404
538	-41.79	20.60	5.0%	850/850	1000	0.3941	1.913913
539	-63.57	20.26	5.0%	76/76	86	0.3932	2.050056
540	-68.38	21.40	5.0%	881/881	1000	0.3833	2.006860
541	27.34	20.87	5.0%	119/119	135	0.3820	1.857266
542	-25.95	22.00	5.0%	892/892	1000	0.3721	2.006811
543	4.58	22.28	5.0%	128/128	139	0.3707	1.980499
544	-79.69	24.79	5.0%	889/889	1000	0.3608	2.092618
545	-15.14	24.87	5.0%	835/835	1000	0.3509	2.070472
546	-39.18	26.76	5.0%	850/850	1000	0.3410	2.094484
547	-43.14	27.60	5.0%	874/874	1000	0.3311	2.088716
548	-21.64	27.91	5.0%	842/842	1000	0.3212	2.098296
549	34.33	28.60	5.0%	117/117	130	0.3199	1.890805
550	31.46	28.23	5.0%	101/101	110	0.3188	2.015213
551	-74.31	28.82	5.0%	893/893	1000	0.3089	2.200392
552	-64.30	30.85	5.0%	846/846	1000	0.2990	2.069074
553	-39.09	32.34	5.0%	850/850	1000	0.2891	2.074233
554	-37.84	33.77	5.0%	808/808	1000	0.2792	2.089280
555	-29.06	35.87	5.0%	852/852	1000	0.2693	2.015395
556	-4.07	37.55	5.0%	799/799	1000	0.2594	2.140471
557	-41.68	37.87	5.0%	809/809	1000	0.2495	2.031187
558	-9.09	38.25	5.0%	844/844	1000	0.2396	2.135389
559	17.49	38.58	5.0%	779/779	1000	0.2297	2.083182
560	-3.95	39.93	5.0%	793/793	1000	0.2198	2.003288
561	-158.23	39.87	5.0%	448/448	598	0.2139	1.971108
562	-193.70	40.21	4.0%	409/409	501	0.2089	1.998481
563	-43.89	41.63	4.0%	839/839	1000	0.1990	2.099272
564	-53.01	40.71	4.0%	846/846	1000	0.1891	2.101420
565	-57.79	40.84	4.0%	812/812	1000	0.1792	2.116747
566	-105.34	42.88	4.0%	855/855	1000	0.1693	2.149631
567	-39.22	42.73	4.0%	920/920	1000	0.1594	2.091721
568	-32.47	44.21	4.0%	867/867	1000	0.1495	2.102244
569	-49.98	42.55	4.0%	886/886	1000	0.1396	2.109497
570	-85.06	43.29	4.0%	869/869	1000	0.1297	1.811279
571	-59.22	43.36	4.0%	946/946	1000	0.1198	2.021152

Continued: Dqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
572	-75.28	42.55	4.0%	902/902	1000	0.1099	1.953389
573	-106.91	44.34	4.0%	847/847	1000	0.1000	1.845580
574	-60.36	44.68	4.0%	849/849	1000	0.0901	1.843238
575	-27.23	46.78	4.0%	861/861	1000	0.0802	1.899847
576	-52.52	44.98	4.0%	815/815	1000	0.0703	1.925485
577	-30.54	44.85	4.0%	813/813	1000	0.0604	1.803491
578	-40.23	45.86	4.0%	873/873	1000	0.0505	1.900650
579	-98.40	44.59	4.0%	762/762	1000	0.0406	1.870649
580	-43.08	45.96	4.0%	722/722	1000	0.0307	1.837314
581	-41.48	47.21	4.0%	711/711	1000	0.0208	1.848014
582	-59.82	48.20	3.0%	737/737	1000	0.0109	1.831935
583	-42.93	47.22	3.0%	714/714	1000	0.0100	1.873329
584	-53.06	48.15	3.0%	758/758	1000	0.0100	1.687419
585	-17.26	48.20	3.0%	718/718	1000	0.0100	1.741891
586	-65.64	46.49	3.0%	756/756	1000	0.0100	1.650325
587	-67.98	48.25	3.0%	705/705	1000	0.0100	1.570983
588	-5.25	48.92	3.0%	745/745	1000	0.0100	1.669104
589	23.75	48.38	3.0%	719/719	1000	0.0100	1.496219
590	-48.76	47.84	3.0%	716/716	1000	0.0100	1.515890
591	-6.34	46.44	3.0%	731/731	1000	0.0100	1.468482
592	-12.93	47.21	3.0%	741/741	1000	0.0100	1.509099
593	25.70	48.36	3.0%	753/753	1000	0.0100	1.448036
594	-18.61	48.14	3.0%	764/764	1000	0.0100	1.343785
595	45.61	46.72	3.0%	807/807	1000	0.0100	1.353234
596	-23.83	47.02	3.0%	791/791	1000	0.0100	1.348726
597	9.31	46.17	2.0%	796/796	1000	0.0100	1.322595
598	230.37	45.78	3.0%	231/231	321	0.0100	1.320313
599	-1.36	45.15	3.0%	879/879	1000	0.0100	1.226314
600	-1.97	45.37	3.0%	889/889	1000	0.0100	1.277864
601	2.62	44.72	3.0%	787/787	1000	0.0100	1.296708
602	11.54	45.17	3.0%	818/818	1000	0.0100	1.181735
603	15.44	46.03	3.0%	822/822	1000	0.0100	1.121776
604	-12.93	44.21	3.0%	857/857	1000	0.0100	1.202125
605	-8.02	43.64	3.0%	885/885	1000	0.0100	1.187926
606	-2.67	43.49	3.0%	841/841	1000	0.0100	1.199419
607	-14.04	42.25	3.0%	776/776	1000	0.0100	1.223444
608	7.30	41.94	3.0%	764/764	1000	0.0100	1.203857
609	-29.50	39.77	3.0%	751/751	1000	0.0100	1.145234
610	-49.97	40.23	3.0%	775/775	1000	0.0100	1.076230
611	24.42	39.71	3.0%	782/782	1000	0.0100	1.049706
612	-33.04	40.06	3.0%	734/734	1000	0.0100	1.121843
613	-6.96	41.29	3.0%	752/752	1000	0.0100	1.090572
614	-38.46	43.03	3.0%	757/757	1000	0.0100	1.053851
615	-11.81	43.72	3.0%	746/746	1000	0.0100	1.011194
616	-28.98	43.26	3.0%	816/816	1000	0.0100	0.975086
617	56.22	43.66	2.0%	786/786	1000	0.0100	1.029162
618	-22.75	44.78	2.0%	837/837	1000	0.0100	1.002989
619	-29.09	46.69	2.0%	845/845	1000	0.0100	0.914043
620	224.62	47.28	3.0%	340/340	456	0.0100	0.896180
621	213.54	46.76	4.0%	627/627	740	0.0100	0.941897
622	-7.05	47.98	4.0%	776/776	1000	0.0100	0.928557
623	156.82	48.94	5.0%	722/722	918	0.0100	0.894135
624	54.82	48.70	5.0%	190/190	230	0.0100	0.852496
625	26.55	48.35	5.0%	846/846	1000	0.0100	0.906508
626	10.01	49.60	5.0%	902/902	1000	0.0100	0.850474
627	0.39	49.97	5.0%	849/849	1000	0.0100	0.820422
628	215.89	51.56	6.0%	503/503	632	0.0100	0.900650
629	-34.40	51.49	6.0%	858/858	1000	0.0100	0.794264
630	-13.81	47.83	6.0%	827/827	1000	0.0100	0.765384
631	-11.66	52.72	6.0%	828/828	1000	0.0100	0.787692
632	22.70	53.10	6.0%	843/843	1000	0.0100	0.769071
633	-21.62	52.36	6.0%	825/825	1000	0.0100	0.768657
634	-26.88	54.79	5.0%	834/834	1000	0.0100	0.718427
635	-13.62	53.15	5.0%	777/777	1000	0.0100	0.666539
636	-11.41	53.95	5.0%	801/801	1000	0.0100	0.640526
637	3.95	52.66	5.0%	796/796	1000	0.0100	0.622717
638	1.99	53.38	5.0%	821/821	1000	0.0100	0.590218
639	-53.18	55.38	5.0%	779/779	1000	0.0100	0.580647
640	-34.87	56.82	5.0%	824/824	1000	0.0100	0.521042
641	19.45	57.12	5.0%	856/856	1000	0.0100	0.572175
642	224.35	59.45	6.0%	352/352	492	0.0100	0.590541
643	277.86	57.24	7.0%	267/267	380	0.0100	0.525553
644	235.28	58.30	8.0%	529/529	718	0.0100	0.491790
645	229.44	59.62	9.0%	463/463	564	0.0100	0.480987

Continued: Dqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
646	232.50	58.55	10.0%	496/496	693	0.0100	0.477190
647	201.71	59.46	11.0%	431/431	560	0.0100	0.459067
648	229.53	59.04	12.0%	327/327	449	0.0100	0.476200
649	231.84	59.23	13.0%	230/230	345	0.0100	0.448630
650	297.13	60.33	14.0%	276/276	365	0.0100	0.471710
651	232.35	60.43	15.0%	276/276	398	0.0100	0.418784
652	239.48	60.48	16.0%	304/304	399	0.0100	0.440203
653	255.88	60.16	17.0%	462/462	605	0.0100	0.417100
654	197.41	64.85	18.0%	594/594	826	0.0100	0.430906
655	257.35	65.17	19.0%	419/419	589	0.0100	0.432282
656	268.94	64.83	20.0%	345/345	461	0.0100	0.397127
657	229.19	66.93	21.0%	317/317	460	0.0100	0.342343
658	197.26	67.24	22.0%	668/668	908	0.0100	0.381648
659	217.66	67.35	22.0%	447/447	521	0.0100	0.368301
660	177.36	67.81	23.0%	741/741	976	0.0100	0.376456
661	241.27	68.27	24.0%	381/381	488	0.0100	0.365892
662	-19.27	69.78	24.0%	870/870	1000	0.0100	0.388007
663	246.76	68.95	25.0%	712/712	961	0.0100	0.364343
664	213.61	68.13	26.0%	535/535	654	0.0100	0.371458
665	0.67	69.28	26.0%	922/922	1000	0.0100	0.342503
666	9.22	67.09	26.0%	879/879	1000	0.0100	0.324959
667	228.01	67.73	27.0%	571/571	705	0.0100	0.316321
668	215.89	68.42	28.0%	373/373	433	0.0100	0.333555
669	222.71	67.22	29.0%	370/370	445	0.0100	0.338031
670	224.40	67.62	30.0%	295/295	373	0.0100	0.328643
671	195.32	66.73	31.0%	625/625	734	0.0100	0.352222
672	7.48	67.10	31.0%	850/850	1000	0.0100	0.329157
673	5.30	66.82	31.0%	856/856	1000	0.0100	0.351434
674	-40.88	65.20	31.0%	893/893	1000	0.0100	0.320473
675	22.52	64.37	31.0%	775/775	1000	0.0100	0.305741
676	31.22	66.07	31.0%	825/825	1000	0.0100	0.321966
677	208.39	66.35	32.0%	703/703	962	0.0100	0.305261
678	20.13	66.36	32.0%	783/783	1000	0.0100	0.301755
679	190.81	68.86	33.0%	709/709	993	0.0100	0.316662
680	225.71	70.00	34.0%	480/480	669	0.0100	0.306582
681	246.17	69.71	34.0%	379/379	545	0.0100	0.320402
682	-0.83	68.20	34.0%	811/811	1000	0.0100	0.339433
683	141.00	67.50	34.0%	766/766	832	0.0100	0.335616
684	-34.15	67.86	34.0%	783/783	1000	0.0100	0.364299
685	53.74	67.02	34.0%	782/782	1000	0.0100	0.352826
686	-95.72	71.04	34.0%	846/846	1000	0.0100	0.351653
687	280.67	69.70	35.0%	415/415	588	0.0100	0.370868
688	-96.02	68.29	35.0%	826/826	1000	0.0100	0.368809
689	-111.93	67.19	35.0%	819/819	1000	0.0100	0.351183
690	-77.72	68.43	35.0%	811/811	1000	0.0100	0.372307
691	26.39	69.94	35.0%	806/806	1000	0.0100	0.404472
692	-86.18	70.58	35.0%	858/858	1000	0.0100	0.409816
693	-2.94	69.44	35.0%	848/848	1000	0.0100	0.421000
694	116.43	71.49	35.0%	664/664	1000	0.0100	0.437022
695	-39.57	69.98	35.0%	816/816	1000	0.0100	0.383624
696	255.35	70.19	36.0%	273/273	355	0.0100	0.373685
697	1.09	72.50	36.0%	748/748	1000	0.0100	0.384038
698	111.27	71.65	35.0%	610/610	1000	0.0100	0.416129
699	-27.01	70.16	35.0%	758/758	1000	0.0100	0.430640
700	218.93	68.88	35.0%	684/684	791	0.0100	0.412118
701	235.12	70.41	35.0%	722/722	895	0.0100	0.397845
702	141.67	67.75	35.0%	606/606	1000	0.0100	0.415722
703	276.60	67.99	36.0%	170/170	245	0.0100	0.414278
704	60.26	68.93	36.0%	712/712	1000	0.0100	0.367725
705	222.13	70.63	37.0%	537/537	765	0.0100	0.429750
706	280.98	71.06	38.0%	377/377	591	0.0100	0.396536
707	216.29	71.52	39.0%	408/408	533	0.0100	0.400818
708	142.54	71.75	39.0%	812/812	1000	0.0100	0.401424
709	259.24	72.33	40.0%	327/327	457	0.0100	0.410128
710	139.30	73.87	40.0%	678/678	1000	0.0100	0.408130
711	61.45	72.05	40.0%	768/768	1000	0.0100	0.406834
712	213.33	74.22	41.0%	491/491	637	0.0100	0.420285
713	163.39	74.01	41.0%	615/615	1000	0.0100	0.419113
714	95.24	76.35	41.0%	584/584	1000	0.0100	0.407543
715	234.99	76.16	42.0%	341/341	527	0.0100	0.415019
716	171.18	75.22	43.0%	680/680	845	0.0100	0.369036
717	232.67	77.69	44.0%	278/278	401	0.0100	0.378550
718	55.23	76.91	44.0%	71/71	84	0.0100	0.368392
719	81.04	77.85	44.0%	638/638	1000	0.0100	0.384007

Continued: Dqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
720	209.54	81.02	44.0%	612/612	781	0.0100	0.392039
721	-1.54	80.65	43.0%	71/71	82	0.0100	0.388405
722	80.70	81.14	43.0%	688/688	1000	0.0100	0.365559
723	4.03	81.35	42.0%	66/66	77	0.0100	0.308220
724	0.30	81.20	42.0%	77/77	97	0.0100	0.363285
725	176.75	82.50	42.0%	573/573	1000	0.0100	0.362866
726	250.69	83.78	43.0%	174/174	248	0.0100	0.355469
727	252.82	82.54	44.0%	469/469	605	0.0100	0.370243
728	-92.59	83.47	43.0%	101/101	125	0.0100	0.367341
729	-17.77	83.32	43.0%	899/899	1000	0.0100	0.351885
730	122.29	83.27	43.0%	574/574	1000	0.0100	0.343333
731	223.17	84.82	44.0%	430/430	572	0.0100	0.383031
732	254.26	83.78	45.0%	182/182	293	0.0100	0.352220
733	303.29	83.04	46.0%	192/192	309	0.0100	0.376986
734	153.98	82.16	46.0%	535/535	1000	0.0100	0.347208
735	268.43	83.49	47.0%	443/443	786	0.0100	0.361721
736	-10.10	84.26	47.0%	69/69	85	0.0100	0.382618
737	253.26	83.80	48.0%	253/253	371	0.0100	0.412595
738	153.58	82.88	48.0%	558/558	1000	0.0100	0.335898
739	254.02	81.81	49.0%	554/554	935	0.0100	0.352583
740	137.12	80.73	49.0%	630/630	1000	0.0100	0.365429
741	127.86	79.30	49.0%	738/738	1000	0.0100	0.344047
742	167.20	82.06	48.0%	653/653	1000	0.0100	0.369556
743	21.51	82.22	47.0%	896/896	1000	0.0100	0.345998
744	217.74	80.34	47.0%	688/688	980	0.0100	0.359434
745	211.55	81.02	47.0%	391/391	611	0.0100	0.383272
746	97.00	79.36	46.0%	689/689	1000	0.0100	0.379160
747	240.44	79.24	46.0%	394/394	725	0.0100	0.390116
748	285.06	77.09	46.0%	329/329	480	0.0100	0.374088
749	244.52	77.26	46.0%	199/199	271	0.0100	0.367763
750	-18.89	77.61	45.0%	122/122	146	0.0100	0.383477
751	-274.52	77.36	44.0%	229/229	245	0.0100	0.395319
752	74.10	75.84	43.0%	727/727	1000	0.0100	0.405882
753	267.41	74.18	43.0%	428/428	528	0.0100	0.397290
754	143.68	70.91	42.0%	760/760	1000	0.0100	0.428368
755	175.89	69.81	41.0%	640/640	1000	0.0100	0.405076
756	255.09	69.97	40.0%	565/565	771	0.0100	0.427481
757	290.80	69.57	40.0%	283/283	392	0.0100	0.429627
758	292.26	68.16	40.0%	430/430	799	0.0100	0.393375
759	177.56	67.73	40.0%	509/509	1000	0.0100	0.362362
760	251.79	65.28	39.0%	648/648	869	0.0100	0.410304
761	262.77	65.93	38.0%	289/289	306	0.0100	0.407441
762	293.93	65.35	39.0%	336/336	649	0.0100	0.360781
763	241.62	64.02	39.0%	415/415	872	0.0100	0.400244
764	244.83	62.40	39.0%	347/347	736	0.0100	0.372195
765	127.36	59.77	39.0%	511/511	1000	0.0100	0.385182
766	64.45	56.89	39.0%	673/673	1000	0.0100	0.381052
767	178.00	56.77	38.0%	702/702	1000	0.0100	0.378778
768	163.90	54.96	37.0%	572/572	1000	0.0100	0.365789
769	121.69	54.72	36.0%	612/612	1000	0.0100	0.410970
770	235.36	53.94	36.0%	372/372	618	0.0100	0.387432
771	187.09	54.79	35.0%	554/554	1000	0.0100	0.365574
772	121.94	52.67	35.0%	536/536	1000	0.0100	0.358968
773	154.46	51.21	35.0%	536/536	1000	0.0100	0.396574
774	285.76	52.42	35.0%	272/272	535	0.0100	0.424074
775	289.85	54.24	36.0%	335/335	586	0.0100	0.347595
776	274.78	56.53	36.0%	242/242	430	0.0100	0.413514
777	290.31	56.36	35.0%	174/174	256	0.0100	0.364317
778	157.66	55.21	35.0%	792/792	1000	0.0100	0.384473
779	243.20	56.73	35.0%	250/250	473	0.0100	0.417663
780	201.75	55.39	34.0%	849/849	921	0.0100	0.370066
781	17.33	54.72	35.0%	99/99	117	0.0100	0.439942
782	52.10	56.70	36.0%	134/134	152	0.0100	0.375428
783	251.39	53.65	36.0%	450/450	613	0.0100	0.421215
784	260.40	54.37	36.0%	196/196	356	0.0100	0.362519
785	29.57	52.79	36.0%	96/96	125	0.0100	0.430293
786	43.62	53.80	36.0%	68/68	104	0.0100	0.296919
787	-8.20	53.08	35.0%	58/58	77	0.0100	0.335160
788	12.64	53.40	35.0%	149/149	193	0.0100	0.352882
789	52.42	52.22	35.0%	99/99	115	0.0100	0.418390
790	284.04	52.28	36.0%	318/318	520	0.0100	0.384871
791	18.53	51.50	36.0%	118/118	136	0.0100	0.423196
792	-68.19	51.49	36.0%	55/55	74	0.0100	0.489325
793	-11.65	53.06	36.0%	47/47	64	0.0100	0.459804

Continued: Dqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
794	-3.86	52.89	36.0%	77/77	110	0.0100	0.384739
795	302.35	47.14	37.0%	266/266	397	0.0100	0.388793
796	-22.98	49.35	36.0%	62/62	83	0.0100	0.319818
797	-17.40	50.33	36.0%	66/66	97	0.0100	0.453751
798	21.90	50.42	36.0%	89/89	101	0.0100	0.446597
799	148.52	47.93	36.0%	797/797	836	0.0100	0.409340
800	41.60	49.12	36.0%	84/84	91	0.0100	0.430006

Table 2: Complete per-iteration training output - Ddqn Original (800 episodes).

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
1	-132.79	0.24	0.0%	51/51	71	0.9993	nan
2	-436.57	0.24	0.0%	66/66	93	0.9984	nan
3	-112.58	0.24	0.0%	104/104	124	0.9971	nan
4	-74.45	0.24	0.0%	67/67	86	0.9963	nan
5	-141.56	0.24	0.0%	104/104	139	0.9949	nan
6	-168.37	0.24	0.0%	82/82	113	0.9938	nan
7	-170.35	0.24	0.0%	69/69	88	0.9929	nan
8	-264.71	0.24	0.0%	43/43	63	0.9923	nan
9	-262.58	0.24	0.0%	88/88	114	0.9912	nan
10	-256.16	-0.01	0.0%	94/94	126	0.9899	2.120335
11	-57.57	-0.34	9.1%	51/51	65	0.9893	2.219106
12	-174.41	-0.08	8.3%	67/67	86	0.9884	1.914462
13	-304.37	0.10	7.7%	68/68	92	0.9875	2.113536
14	-285.82	0.40	7.1%	74/74	102	0.9865	1.577444
15	-245.45	0.48	6.7%	86/86	122	0.9853	1.506126
16	-108.64	0.66	6.2%	65/65	84	0.9845	1.834367
17	-214.15	0.59	5.9%	86/86	104	0.9834	1.396790
18	-182.99	0.72	5.6%	62/62	85	0.9826	1.323052
19	-78.73	0.56	10.5%	81/81	104	0.9816	1.397083
20	-76.50	0.88	10.0%	57/57	74	0.9808	1.512228
21	-187.82	0.88	9.5%	51/51	71	0.9801	1.510924
22	-378.98	1.17	9.1%	80/80	104	0.9791	1.423495
23	-350.93	0.90	8.7%	69/69	87	0.9782	1.305524
24	-26.12	1.28	8.3%	75/75	97	0.9773	1.054310
25	-79.35	1.12	8.0%	76/76	99	0.9763	1.407919
26	-174.05	1.43	7.7%	50/50	61	0.9757	1.117375
27	-95.26	1.55	7.4%	58/58	83	0.9749	1.390071
28	-173.95	1.36	7.1%	92/92	120	0.9737	1.169645
29	-102.62	1.55	6.9%	47/47	67	0.9730	0.972897
30	-240.45	1.57	6.7%	67/67	87	0.9722	1.196482
31	-164.48	1.62	6.5%	118/118	146	0.9707	1.193377
32	-89.86	1.81	6.2%	52/52	74	0.9700	1.233512
33	-200.89	2.05	6.1%	84/84	113	0.9689	1.086943
34	-20.86	2.06	5.9%	98/98	134	0.9675	1.227838
35	-251.55	2.22	5.7%	86/86	110	0.9665	1.165912
36	-335.83	2.35	5.6%	84/84	106	0.9654	1.134788
37	-106.60	2.67	5.4%	64/64	74	0.9647	1.179391
38	-112.47	2.86	5.3%	59/59	71	0.9640	1.299894
39	-88.76	2.68	5.1%	65/65	89	0.9631	1.139534
40	-161.58	2.57	5.0%	88/88	113	0.9620	1.121565
41	-195.98	2.53	4.9%	88/88	120	0.9608	1.115067
42	-187.76	3.46	4.8%	51/51	71	0.9601	1.260285
43	-166.05	3.44	4.7%	98/98	127	0.9588	1.306968
44	-137.57	3.44	4.5%	98/98	126	0.9576	1.069580
45	-80.07	3.37	4.4%	48/48	62	0.9570	1.175972
46	-168.19	3.27	4.3%	93/93	110	0.9559	1.162469
47	-114.55	4.23	4.3%	86/86	107	0.9548	1.319227
48	-217.08	4.17	4.2%	73/73	94	0.9539	1.136204
49	-34.15	4.21	4.1%	91/91	119	0.9527	1.359850
50	-233.19	4.24	4.0%	56/56	70	0.9520	1.247217
51	28.87	3.84	3.9%	94/94	130	0.9507	1.252429
52	-271.32	5.02	3.8%	83/83	111	0.9496	1.514646
53	-162.56	4.97	3.8%	97/97	132	0.9483	1.252136
54	-84.39	4.74	3.7%	52/52	65	0.9477	1.245617
55	-196.19	4.88	3.6%	65/65	88	0.9468	1.228242
56	-341.33	4.93	3.6%	85/85	114	0.9457	1.161186
57	-164.49	5.93	3.5%	47/47	64	0.9450	1.687367
58	-116.90	5.61	3.4%	95/95	133	0.9437	1.253310
59	-79.30	5.75	3.4%	40/40	58	0.9432	1.391669
60	-126.07	5.95	3.3%	77/77	99	0.9422	1.305260
61	-299.60	5.96	3.3%	53/53	71	0.9415	1.438606
62	-133.62	5.76	3.2%	56/56	76	0.9407	1.329350
63	-186.20	6.80	3.2%	82/82	113	0.9396	1.323877
64	-130.17	6.80	3.1%	101/101	128	0.9383	1.441404
65	-423.23	7.07	3.1%	76/76	108	0.9373	1.366945
66	-91.18	6.65	3.0%	59/59	81	0.9365	1.127322
67	-105.63	8.81	3.0%	69/69	92	0.9356	1.242958
68	-95.78	8.13	2.9%	87/87	109	0.9345	1.609852
69	-106.21	8.19	2.9%	65/65	93	0.9336	1.446459
70	-122.57	8.16	2.9%	74/74	101	0.9326	1.382510
71	-93.36	8.02	2.8%	107/107	138	0.9312	1.441290
72	-127.52	9.83	2.8%	77/77	103	0.9302	1.613293

Continued: Ddqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
73	-127.56	9.66	2.7%	45/45	72	0.9295	1.455809
74	-81.81	9.60	2.7%	61/61	78	0.9287	1.429500
75	-47.10	9.68	2.7%	61/61	80	0.9279	1.476662
76	-118.27	9.36	2.6%	77/77	90	0.9270	1.550372
77	-107.49	9.35	2.6%	64/64	83	0.9262	1.701725
78	-136.67	10.49	2.6%	68/68	92	0.9253	1.688008
79	-37.15	10.54	2.5%	57/57	74	0.9245	1.706836
80	-96.75	10.66	2.5%	68/68	87	0.9237	1.462011
81	-79.60	10.81	2.5%	55/55	68	0.9230	1.640296
82	-116.14	10.59	2.4%	111/111	136	0.9217	1.422630
83	-126.90	12.03	2.4%	81/81	96	0.9207	1.646084
84	-227.55	11.83	2.4%	59/59	73	0.9200	1.706392
85	-112.42	11.61	2.4%	69/69	89	0.9191	1.448884
86	-131.70	11.82	2.3%	70/70	93	0.9182	1.679928
87	-289.43	11.79	2.3%	81/81	98	0.9172	1.430375
88	-223.72	11.50	2.3%	48/48	65	0.9166	1.395078
89	-94.22	12.18	2.2%	55/55	75	0.9158	1.606973
90	-149.36	12.72	2.2%	52/52	72	0.9151	1.489153
91	-202.71	12.54	2.2%	75/75	103	0.9141	1.410939
92	-213.81	12.77	2.2%	75/75	84	0.9133	1.626074
93	-228.83	12.52	2.2%	99/99	129	0.9120	1.504622
94	-74.44	14.73	2.1%	83/83	117	0.9108	1.445100
95	-98.05	13.76	2.1%	94/94	114	0.9097	1.537599
96	-266.10	13.37	2.1%	103/103	135	0.9084	1.551501
97	-53.59	13.82	3.1%	93/93	119	0.9072	1.438574
98	-74.49	13.61	3.1%	57/57	77	0.9064	1.508911
99	-75.28	14.53	3.0%	61/61	73	0.9057	1.825171
100	-152.66	14.82	3.0%	90/90	114	0.9046	1.534334
101	-70.13	15.19	3.0%	50/50	67	0.9039	1.581905
102	-67.12	15.14	3.0%	92/92	111	0.9028	1.797954
103	-157.19	14.91	3.0%	85/85	104	0.9018	1.535515
104	-43.62	14.56	3.0%	52/52	69	0.9011	1.657230
105	-194.52	15.37	3.0%	91/91	115	0.9000	1.561399
106	-176.32	15.80	3.0%	98/98	118	0.8988	1.472515
107	-103.44	15.49	3.0%	51/51	61	0.8982	1.533862
108	-57.73	15.39	3.0%	87/87	100	0.8972	1.625300
109	-308.21	15.75	3.0%	92/92	115	0.8961	1.445679
110	-324.36	16.61	3.0%	79/79	96	0.8951	1.843895
111	24.92	16.94	2.0%	56/56	72	0.8944	1.461619
112	-72.49	16.97	2.0%	83/83	112	0.8933	1.531529
113	13.36	16.65	2.0%	77/77	99	0.8923	1.557509
114	-66.01	17.83	2.0%	130/130	156	0.8908	1.766659
115	-404.98	17.89	2.0%	96/96	115	0.8896	1.536547
116	-52.11	18.25	3.0%	93/93	127	0.8884	1.633969
117	-81.20	17.78	3.0%	82/82	102	0.8873	1.653379
118	-37.31	17.72	3.0%	55/55	71	0.8866	1.771166
119	-137.31	18.53	2.0%	66/66	85	0.8858	1.614253
120	-126.77	18.69	2.0%	78/78	106	0.8848	1.401425
121	-46.26	18.49	2.0%	56/56	65	0.8841	1.445665
122	-27.02	18.16	2.0%	108/108	140	0.8827	1.784458
123	-90.16	18.53	2.0%	52/52	69	0.8820	1.558937
124	-92.72	18.74	2.0%	51/51	67	0.8814	1.737782
125	-86.18	19.56	2.0%	81/81	95	0.8804	1.734063
126	-99.02	19.46	2.0%	95/95	129	0.8792	1.630488
127	-69.69	19.73	2.0%	121/121	142	0.8778	1.599967
128	-76.96	19.40	2.0%	74/74	101	0.8768	1.553383
129	-81.73	20.37	2.0%	101/101	129	0.8755	1.790738
130	-96.66	20.16	3.0%	71/71	89	0.8746	1.744820
131	-95.86	20.35	3.0%	66/66	87	0.8737	1.453757
132	-73.10	20.15	3.0%	75/75	91	0.8728	1.656667
133	-164.32	19.96	3.0%	63/63	85	0.8720	1.541116
134	-101.50	20.24	3.0%	69/69	90	0.8711	1.603781
135	-152.84	20.25	3.0%	82/82	106	0.8701	1.791160
136	-113.91	20.25	3.0%	57/57	72	0.8693	1.600377
137	-226.59	20.53	3.0%	82/82	106	0.8683	1.199955
138	-164.72	20.29	3.0%	94/94	130	0.8670	1.681583
139	-75.91	21.12	3.0%	59/59	73	0.8663	1.615417
140	-56.98	21.00	3.0%	92/92	121	0.8651	1.542354
141	-86.85	20.88	3.0%	62/62	82	0.8643	1.500009
142	-165.51	20.80	3.0%	108/108	140	0.8629	1.421913
143	-113.13	21.11	3.0%	68/68	89	0.8620	1.668391
144	-99.97	21.31	3.0%	58/58	72	0.8613	1.823670
145	-73.07	22.00	3.0%	94/94	116	0.8601	1.417464
146	-75.84	21.87	3.0%	67/67	98	0.8592	1.449709

Continued: Ddqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
147	-313.59	22.12	3.0%	64/64	79	0.8584	1.605624
148	-64.80	21.91	3.0%	87/87	120	0.8572	2.028254
149	-96.17	21.72	3.0%	49/49	70	0.8565	1.528067
150	-62.36	22.42	3.0%	59/59	68	0.8558	1.169085
151	-95.48	22.26	3.0%	71/71	89	0.8550	1.404538
152	-24.26	22.35	3.0%	66/66	87	0.8541	1.536002
153	-12.40	22.49	3.0%	106/106	139	0.8527	1.558118
154	-175.13	22.33	3.0%	67/67	87	0.8519	1.534283
155	50.72	23.23	3.0%	68/68	83	0.8510	1.629656
156	-264.98	22.92	3.0%	75/75	93	0.8501	1.780934
157	-55.40	23.01	3.0%	58/58	70	0.8494	1.368846
158	-42.40	22.43	3.0%	47/47	65	0.8488	1.524035
159	-78.32	23.01	3.0%	57/57	72	0.8481	1.214622
160	-63.18	22.24	3.0%	50/50	64	0.8474	1.381111
161	-77.00	22.76	3.0%	57/57	77	0.8467	1.485037
162	-98.11	22.67	3.0%	78/78	101	0.8457	1.707239
163	-111.30	23.03	3.0%	65/65	82	0.8449	1.611874
164	-99.05	22.70	3.0%	52/52	65	0.8442	1.507015
165	-117.03	22.94	3.0%	59/59	81	0.8434	1.931908
166	-85.42	22.84	3.0%	84/84	115	0.8423	1.333633
167	-84.31	23.39	4.0%	81/81	106	0.8412	1.840733
168	-171.72	23.74	4.0%	55/55	71	0.8405	1.533223
169	-80.41	23.28	4.0%	94/94	124	0.8393	1.667692
170	-80.03	23.23	4.0%	80/80	97	0.8383	1.626951
171	-127.49	22.80	4.0%	85/85	106	0.8373	1.520922
172	-91.73	22.99	4.0%	55/55	68	0.8366	1.379708
173	-94.09	23.17	4.0%	53/53	65	0.8360	1.793131
174	-87.93	22.94	4.0%	59/59	81	0.8352	1.329110
175	-103.03	22.73	4.0%	85/85	105	0.8341	2.096388
176	-60.17	23.45	4.0%	92/92	112	0.8330	1.330064
177	-101.94	23.18	4.0%	72/72	95	0.8321	1.361179
178	-114.34	23.17	4.0%	48/48	70	0.8314	1.915216
179	-38.74	23.62	4.0%	99/99	124	0.8302	1.448774
180	-220.51	23.62	4.0%	85/85	104	0.8291	1.347604
181	-100.50	22.91	4.0%	59/59	78	0.8284	1.630677
182	-240.57	23.66	4.0%	85/85	108	0.8273	1.944989
183	-59.38	23.49	4.0%	86/86	115	0.8261	1.889959
184	-83.01	23.21	4.0%	68/68	80	0.8254	1.609631
185	-67.44	23.22	4.0%	69/69	88	0.8245	1.627768
186	-68.60	23.26	4.0%	85/85	113	0.8234	1.229601
187	-341.30	23.16	4.0%	79/79	99	0.8224	1.588216
188	-57.41	23.94	4.0%	79/79	104	0.8214	1.921886
189	-118.86	23.32	4.0%	112/112	132	0.8200	1.794255
190	-86.33	23.25	4.0%	60/60	73	0.8193	1.838684
191	-139.63	23.11	4.0%	66/66	85	0.8185	1.642867
192	-136.28	23.31	4.0%	98/98	117	0.8173	1.587177
193	-77.88	23.50	4.0%	63/63	77	0.8166	1.877965
194	-109.58	23.50	4.0%	110/110	134	0.8152	1.514314
195	-82.74	23.35	4.0%	84/84	112	0.8141	1.677510
196	-68.88	23.52	4.0%	53/53	70	0.8134	1.638373
197	-87.80	23.57	3.0%	68/68	88	0.8126	1.703030
198	-55.56	22.43	3.0%	92/92	108	0.8115	1.688982
199	-90.59	23.26	3.0%	57/57	71	0.8108	1.817013
200	-128.54	22.85	3.0%	104/104	131	0.8095	1.963644
201	-74.98	22.75	3.0%	88/88	120	0.8083	1.608513
202	-88.61	22.66	3.0%	66/66	89	0.8074	1.615185
203	-89.73	22.40	3.0%	63/63	79	0.8066	1.945853
204	-94.13	22.68	3.0%	69/69	87	0.8058	1.464193
205	-97.33	22.90	3.0%	100/100	121	0.8046	1.530377
206	-40.22	22.44	3.0%	55/55	65	0.8039	1.459274
207	-96.79	22.58	3.0%	75/75	97	0.8030	1.827271
208	-114.24	22.98	3.0%	65/65	78	0.8022	1.905567
209	-62.96	22.53	3.0%	48/48	66	0.8016	1.530909
210	-108.10	22.47	3.0%	91/91	113	0.8004	1.350212
211	-54.28	22.23	3.0%	77/77	95	0.7995	1.466489
212	-59.19	22.49	3.0%	59/59	68	0.7988	1.937802
213	-93.78	22.24	3.0%	73/73	102	0.7978	1.360372
214	-123.44	22.05	3.0%	68/68	89	0.7969	1.419657
215	-60.30	22.24	3.0%	56/56	70	0.7962	2.418931
216	-55.24	21.63	2.0%	66/66	85	0.7954	1.573498
217	-88.85	22.36	2.0%	82/82	104	0.7944	1.711153
218	-85.46	22.44	2.0%	70/70	92	0.7935	1.418986
219	-102.79	21.78	2.0%	69/69	87	0.7926	1.606614
220	-76.43	21.89	2.0%	71/71	90	0.7917	1.648639

Continued: Ddqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
221	-130.51	22.22	2.0%	66/66	84	0.7909	1.918015
222	-38.50	22.55	2.0%	50/50	68	0.7902	1.825389
223	-92.98	22.27	2.0%	51/51	61	0.7896	1.463219
224	-85.47	21.99	2.0%	92/92	111	0.7885	1.358225
225	-87.43	22.31	2.0%	92/92	114	0.7874	1.574465
226	-98.99	21.64	2.0%	48/48	64	0.7867	1.453475
227	-74.06	21.60	2.0%	54/54	74	0.7860	1.560851
228	-76.55	21.36	2.0%	103/103	128	0.7847	1.942909
229	-90.28	20.97	2.0%	79/79	99	0.7838	1.477824
230	-173.46	21.75	1.0%	83/83	107	0.7827	1.569286
231	-82.20	21.11	1.0%	51/51	66	0.7820	1.548526
232	-49.50	21.21	1.0%	66/66	79	0.7813	1.632938
233	-41.15	21.30	1.0%	66/66	84	0.7804	1.885037
234	-71.65	21.15	1.0%	64/64	82	0.7796	1.572154
235	-118.99	21.12	1.0%	65/65	87	0.7788	1.676862
236	-123.46	21.06	1.0%	69/69	84	0.7779	1.493541
237	-90.38	20.56	1.0%	66/66	79	0.7771	1.538128
238	-79.19	20.90	1.0%	76/76	99	0.7762	1.840692
239	-192.68	20.40	1.0%	111/111	136	0.7748	1.693084
240	-75.47	20.06	1.0%	90/90	108	0.7737	1.715704
241	6.89	20.41	1.0%	59/59	71	0.7730	1.643630
242	-115.88	19.34	1.0%	80/80	97	0.7721	1.745693
243	-90.31	19.89	1.0%	77/77	98	0.7711	1.586654
244	-97.39	19.99	1.0%	89/89	106	0.7701	1.742659
245	-48.02	19.77	2.0%	56/56	70	0.7694	2.027872
246	-108.10	19.46	2.0%	61/61	80	0.7686	1.780995
247	-79.19	19.56	2.0%	52/52	61	0.7680	1.770342
248	-47.41	19.59	2.0%	76/76	101	0.7670	1.832935
249	-92.95	19.28	2.0%	69/69	88	0.7661	1.691975
250	-101.24	18.80	2.0%	80/80	102	0.7651	1.646006
251	-56.05	20.03	2.0%	63/63	80	0.7643	2.199724
252	-140.06	18.66	2.0%	70/70	86	0.7634	1.938795
253	-88.24	19.41	2.0%	47/47	56	0.7629	1.424933
254	-80.40	17.91	2.0%	96/96	120	0.7617	2.191430
255	-8.69	17.66	2.0%	113/113	135	0.7604	1.628138
256	-16.71	18.00	3.0%	97/97	125	0.7591	1.810280
257	-122.36	18.30	3.0%	83/83	111	0.7580	1.982012
258	-128.90	17.04	3.0%	68/68	83	0.7572	1.611440
259	-30.31	17.09	3.0%	77/77	101	0.7562	1.914634
260	-107.14	17.53	3.0%	73/73	93	0.7553	1.842728
261	-45.17	16.66	4.0%	56/56	72	0.7546	1.218604
262	-47.41	16.72	4.0%	77/77	97	0.7536	1.739137
263	-32.66	16.80	4.0%	59/59	71	0.7529	1.948145
264	-195.18	16.19	4.0%	91/91	109	0.7518	1.685366
265	-62.56	15.76	4.0%	110/110	131	0.7505	1.996075
266	-58.36	16.44	4.0%	110/110	129	0.7493	2.084319
267	-201.88	15.95	3.0%	84/84	103	0.7482	1.955176
268	-16.97	15.94	4.0%	56/56	70	0.7476	1.549837
269	-101.00	15.61	4.0%	64/64	84	0.7467	1.876105
270	-186.46	15.01	4.0%	110/110	145	0.7453	1.695637
271	-76.39	15.82	4.0%	70/70	96	0.7443	1.910067
272	-64.06	15.55	4.0%	80/80	94	0.7434	1.909203
273	-75.00	14.66	4.0%	90/90	114	0.7423	1.925705
274	-126.89	15.36	4.0%	82/82	100	0.7413	2.129779
275	-67.96	15.22	4.0%	63/63	78	0.7405	1.629749
276	-62.30	15.09	4.0%	72/72	85	0.7397	1.814668
277	-84.02	15.04	4.0%	82/82	102	0.7387	1.587045
278	-108.53	14.85	4.0%	62/62	74	0.7379	2.117667
279	-35.39	13.71	4.0%	100/100	122	0.7367	1.931841
280	-60.98	14.16	4.0%	54/54	66	0.7361	1.745285
281	-78.80	14.39	4.0%	73/73	92	0.7352	1.854308
282	-80.80	13.99	4.0%	79/79	98	0.7342	1.829802
283	-79.48	14.25	4.0%	92/92	109	0.7331	2.006370
284	-43.66	13.03	4.0%	102/102	128	0.7318	1.859217
285	-80.84	14.10	4.0%	68/68	84	0.7310	1.822400
286	-111.47	13.50	4.0%	73/73	93	0.7301	2.224558
287	-58.56	13.73	4.0%	115/115	139	0.7287	1.798636
288	-64.27	13.78	4.0%	53/53	72	0.7280	1.425495
289	-72.93	12.58	4.0%	66/66	82	0.7272	1.561913
290	-58.16	12.01	4.0%	80/80	99	0.7262	2.114531
291	-73.43	12.46	4.0%	103/103	122	0.7250	1.829760
292	-74.10	11.98	4.0%	70/70	84	0.7242	1.571418
293	-76.66	12.45	4.0%	83/83	101	0.7232	1.897919
294	-92.52	11.13	4.0%	74/74	101	0.7222	1.707758

Continued: Ddqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
295	-142.41	11.85	4.0%	75/75	98	0.7212	1.615216
296	-60.58	11.02	4.0%	85/85	113	0.7201	2.167725
297	-71.40	11.98	4.0%	69/69	82	0.7193	1.847643
298	-109.24	11.51	4.0%	69/69	85	0.7184	1.830173
299	-64.14	10.80	4.0%	77/77	87	0.7176	1.904852
300	-56.66	11.07	4.0%	88/88	113	0.7164	1.869629
301	-171.07	10.73	4.0%	85/85	100	0.7155	1.664337
302	-112.99	10.79	4.0%	54/54	66	0.7148	2.081488
303	-13.65	10.12	4.0%	102/102	125	0.7136	2.099709
304	-45.82	10.92	4.0%	47/47	66	0.7129	2.100694
305	-57.84	9.91	4.0%	106/106	122	0.7117	2.326991
306	-87.39	10.69	4.0%	82/82	97	0.7107	2.321657
307	-5.04	10.19	4.0%	115/115	136	0.7094	2.053267
308	-106.27	10.18	4.0%	73/73	90	0.7085	2.246442
309	-44.65	9.33	5.0%	65/65	79	0.7077	2.178892
310	-146.25	9.64	5.0%	62/62	73	0.7070	2.014849
311	-62.57	9.71	5.0%	89/89	120	0.7058	2.219280
312	-16.65	9.74	5.0%	57/57	67	0.7051	1.895181
313	-72.29	9.56	5.0%	119/119	138	0.7038	2.014676
314	-53.83	8.85	5.0%	54/54	75	0.7030	1.774247
315	-53.47	9.08	6.0%	82/82	105	0.7020	2.369001
316	-49.49	9.19	6.0%	108/108	134	0.7007	2.077648
317	-59.33	9.07	6.0%	63/63	79	0.6999	1.643977
318	-130.84	9.66	6.0%	74/74	94	0.6990	1.792806
319	-88.75	9.74	6.0%	65/65	79	0.6982	2.115097
320	-95.47	8.36	6.0%	72/72	87	0.6973	2.214764
321	-118.42	8.14	6.0%	86/86	100	0.6963	1.821083
322	-86.43	7.97	6.0%	73/73	91	0.6954	2.149022
323	-61.43	8.58	6.0%	75/75	97	0.6945	2.064251
324	-6.13	8.72	6.0%	111/111	136	0.6931	2.000525
325	-105.06	8.00	6.0%	51/51	73	0.6924	1.858976
326	-61.25	8.07	6.0%	55/55	73	0.6917	2.083863
327	-80.20	8.79	6.0%	67/67	80	0.6909	2.067063
328	-13.05	8.72	6.0%	89/89	109	0.6898	2.133909
329	-19.79	8.52	6.0%	67/67	84	0.6890	1.900431
330	-39.29	7.40	7.0%	91/91	113	0.6879	1.811281
331	-70.07	7.46	7.0%	62/62	79	0.6871	1.612279
332	-23.44	7.25	7.0%	97/97	127	0.6858	2.061373
333	-49.86	6.86	7.0%	82/82	100	0.6848	2.448787
334	-9.82	7.63	8.0%	83/83	105	0.6838	2.120857
335	-57.15	7.69	8.0%	88/88	111	0.6827	1.868202
336	-50.63	6.93	8.0%	65/65	75	0.6819	1.942321
337	-46.28	7.13	8.0%	76/76	91	0.6810	1.983668
338	-68.98	7.27	8.0%	57/57	71	0.6803	2.177882
339	-60.69	6.83	8.0%	110/110	139	0.6790	1.939929
340	-75.14	7.07	8.0%	84/84	99	0.6780	2.198638
341	-66.00	7.32	8.0%	100/100	120	0.6768	1.961078
342	-107.65	6.57	8.0%	79/79	103	0.6758	2.374653
343	-26.84	6.95	8.0%	63/63	74	0.6750	1.851807
344	-121.83	6.99	8.0%	99/99	113	0.6739	1.835206
345	-63.09	5.97	7.0%	71/71	84	0.6731	2.068032
346	-59.87	6.66	7.0%	88/88	109	0.6720	2.356334
347	-39.35	7.06	7.0%	69/69	84	0.6712	2.428952
348	-66.57	6.37	7.0%	60/60	74	0.6704	2.440426
349	-62.17	6.46	7.0%	49/49	63	0.6698	2.015766
350	-36.60	6.72	7.0%	84/84	108	0.6688	2.320975
351	-117.73	6.83	7.0%	59/59	72	0.6680	2.276506
352	-4.27	6.27	8.0%	80/80	105	0.6670	2.214620
353	-95.30	6.99	8.0%	66/66	89	0.6661	1.993103
354	-25.01	6.95	8.0%	100/100	115	0.6650	2.191291
355	13.25	6.56	8.0%	81/81	97	0.6640	1.943993
356	-61.11	6.23	7.0%	55/55	64	0.6634	2.580250
357	-54.08	6.91	7.0%	96/96	124	0.6622	2.086806
358	-5.32	6.39	8.0%	71/71	89	0.6613	1.856523
359	-79.98	6.08	8.0%	72/72	86	0.6604	2.185174
360	-28.27	6.46	8.0%	86/86	103	0.6594	2.004092
361	-9.16	6.66	7.0%	69/69	84	0.6586	2.342319
362	-56.36	6.21	7.0%	78/78	100	0.6576	2.195731
363	-53.97	5.50	7.0%	88/88	113	0.6565	1.743275
364	-51.87	5.10	7.0%	68/68	78	0.6557	1.681448
365	-48.79	5.26	7.0%	102/102	132	0.6544	2.412744
366	-61.93	4.88	7.0%	109/109	124	0.6532	2.202618
367	-44.25	5.02	7.0%	114/114	133	0.6518	2.222927
368	-112.75	5.05	6.0%	94/94	111	0.6507	2.015333

Continued: Ddqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
369	-18.72	5.92	6.0%	59/59	74	0.6500	2.508833
370	-39.76	5.10	6.0%	94/94	113	0.6489	2.485069
371	-39.89	5.31	7.0%	95/95	106	0.6478	1.848503
372	-56.18	4.40	8.0%	85/85	112	0.6467	2.331068
373	-6.97	4.39	8.0%	87/87	102	0.6457	2.007565
374	-132.37	5.41	8.0%	74/74	87	0.6449	1.761904
375	-29.39	5.02	8.0%	107/107	129	0.6436	2.109115
376	-107.56	4.90	8.0%	72/72	88	0.6427	2.242170
377	-52.50	5.02	8.0%	67/67	81	0.6419	2.057042
378	-190.88	4.87	8.0%	82/82	101	0.6409	2.231789
379	-107.38	4.26	8.0%	91/91	118	0.6397	2.007592
380	-74.29	4.92	8.0%	75/75	97	0.6388	2.034870
381	-36.53	4.94	8.0%	82/82	109	0.6377	2.200733
382	-72.88	4.63	8.0%	62/62	72	0.6370	1.857883
383	-33.85	5.80	8.0%	103/103	132	0.6357	2.039110
384	-87.97	4.63	8.0%	106/106	128	0.6344	2.187184
385	-83.57	5.95	8.0%	65/65	80	0.6336	2.452026
386	-19.39	4.98	8.0%	86/86	113	0.6325	2.045042
387	-59.90	5.37	8.0%	90/90	119	0.6313	2.061913
388	-46.12	5.63	8.0%	79/79	94	0.6304	1.938592
389	-98.79	4.58	8.0%	117/117	141	0.6290	1.893088
390	-86.80	6.08	8.0%	105/105	127	0.6278	2.271907
391	-71.58	4.64	8.0%	82/82	105	0.6267	2.489404
392	-49.57	5.76	8.0%	69/69	78	0.6259	2.148888
393	31.25	5.25	8.0%	127/127	152	0.6244	1.902861
394	-98.62	5.09	8.0%	77/77	98	0.6235	2.108707
395	-38.57	4.71	8.0%	84/84	106	0.6224	2.065863
396	-62.62	5.16	8.0%	57/57	71	0.6217	2.391584
397	-15.84	5.42	8.0%	808/808	1000	0.6118	2.141872
398	-17.99	4.90	8.0%	87/87	101	0.6108	2.347038
399	-49.77	4.39	8.0%	56/56	79	0.6100	2.297859
400	-48.85	6.10	8.0%	96/96	111	0.6089	2.386700
401	-70.90	5.11	8.0%	48/48	59	0.6083	2.658266
402	-69.22	6.22	8.0%	84/84	114	0.6072	2.199284
403	-37.32	5.89	8.0%	75/75	82	0.6064	2.527638
404	23.05	6.73	8.0%	100/100	114	0.6053	2.422647
405	-17.29	6.21	8.0%	60/60	78	0.6045	2.704886
406	-45.18	6.08	8.0%	71/71	85	0.6037	2.169572
407	-20.39	6.91	8.0%	85/85	102	0.6027	2.138558
408	-113.95	5.99	8.0%	128/128	158	0.6011	2.521910
409	-22.18	6.31	7.0%	137/137	170	0.5994	2.474915
410	-107.59	6.92	7.0%	81/81	106	0.5984	2.091849
411	-85.42	6.17	7.0%	85/85	106	0.5973	2.597854
412	-25.84	6.74	7.0%	109/109	120	0.5961	2.350578
413	-73.85	6.80	7.0%	94/94	117	0.5950	2.164478
414	-29.28	7.95	7.0%	101/101	116	0.5938	1.999313
415	-97.25	7.48	6.0%	72/72	80	0.5930	1.644205
416	-71.11	7.60	6.0%	69/69	85	0.5922	2.326913
417	-63.42	7.68	6.0%	72/72	88	0.5913	2.477731
418	-26.25	8.40	6.0%	71/71	87	0.5904	2.214476
419	-33.68	7.78	6.0%	112/112	137	0.5891	2.150022
420	-21.11	7.68	6.0%	102/102	134	0.5878	2.424036
421	-74.47	7.00	6.0%	82/82	108	0.5867	2.089931
422	-21.17	7.74	6.0%	54/54	68	0.5860	1.892065
423	22.87	7.87	6.0%	100/100	111	0.5849	2.008498
424	-41.85	7.99	6.0%	55/55	71	0.5842	1.752210
425	-91.42	8.35	6.0%	694/694	809	0.5762	2.094000
426	-102.32	7.90	6.0%	65/65	75	0.5755	2.195495
427	2.29	9.14	6.0%	109/109	137	0.5741	2.210074
428	17.03	9.59	6.0%	128/128	161	0.5725	2.244067
429	-46.40	9.71	6.0%	71/71	91	0.5716	2.266600
430	-19.63	9.50	6.0%	71/71	89	0.5707	2.049492
431	-57.21	9.60	6.0%	70/70	76	0.5700	2.355873
432	13.55	10.44	6.0%	140/140	165	0.5684	2.148068
433	-54.14	10.43	6.0%	66/66	75	0.5676	2.616559
434	20.49	9.62	5.0%	108/108	138	0.5662	2.014212
435	-200.49	9.82	5.0%	151/151	176	0.5645	2.032708
436	-5.57	10.21	5.0%	134/134	160	0.5629	2.489259
437	-58.44	10.68	5.0%	94/94	121	0.5617	2.230405
438	-35.48	10.18	5.0%	77/77	95	0.5608	1.824069
439	-97.00	10.40	5.0%	90/90	109	0.5597	1.928563
440	10.52	11.04	5.0%	137/137	157	0.5581	1.852123
441	-62.32	11.14	5.0%	101/101	121	0.5569	2.072175
442	-0.98	11.31	5.0%	190/190	227	0.5547	1.988084

Continued: Ddqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
443	-74.80	11.72	5.0%	99/99	125	0.5535	1.981220
444	-83.15	11.74	5.0%	90/90	104	0.5524	2.189272
445	-13.21	12.31	5.0%	84/84	99	0.5515	2.012522
446	-24.29	12.46	5.0%	118/118	139	0.5501	2.053376
447	1.84	12.50	5.0%	91/91	109	0.5490	1.963336
448	-79.27	11.45	5.0%	96/96	116	0.5478	1.996379
449	-4.97	11.71	5.0%	76/76	91	0.5469	2.264469
450	-28.58	12.60	5.0%	859/859	1000	0.5370	2.156582
451	11.25	13.45	6.0%	88/88	102	0.5360	1.936949
452	26.05	12.94	5.0%	107/107	126	0.5348	2.270718
453	-41.64	13.94	5.0%	69/69	79	0.5340	1.776997
454	12.33	13.79	5.0%	281/281	339	0.5307	2.200667
455	-105.58	13.93	5.0%	94/94	104	0.5296	2.053877
456	-36.74	15.20	5.0%	102/102	131	0.5283	2.195576
457	11.04	15.26	5.0%	105/105	129	0.5270	2.373856
458	-22.59	14.68	4.0%	92/92	126	0.5258	2.325152
459	-20.95	15.04	5.0%	96/96	112	0.5247	1.994342
460	-45.81	15.41	5.0%	93/93	107	0.5236	2.165225
461	-23.09	14.22	5.0%	96/96	114	0.5225	2.237493
462	-26.16	16.22	5.0%	66/66	78	0.5217	2.724532
463	-50.97	14.69	5.0%	82/82	102	0.5207	2.362777
464	3.87	16.00	5.0%	78/78	102	0.5197	2.276784
465	-37.74	15.07	5.0%	69/69	77	0.5189	2.034264
466	-4.82	15.33	5.0%	86/86	98	0.5180	2.096598
467	-63.38	15.87	5.0%	93/93	114	0.5169	2.060322
468	-50.25	15.16	5.0%	80/80	91	0.5159	2.116940
469	14.58	16.05	6.0%	94/94	109	0.5149	2.304878
470	-20.16	16.37	6.0%	77/77	89	0.5140	2.409199
471	-30.08	16.62	5.0%	94/94	110	0.5129	1.995865
472	13.89	15.92	5.0%	77/77	92	0.5120	2.055146
473	42.65	15.51	5.0%	128/128	171	0.5103	2.199956
474	7.12	16.62	5.0%	134/134	158	0.5087	2.068596
475	3.60	16.48	5.0%	104/104	142	0.5073	1.974928
476	-3.76	15.35	5.0%	83/83	96	0.5064	1.978720
477	30.11	15.95	5.0%	69/69	81	0.5056	1.990688
478	19.81	16.27	6.0%	95/95	117	0.5044	1.919618
479	-117.19	16.41	6.0%	124/124	138	0.5030	2.265060
480	15.62	16.71	6.0%	101/101	124	0.5018	1.941855
481	-25.28	16.80	6.0%	64/64	76	0.5011	1.898428
482	17.60	17.63	6.0%	120/120	150	0.4996	2.233117
483	-12.03	17.88	6.0%	94/94	110	0.4985	2.028969
484	15.67	17.71	6.0%	107/107	126	0.4972	2.051550
485	-65.98	17.25	6.0%	66/66	81	0.4964	1.984363
486	-3.28	17.02	6.0%	91/91	115	0.4953	2.125995
487	-23.27	17.84	6.0%	142/142	165	0.4937	2.245756
488	-49.45	17.53	6.0%	100/100	118	0.4925	2.092334
489	-64.86	17.23	6.0%	128/128	154	0.4910	2.039883
490	-31.06	17.06	6.0%	102/102	114	0.4899	1.911560
491	-30.37	17.99	6.0%	93/93	105	0.4888	2.225257
492	7.07	17.72	7.0%	98/98	126	0.4876	2.066245
493	-43.08	18.14	7.0%	93/93	107	0.4865	2.078609
494	-9.62	17.88	8.0%	102/102	122	0.4853	2.219338
495	-21.99	19.68	8.0%	118/118	137	0.4839	2.030998
496	-13.70	19.93	8.0%	91/91	111	0.4828	1.974329
497	-37.65	19.81	8.0%	102/102	117	0.4817	2.269247
498	-37.83	19.74	8.0%	99/99	123	0.4805	2.110061
499	-65.92	20.77	8.0%	119/119	141	0.4791	2.077336
500	-8.62	20.23	8.0%	107/107	127	0.4778	2.227235
501	-48.06	20.51	8.0%	63/63	77	0.4771	2.487910
502	38.95	20.86	8.0%	118/118	144	0.4756	2.283982
503	-114.60	22.51	8.0%	771/771	891	0.4668	2.087320
504	8.23	24.01	8.0%	874/874	1000	0.4569	2.113234
505	26.40	23.70	8.0%	118/118	144	0.4555	1.906813
506	-50.34	24.68	8.0%	77/77	88	0.4546	2.321437
507	-16.42	23.59	8.0%	97/97	106	0.4536	1.910728
508	10.13	25.73	8.0%	898/898	1000	0.4437	2.093337
509	38.26	26.87	8.0%	111/111	137	0.4423	2.256134
510	5.67	26.18	8.0%	107/107	120	0.4411	2.023076
511	6.00	27.19	8.0%	70/70	87	0.4403	2.393104
512	-4.38	27.04	8.0%	127/127	149	0.4388	2.309848
513	44.51	26.61	9.0%	114/114	127	0.4375	2.415775
514	10.90	29.23	9.0%	893/893	1000	0.4276	2.156320
515	-162.42	30.79	9.0%	636/636	739	0.4203	2.239384
516	27.71	30.72	10.0%	135/135	159	0.4187	2.291326

Continued: Ddqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
517	45.34	31.02	10.0%	107/107	119	0.4176	1.817282
518	-46.63	30.68	10.0%	82/82	93	0.4166	2.107507
519	20.80	31.43	10.0%	131/131	161	0.4150	2.255856
520	-32.67	32.82	10.0%	861/861	1000	0.4051	2.185491
521	2.20	32.38	10.0%	84/84	95	0.4042	2.171878
522	-16.22	33.10	10.0%	871/871	1000	0.3943	2.224746
523	-60.50	33.44	10.0%	71/71	87	0.3934	2.170188
524	27.40	33.36	11.0%	118/118	134	0.3921	2.384510
525	-39.38	34.43	11.0%	143/143	180	0.3903	2.069836
526	-59.15	35.41	11.0%	853/853	1000	0.3804	2.182550
527	20.49	35.41	11.0%	145/145	163	0.3788	2.301114
528	31.60	36.23	11.0%	861/861	1000	0.3689	2.178501
529	-81.61	35.91	11.0%	209/209	236	0.3666	2.134724
530	-0.14	36.17	10.0%	104/104	117	0.3654	2.115758
531	-49.79	35.90	10.0%	867/867	1000	0.3555	2.226328
532	-150.03	36.67	10.0%	214/214	262	0.3529	2.199675
533	-25.33	36.36	10.0%	898/898	1000	0.3430	2.163188
534	-5.61	37.76	10.0%	841/841	1000	0.3331	2.231871
535	37.71	37.06	10.0%	85/85	97	0.3322	2.234358
536	-19.50	37.65	10.0%	897/897	1000	0.3223	2.229193
537	3.17	37.46	10.0%	121/121	132	0.3210	1.913514
538	23.78	37.05	10.0%	109/109	134	0.3196	2.288059
539	52.86	37.46	10.0%	104/104	114	0.3185	1.970167
540	-10.31	36.99	10.0%	891/891	1000	0.3086	2.273201
541	46.55	37.26	10.0%	116/116	132	0.3073	2.167633
542	-99.89	37.56	10.0%	145/145	163	0.3057	2.014391
543	20.75	36.20	10.0%	144/144	169	0.3040	2.242050
544	-130.72	36.54	10.0%	617/617	657	0.2975	2.279354
545	-98.96	36.56	10.0%	292/292	322	0.2943	2.257443
546	-43.91	36.03	10.0%	494/494	536	0.2890	2.144823
547	22.96	36.76	10.0%	822/822	1000	0.2791	2.268904
548	-6.92	36.57	10.0%	883/883	1000	0.2692	2.244905
549	-10.92	35.97	10.0%	869/869	1000	0.2593	2.169140
550	45.43	33.72	10.0%	868/868	1000	0.2494	2.263227
551	-100.42	35.26	9.0%	342/342	385	0.2456	2.256447
552	-181.57	34.90	9.0%	626/626	688	0.2388	2.245526
553	-53.41	34.75	9.0%	867/867	1000	0.2289	2.213559
554	-34.84	34.29	9.0%	856/856	1000	0.2190	2.225482
555	13.16	35.19	9.0%	854/854	1000	0.2091	2.117561
556	-1.31	34.06	9.0%	836/836	1000	0.1992	2.135824
557	-40.99	34.83	9.0%	799/799	1000	0.1893	2.177932
558	26.23	35.90	9.0%	851/851	1000	0.1794	2.174556
559	10.20	36.11	8.0%	834/834	1000	0.1695	2.159282
560	-6.65	36.45	8.0%	833/833	1000	0.1596	2.060927
561	5.34	38.04	8.0%	842/842	1000	0.1497	2.053427
562	-31.43	38.88	8.0%	859/859	1000	0.1398	2.101515
563	-72.09	39.87	8.0%	829/829	1000	0.1299	2.029103
564	-50.12	39.05	8.0%	823/823	1000	0.1200	1.954713
565	-44.87	38.47	8.0%	865/865	1000	0.1101	2.220863
566	-8.51	39.99	8.0%	790/790	1000	0.1002	2.019323
567	-2.48	39.89	8.0%	843/843	1000	0.0903	1.910393
568	-6.61	40.47	8.0%	911/911	1000	0.0804	1.895835
569	-29.68	40.71	7.0%	867/867	1000	0.0705	1.907286
570	-53.29	41.67	7.0%	868/868	1000	0.0606	1.954839
571	-3.46	42.87	7.0%	868/868	1000	0.0507	1.937218
572	-20.08	41.58	6.0%	866/866	1000	0.0408	1.899397
573	-56.71	42.98	6.0%	868/868	1000	0.0309	1.849679
574	-10.37	42.22	6.0%	840/840	1000	0.0210	1.859757
575	93.39	45.70	6.0%	797/797	1000	0.0111	1.854064
576	46.06	44.65	6.0%	833/833	1000	0.0100	1.866808
577	279.21	45.34	7.0%	364/364	467	0.0100	1.794767
578	40.52	45.07	6.0%	823/823	1000	0.0100	1.919856
579	-22.69	46.19	6.0%	819/819	1000	0.0100	1.703514
580	10.40	45.46	6.0%	829/829	1000	0.0100	1.702715
581	215.74	46.02	7.0%	489/489	590	0.0100	1.656326
582	31.96	46.76	7.0%	801/801	1000	0.0100	1.550960
583	229.50	47.33	8.0%	320/320	452	0.0100	1.560385
584	166.03	49.63	9.0%	749/749	920	0.0100	1.570260
585	27.11	46.93	9.0%	821/821	1000	0.0100	1.480945
586	-49.96	49.36	9.0%	839/839	1000	0.0100	1.453697
587	-42.76	48.62	9.0%	848/848	1000	0.0100	1.436978
588	5.90	48.20	9.0%	792/792	1000	0.0100	1.475060
589	-32.12	47.59	9.0%	689/689	1000	0.0100	1.402095
590	-36.29	47.38	9.0%	780/780	1000	0.0100	1.354233

Continued: Ddqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
591	-3.37	48.50	9.0%	786/786	1000	0.0100	1.354786
592	-16.08	50.67	8.0%	755/755	1000	0.0100	1.342594
593	25.45	51.24	8.0%	793/793	1000	0.0100	1.283155
594	-24.48	51.61	7.0%	728/728	1000	0.0100	1.285044
595	16.75	51.50	7.0%	872/872	1000	0.0100	1.255648
596	-18.57	51.41	7.0%	841/841	1000	0.0100	1.194250
597	16.71	54.87	7.0%	806/806	1000	0.0100	1.240567
598	-49.56	53.35	7.0%	839/839	1000	0.0100	1.223573
599	29.34	52.62	7.0%	790/790	1000	0.0100	1.166419
600	-0.18	51.99	7.0%	827/827	1000	0.0100	1.125839
601	-8.76	49.65	7.0%	828/828	1000	0.0100	1.070030
602	2.89	49.41	7.0%	800/800	1000	0.0100	1.115206
603	-53.39	46.43	7.0%	797/797	1000	0.0100	1.072516
604	-24.98	47.51	7.0%	828/828	1000	0.0100	1.047292
605	-15.89	46.49	7.0%	855/855	1000	0.0100	1.029109
606	-3.55	45.93	7.0%	817/817	1000	0.0100	1.027574
607	-15.83	45.91	7.0%	786/786	1000	0.0100	0.972385
608	13.01	45.24	7.0%	793/793	1000	0.0100	0.959675
609	-10.00	46.71	7.0%	806/806	1000	0.0100	0.973539
610	-44.97	47.25	7.0%	819/819	1000	0.0100	0.946316
611	69.18	46.10	7.0%	800/800	1000	0.0100	0.887567
612	43.50	45.89	7.0%	803/803	1000	0.0100	0.881577
613	-3.20	46.00	6.0%	837/837	1000	0.0100	0.880103
614	-49.07	45.57	6.0%	825/825	1000	0.0100	0.843194
615	147.08	46.57	7.0%	799/799	987	0.0100	0.806115
616	261.05	47.59	7.0%	282/282	361	0.0100	0.730023
617	259.18	48.49	8.0%	248/248	349	0.0100	0.823380
618	242.93	48.73	9.0%	154/154	219	0.0100	0.707065
619	238.38	49.71	10.0%	326/326	463	0.0100	0.687388
620	-35.25	50.30	10.0%	843/843	1000	0.0100	0.743365
621	10.59	48.91	10.0%	844/844	1000	0.0100	0.713110
622	245.66	49.91	11.0%	315/315	451	0.0100	0.641609
623	210.92	52.25	12.0%	685/685	851	0.0100	0.669727
624	256.27	52.83	12.0%	359/359	459	0.0100	0.724296
625	269.23	52.65	13.0%	392/392	490	0.0100	0.644377
626	288.61	51.56	14.0%	491/491	698	0.0100	0.694258
627	291.72	51.69	15.0%	174/174	228	0.0100	0.678116
628	52.12	51.26	15.0%	823/823	1000	0.0100	0.623395
629	224.09	52.31	16.0%	341/341	419	0.0100	0.591657
630	275.54	52.14	17.0%	147/147	219	0.0100	0.636184
631	11.78	52.03	17.0%	804/804	1000	0.0100	0.629307
632	196.23	51.99	18.0%	703/703	955	0.0100	0.614482
633	220.31	50.06	19.0%	427/427	565	0.0100	0.596930
634	222.98	49.12	20.0%	301/301	429	0.0100	0.561561
635	252.82	49.27	21.0%	220/220	299	0.0100	0.621492
636	248.96	51.97	22.0%	351/351	561	0.0100	0.540351
637	251.28	50.59	23.0%	370/370	490	0.0100	0.528677
638	255.81	49.44	24.0%	284/284	391	0.0100	0.525225
639	239.22	50.13	25.0%	150/150	223	0.0100	0.497624
640	226.91	49.85	26.0%	489/489	741	0.0100	0.509268
641	191.10	50.10	27.0%	533/533	746	0.0100	0.511242
642	272.12	51.89	28.0%	127/127	206	0.0100	0.519933
643	189.38	52.01	29.0%	624/624	767	0.0100	0.496559
644	182.78	49.92	29.0%	618/618	1000	0.0100	0.458069
645	67.51	48.43	29.0%	798/798	1000	0.0100	0.446498
646	289.87	48.55	30.0%	252/252	395	0.0100	0.404672
647	258.42	50.01	31.0%	210/210	420	0.0100	0.425631
648	242.94	49.90	32.0%	336/336	448	0.0100	0.445050
649	216.39	51.34	33.0%	345/345	428	0.0100	0.408972
650	314.69	50.54	34.0%	200/200	250	0.0100	0.424440
651	256.38	50.14	35.0%	236/236	384	0.0100	0.392141
652	249.92	49.74	35.0%	337/337	558	0.0100	0.449590
653	131.79	48.42	35.0%	765/765	1000	0.0100	0.404520
654	-59.68	51.97	35.0%	884/884	1000	0.0100	0.395341
655	276.78	52.49	36.0%	321/321	409	0.0100	0.399780
656	26.91	52.16	36.0%	772/772	1000	0.0100	0.414017
657	178.39	57.54	37.0%	609/609	850	0.0100	0.391655
658	-30.14	58.60	37.0%	749/749	1000	0.0100	0.371682
659	280.48	59.09	38.0%	357/357	495	0.0100	0.330422
660	145.46	59.99	39.0%	700/700	929	0.0100	0.370412
661	14.89	59.61	39.0%	784/784	1000	0.0100	0.363763
662	-47.63	62.39	39.0%	739/739	1000	0.0100	0.339301
663	188.89	64.66	39.0%	615/615	1000	0.0100	0.316752
664	111.95	66.01	39.0%	667/667	1000	0.0100	0.338865

Continued: Ddqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
665	254.99	63.95	40.0%	352/352	439	0.0100	0.348107
666	132.93	66.21	40.0%	681/681	1000	0.0100	0.336823
667	295.69	67.69	41.0%	380/380	608	0.0100	0.335736
668	248.44	68.26	42.0%	324/324	616	0.0100	0.322013
669	133.82	68.65	42.0%	610/610	1000	0.0100	0.318516
670	222.13	70.49	43.0%	433/433	720	0.0100	0.334718
671	279.93	70.49	44.0%	250/250	430	0.0100	0.291869
672	239.14	71.36	45.0%	417/417	556	0.0100	0.313304
673	253.23	71.77	46.0%	452/452	772	0.0100	0.306655
674	98.42	72.78	46.0%	689/689	1000	0.0100	0.283330
675	116.02	72.51	46.0%	718/718	1000	0.0100	0.298865
676	256.70	73.46	47.0%	525/525	732	0.0100	0.285059
677	245.13	74.11	47.0%	595/595	933	0.0100	0.291244
678	244.62	74.26	48.0%	242/242	352	0.0100	0.308327
679	268.58	75.58	49.0%	370/370	537	0.0100	0.311488
680	278.73	75.77	50.0%	318/318	582	0.0100	0.297741
681	169.61	75.72	49.0%	543/543	1000	0.0100	0.307440
682	252.92	78.05	50.0%	348/348	579	0.0100	0.308810
683	264.94	77.18	50.0%	370/370	527	0.0100	0.320620
684	210.52	75.68	50.0%	367/367	526	0.0100	0.305796
685	-525.38	74.99	50.0%	90/90	98	0.0100	0.313191
686	249.79	68.62	51.0%	566/566	997	0.0100	0.346325
687	-115.47	66.16	51.0%	378/378	563	0.0100	0.340552
688	241.25	66.41	52.0%	243/243	406	0.0100	0.332790
689	244.74	67.32	53.0%	333/333	578	0.0100	0.342254
690	263.75	66.66	54.0%	265/265	460	0.0100	0.293044
691	120.35	67.95	54.0%	611/611	1000	0.0100	0.338854
692	251.80	66.57	55.0%	264/264	456	0.0100	0.323626
693	271.22	66.59	56.0%	315/315	399	0.0100	0.338996
694	250.10	68.35	57.0%	340/340	443	0.0100	0.327743
695	249.51	65.81	58.0%	303/303	441	0.0100	0.344796
696	230.55	66.76	59.0%	485/485	809	0.0100	0.366118
697	237.70	67.31	60.0%	251/251	444	0.0100	0.344531
698	237.26	66.58	61.0%	271/271	404	0.0100	0.363775
699	241.79	66.19	62.0%	223/223	306	0.0100	0.335062
700	69.98	63.61	62.0%	841/841	1000	0.0100	0.378916
701	98.50	65.63	62.0%	757/757	1000	0.0100	0.401114
702	193.37	65.85	63.0%	628/628	782	0.0100	0.350956
703	276.74	64.24	64.0%	216/216	317	0.0100	0.331195
704	262.51	65.60	65.0%	172/172	272	0.0100	0.360370
705	262.89	65.55	66.0%	232/232	360	0.0100	0.346361
706	264.71	67.21	67.0%	361/361	666	0.0100	0.378309
707	224.08	67.37	68.0%	334/334	471	0.0100	0.395763
708	26.02	67.68	68.0%	845/845	1000	0.0100	0.364597
709	277.00	69.06	69.0%	271/271	448	0.0100	0.350848
710	250.68	68.64	70.0%	429/429	643	0.0100	0.341454
711	211.74	69.13	71.0%	279/279	412	0.0100	0.368164
712	244.99	68.60	72.0%	311/311	529	0.0100	0.333746
713	68.46	69.10	72.0%	751/751	1000	0.0100	0.308837
714	237.29	68.46	73.0%	252/252	503	0.0100	0.326201
715	262.07	68.81	73.0%	145/145	218	0.0100	0.384118
716	272.43	68.97	73.0%	239/239	433	0.0100	0.324413
717	215.26	68.50	73.0%	424/424	600	0.0100	0.314506
718	279.99	70.79	73.0%	296/296	507	0.0100	0.337828
719	236.44	71.02	73.0%	276/276	465	0.0100	0.344704
720	262.66	70.34	74.0%	239/239	357	0.0100	0.332580
721	265.86	70.74	75.0%	308/308	471	0.0100	0.339727
722	270.86	70.34	75.0%	121/121	232	0.0100	0.367570
723	155.13	69.47	74.0%	495/495	1000	0.0100	0.354664
724	174.64	69.53	73.0%	442/442	1000	0.0100	0.332997
725	270.59	69.11	73.0%	329/329	639	0.0100	0.354424
726	259.51	69.30	73.0%	100/100	180	0.0100	0.308646
727	291.76	68.65	73.0%	212/212	292	0.0100	0.347418
728	241.37	69.84	74.0%	342/342	593	0.0100	0.361483
729	151.33	68.63	73.0%	617/617	1000	0.0100	0.357037
730	249.77	71.64	73.0%	418/418	968	0.0100	0.341380
731	262.46	71.64	74.0%	224/224	457	0.0100	0.334234
732	264.95	69.40	74.0%	164/164	297	0.0100	0.382648
733	291.73	68.42	74.0%	408/408	803	0.0100	0.351936
734	285.98	69.73	74.0%	165/165	275	0.0100	0.404427
735	286.80	69.63	74.0%	187/187	324	0.0100	0.349573
736	271.30	73.29	74.0%	258/258	450	0.0100	0.344386
737	257.75	72.72	74.0%	288/288	394	0.0100	0.385817
738	285.14	71.55	74.0%	188/188	362	0.0100	0.409160

Continued: Ddqn Original complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
739	54.23	71.87	73.0%	177/177	223	0.0100	0.354416
740	259.58	74.50	73.0%	250/250	452	0.0100	0.389307
741	56.92	73.99	72.0%	150/150	182	0.0100	0.426513
742	304.18	74.29	72.0%	224/224	371	0.0100	0.366603
743	272.01	76.48	72.0%	228/228	351	0.0100	0.353508
744	284.34	75.41	73.0%	146/146	275	0.0100	0.379896
745	36.80	76.76	73.0%	97/97	147	0.0100	0.458610
746	268.41	74.57	73.0%	202/202	292	0.0100	0.419558
747	267.58	74.33	73.0%	217/217	322	0.0100	0.319450
748	258.64	76.34	73.0%	559/559	739	0.0100	0.431336
749	120.02	78.63	72.0%	961/961	1000	0.0100	0.395125
750	289.62	79.62	71.0%	305/305	487	0.0100	0.361823
751	191.84	77.64	70.0%	543/543	1000	0.0100	0.377396
752	257.07	77.53	71.0%	192/192	368	0.0100	0.341847
753	304.27	78.50	72.0%	271/271	481	0.0100	0.352286
754	134.27	82.00	72.0%	755/755	1000	0.0100	0.410839
755	295.81	83.33	72.0%	197/197	332	0.0100	0.375907
756	299.59	81.74	73.0%	164/164	274	0.0100	0.383154
757	285.82	81.83	73.0%	273/273	333	0.0100	0.376212
758	267.59	80.77	74.0%	383/383	555	0.0100	0.389588
759	276.13	82.80	74.0%	266/266	367	0.0100	0.402467
760	290.45	79.48	74.0%	117/117	199	0.0100	0.432836
761	279.28	80.30	74.0%	333/333	364	0.0100	0.388077
762	289.26	80.76	74.0%	439/439	493	0.0100	0.410146
763	249.37	79.86	75.0%	265/265	476	0.0100	0.426374
764	260.42	81.52	76.0%	226/226	400	0.0100	0.440993
765	257.91	80.75	76.0%	205/205	293	0.0100	0.419717
766	257.77	81.96	77.0%	252/252	413	0.0100	0.410287
767	291.44	83.20	77.0%	255/255	422	0.0100	0.424103
768	181.34	83.07	76.0%	498/498	1000	0.0100	0.430741
769	29.81	84.84	77.0%	84/84	117	0.0100	0.499755
770	269.73	84.44	77.0%	445/445	913	0.0100	0.456341
771	31.26	84.67	76.0%	90/90	103	0.0100	0.345780
772	245.98	85.58	76.0%	203/203	395	0.0100	0.433171
773	265.51	84.47	76.0%	298/298	515	0.0100	0.436458
774	241.79	85.18	76.0%	220/220	226	0.0100	0.432231
775	289.57	84.37	77.0%	184/184	266	0.0100	0.479042
776	64.46	84.32	76.0%	99/99	125	0.0100	0.423974
777	-100.08	84.73	75.0%	338/338	425	0.0100	0.403988
778	281.38	83.15	75.0%	178/178	246	0.0100	0.412259
779	252.33	84.16	75.0%	172/172	260	0.0100	0.451276
780	270.02	83.63	75.0%	166/166	285	0.0100	0.462366
781	237.57	84.43	76.0%	341/341	772	0.0100	0.449872
782	25.51	83.21	75.0%	93/93	125	0.0100	0.406957
783	289.54	83.54	75.0%	151/151	217	0.0100	0.432075
784	275.98	84.43	75.0%	207/207	365	0.0100	0.423078
785	79.10	82.19	75.0%	748/748	1000	0.0100	0.439670
786	275.00	83.57	75.0%	226/226	499	0.0100	0.443625
787	292.45	84.77	76.0%	368/368	813	0.0100	0.478075
788	-130.17	82.45	75.0%	720/720	1000	0.0100	0.428137
789	268.88	82.74	75.0%	161/161	229	0.0100	0.372592
790	290.75	82.81	75.0%	171/171	286	0.0100	0.416400
791	266.63	81.88	76.0%	376/376	705	0.0100	0.423324
792	302.80	83.01	76.0%	241/241	350	0.0100	0.435485
793	302.94	81.78	75.0%	142/142	203	0.0100	0.494605
794	236.87	80.57	74.0%	219/219	255	0.0100	0.488975
795	239.12	82.76	74.0%	655/655	843	0.0100	0.405301
796	48.49	81.60	73.0%	108/108	132	0.0100	0.571625
797	254.51	80.64	73.0%	227/227	264	0.0100	0.463720
798	-56.47	80.28	72.0%	673/673	1000	0.0100	0.462566
799	51.85	81.21	71.0%	142/142	167	0.0100	0.469429
800	23.81	80.51	71.0%	92/92	100	0.0100	0.487393

Table 3: Complete per-iteration training output - Dqn Modified (800 episodes).

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
1	-147.06	0.24	0.0%	50/40	70	0.9993	nan
2	-359.38	0.24	0.0%	67/61	93	0.9984	nan
3	-158.89	0.24	0.0%	102/86	122	0.9972	nan
4	-141.98	0.24	0.0%	67/58	85	0.9963	nan
5	-173.75	0.24	0.0%	84/72	117	0.9952	nan
6	-329.77	0.24	0.0%	78/67	104	0.9941	nan
7	-327.33	0.24	0.0%	57/47	79	0.9934	nan
8	-179.77	0.24	0.0%	58/47	70	0.9927	nan
9	-351.41	0.24	0.0%	79/63	108	0.9916	nan
10	-298.78	0.24	0.0%	90/80	118	0.9904	nan
11	-78.47	-0.36	0.0%	51/43	68	0.9898	2.378366
12	-159.12	-0.58	0.0%	58/45	78	0.9890	2.183941
13	-116.79	-0.43	0.0%	74/67	96	0.9880	2.109947
14	-90.14	-0.15	0.0%	63/49	84	0.9872	1.890186
15	-168.22	0.23	0.0%	80/66	115	0.9861	1.631696
16	-243.62	0.20	0.0%	54/41	74	0.9853	1.630903
17	-280.66	0.14	0.0%	77/61	99	0.9844	1.782376
18	-175.90	0.30	0.0%	76/66	92	0.9834	1.357518
19	-160.71	0.18	0.0%	69/62	93	0.9825	1.550764
20	-33.68	0.12	5.0%	51/44	68	0.9819	1.540087
21	-159.27	0.14	4.8%	48/33	64	0.9812	1.444286
22	-153.45	0.46	4.5%	68/55	87	0.9804	1.372086
23	-175.76	0.17	4.3%	54/49	68	0.9797	1.640241
24	-132.43	0.25	4.2%	54/46	75	0.9789	1.242995
25	-120.53	0.37	4.0%	63/58	78	0.9782	1.178728
26	-93.94	0.27	3.8%	49/38	65	0.9775	1.211328
27	-295.67	0.43	3.7%	62/50	76	0.9768	1.148219
28	-355.29	0.39	3.6%	81/71	105	0.9757	1.296328
29	-108.34	0.75	3.4%	48/41	63	0.9751	2.050307
30	-192.05	0.45	3.3%	49/43	74	0.9744	1.610327
31	-336.50	1.01	3.2%	104/96	137	0.9730	1.229883
32	-135.93	0.77	3.1%	69/64	89	0.9721	1.376396
33	-98.78	0.63	3.0%	96/85	123	0.9709	1.291586
34	-253.02	0.40	2.9%	79/68	103	0.9699	1.638038
35	-152.95	0.51	2.9%	81/69	113	0.9688	1.408279
36	-207.13	0.66	2.8%	88/83	120	0.9676	1.354004
37	-597.02	0.50	2.7%	77/60	98	0.9666	1.301109
38	-185.11	0.50	2.6%	46/33	58	0.9661	1.548023
39	-124.51	1.23	2.6%	59/54	76	0.9653	1.571273
40	-182.35	0.96	2.5%	94/75	113	0.9642	1.588046
41	-101.52	0.71	2.4%	86/75	114	0.9631	1.373784
42	-104.34	0.80	2.4%	57/50	75	0.9623	1.223945
43	-194.28	0.89	2.3%	81/69	109	0.9612	1.665583
44	-134.75	0.81	2.3%	75/68	104	0.9602	1.311982
45	-129.05	0.87	2.2%	46/40	62	0.9596	1.577818
46	-138.97	0.98	2.2%	68/57	83	0.9588	1.610268
47	-170.58	0.90	2.1%	76/69	96	0.9578	1.378401
48	-152.79	1.21	2.1%	70/54	92	0.9569	1.474257
49	-193.73	1.21	2.0%	94/78	112	0.9558	1.290625
50	-96.76	1.11	2.0%	61/54	74	0.9551	1.377693
51	-171.03	1.43	2.0%	78/69	102	0.9541	1.497369
52	-223.71	0.98	1.9%	84/72	113	0.9529	1.477513
53	-360.34	1.03	1.9%	79/66	96	0.9520	1.345634
54	-138.15	1.33	1.9%	41/33	58	0.9514	1.390091
55	-33.19	1.57	3.6%	62/55	83	0.9506	1.574210
56	-163.62	1.04	3.6%	89/86	118	0.9494	1.532085
57	-111.22	1.00	3.5%	47/39	65	0.9488	1.452840
58	-51.10	1.18	5.2%	83/70	111	0.9477	1.566933
59	-336.09	1.37	5.1%	60/50	79	0.9469	1.480893
60	-371.88	1.28	5.0%	71/61	99	0.9459	1.343086
61	-116.42	1.75	4.9%	56/44	71	0.9452	1.757143
62	-95.35	1.60	4.8%	44/36	65	0.9446	1.680483
63	-277.99	1.86	4.8%	77/67	108	0.9435	1.714475
64	-290.81	1.63	4.7%	77/71	97	0.9425	1.589762
65	-203.93	1.68	4.6%	57/49	82	0.9417	1.496282
66	-153.73	1.66	4.5%	54/43	73	0.9410	1.655809
67	-263.87	2.30	4.5%	59/50	86	0.9402	1.773135
68	-277.42	1.89	4.4%	78/70	97	0.9392	1.536151
69	-159.74	2.00	4.3%	84/73	108	0.9381	1.571837
70	-302.09	1.96	4.3%	70/63	99	0.9371	1.592491
71	-21.41	2.19	4.2%	73/66	103	0.9361	1.558594
72	-46.53	2.41	5.6%	75/59	102	0.9351	1.847060

Continued: Dqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
73	-213.67	2.68	5.5%	69/60	82	0.9343	1.756783
74	-102.19	2.20	5.4%	42/34	64	0.9337	1.832223
75	-110.53	2.36	5.3%	51/44	65	0.9330	1.860790
76	-162.99	2.58	5.3%	61/49	87	0.9322	1.976125
77	-188.42	2.62	5.2%	71/57	87	0.9313	1.537485
78	-128.26	3.20	5.1%	62/56	83	0.9305	1.933873
79	-96.99	3.08	5.1%	46/39	65	0.9298	2.002488
80	-119.17	3.04	5.0%	49/41	72	0.9291	1.687444
81	-85.53	2.88	4.9%	51/40	62	0.9285	1.919448
82	-112.73	3.18	4.9%	97/76	125	0.9273	1.910466
83	-163.70	3.13	4.8%	55/48	71	0.9266	1.714962
84	4.12	2.95	4.8%	61/52	78	0.9258	1.964801
85	-93.72	3.61	4.7%	63/51	86	0.9249	2.072651
86	-139.00	3.44	4.7%	57/48	77	0.9242	1.913164
87	-167.14	3.58	4.6%	81/64	99	0.9232	2.073997
88	-115.04	3.48	4.5%	59/53	65	0.9226	1.993013
89	-113.09	4.10	4.5%	48/42	62	0.9219	2.352776
90	-312.91	3.69	4.4%	60/50	75	0.9212	1.715807
91	-293.90	3.79	4.4%	75/65	95	0.9203	1.961326
92	-133.27	4.19	4.3%	64/57	79	0.9195	1.755656
93	-113.86	4.42	4.3%	69/61	89	0.9186	2.478885
94	-68.20	3.87	4.3%	91/82	112	0.9175	2.078659
95	-131.16	4.00	4.2%	83/75	109	0.9164	1.932215
96	-129.84	5.22	4.2%	88/80	121	0.9152	2.351376
97	-98.31	4.58	4.1%	68/59	98	0.9142	1.944437
98	-111.69	5.19	4.1%	60/51	69	0.9136	2.082340
99	-86.88	4.90	4.0%	59/52	72	0.9129	1.942084
100	-163.44	4.96	4.0%	86/75	111	0.9118	1.828488
101	-51.32	4.79	4.0%	47/41	63	0.9111	2.425658
102	-114.04	5.14	4.0%	70/62	91	0.9102	1.969920
103	-222.30	5.19	4.0%	99/83	124	0.9090	1.948536
104	-93.57	5.28	4.0%	47/44	61	0.9084	1.799621
105	-163.92	5.38	4.0%	101/87	126	0.9071	2.047658
106	-197.71	5.11	4.0%	79/68	106	0.9061	2.087507
107	-92.71	5.60	4.0%	53/46	69	0.9054	2.108172
108	-90.17	5.70	4.0%	73/64	87	0.9046	2.133686
109	-80.04	5.57	4.0%	89/70	116	0.9034	1.907772
110	-119.08	5.50	4.0%	77/63	93	0.9025	1.746923
111	-90.69	5.72	4.0%	53/48	64	0.9019	2.234248
112	-122.88	5.90	4.0%	106/85	132	0.9005	1.972350
113	-117.97	6.17	4.0%	52/46	69	0.8999	1.964930
114	-189.60	6.20	4.0%	110/95	135	0.8985	1.901980
115	-103.38	5.88	4.0%	85/71	100	0.8975	1.963505
116	-161.77	5.95	4.0%	96/84	117	0.8964	2.004211
117	-105.84	6.40	4.0%	68/60	85	0.8955	1.971135
118	-22.83	6.27	5.0%	57/48	73	0.8948	2.431992
119	-208.90	6.39	5.0%	64/58	81	0.8940	1.884941
120	-123.52	6.32	4.0%	70/59	94	0.8931	2.069558
121	-137.95	6.20	4.0%	49/42	60	0.8925	1.947342
122	-106.61	6.70	4.0%	91/73	109	0.8914	1.761995
123	-109.22	6.83	4.0%	54/49	66	0.8908	2.351755
124	-119.28	6.91	4.0%	60/49	71	0.8901	1.659916
125	-115.19	6.68	4.0%	72/59	90	0.8892	1.718640
126	-121.10	7.17	4.0%	85/72	112	0.8881	1.674529
127	-153.13	7.04	4.0%	75/60	100	0.8871	1.682289
128	-132.46	6.76	4.0%	87/72	108	0.8860	2.223072
129	-153.25	6.66	4.0%	70/60	95	0.8851	1.637152
130	-147.94	7.23	4.0%	71/58	85	0.8842	1.821706
131	-132.57	6.81	4.0%	79/74	97	0.8832	1.514281
132	-183.28	7.34	4.0%	65/58	86	0.8824	2.043517
133	-105.22	6.99	4.0%	55/49	75	0.8817	1.850414
134	-126.42	7.88	4.0%	76/62	92	0.8807	1.713837
135	-153.82	7.46	4.0%	68/59	92	0.8798	1.727949
136	-99.62	8.00	4.0%	55/43	69	0.8792	1.523598
137	-147.77	8.05	4.0%	124/105	147	0.8777	1.751040
138	-153.68	7.59	4.0%	75/66	101	0.8767	1.526639
139	-101.65	7.95	4.0%	49/43	65	0.8761	1.884556
140	-79.92	7.96	4.0%	84/69	114	0.8749	1.713260
141	-102.57	8.03	4.0%	63/51	74	0.8742	1.446591
142	-115.54	7.85	4.0%	110/91	138	0.8728	1.840938
143	-243.08	8.18	4.0%	59/50	81	0.8720	1.465967
144	-113.72	8.97	4.0%	63/55	82	0.8712	1.808377
145	-184.22	8.85	4.0%	90/75	120	0.8700	1.625598
146	-138.19	8.98	4.0%	66/54	84	0.8692	2.102639

Continued: Dqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
147	-125.32	8.84	4.0%	60/49	76	0.8684	1.800810
148	-102.88	8.81	4.0%	84/74	120	0.8673	1.588386
149	-94.46	8.59	4.0%	49/41	64	0.8666	1.611297
150	-63.85	9.46	4.0%	60/54	78	0.8658	1.657727
151	-90.26	8.62	4.0%	63/49	79	0.8651	1.500743
152	-122.11	8.75	4.0%	60/47	80	0.8643	1.949000
153	-190.20	9.03	4.0%	90/72	121	0.8631	2.283022
154	-166.08	8.96	4.0%	73/60	100	0.8621	1.467445
155	-75.74	9.02	3.0%	56/49	70	0.8614	1.868852
156	-119.97	9.07	3.0%	67/60	86	0.8605	2.017039
157	-94.55	9.03	3.0%	55/45	67	0.8599	1.559056
158	-76.66	9.05	2.0%	39/34	62	0.8593	2.128641
159	-108.50	9.12	2.0%	57/44	72	0.8585	1.759357
160	-95.90	9.12	2.0%	46/41	61	0.8579	1.307643
161	-106.12	9.20	2.0%	56/52	75	0.8572	1.611251
162	-170.42	8.76	2.0%	71/65	98	0.8562	1.751270
163	-34.03	9.01	3.0%	79/73	93	0.8553	1.826123
164	-84.22	8.87	3.0%	54/44	68	0.8546	1.247353
165	-123.47	9.72	3.0%	61/56	80	0.8538	2.128737
166	-58.24	8.79	3.0%	98/83	128	0.8526	1.789866
167	-121.79	9.00	3.0%	70/60	90	0.8517	1.494567
168	-122.01	8.70	3.0%	57/48	73	0.8510	1.991922
169	-104.64	9.37	3.0%	100/83	117	0.8498	1.910807
170	-125.75	9.15	3.0%	64/50	90	0.8489	1.907880
171	-145.90	9.29	3.0%	74/64	99	0.8479	1.961952
172	-85.86	8.82	2.0%	49/43	65	0.8473	1.677778
173	-31.91	9.37	3.0%	48/41	68	0.8466	1.619263
174	-111.72	9.36	3.0%	64/60	84	0.8458	1.880220
175	-148.14	8.94	3.0%	83/75	106	0.8447	1.697668
176	-137.17	9.46	3.0%	65/59	83	0.8439	1.531075
177	-116.47	9.44	3.0%	64/58	89	0.8430	1.486618
178	-110.02	9.32	3.0%	55/45	69	0.8424	1.354746
179	-122.10	9.57	3.0%	69/51	93	0.8414	1.695863
180	-11.22	9.71	3.0%	80/65	106	0.8404	1.592606
181	-154.61	8.97	3.0%	61/54	82	0.8396	2.032694
182	-156.98	8.89	3.0%	86/73	107	0.8385	1.768439
183	-152.01	9.32	3.0%	64/56	86	0.8377	1.517448
184	-93.24	9.14	3.0%	56/46	77	0.8369	1.173172
185	-159.42	9.31	3.0%	64/55	77	0.8361	1.889340
186	-120.71	9.66	3.0%	80/72	111	0.8350	1.745212
187	-146.76	9.25	3.0%	83/71	99	0.8341	1.534703
188	-101.46	9.27	3.0%	79/63	102	0.8330	1.859385
189	-159.11	9.10	3.0%	68/67	96	0.8321	1.473881
190	-111.21	9.07	3.0%	53/46	76	0.8313	1.586508
191	-111.18	9.22	3.0%	61/57	80	0.8306	1.963924
192	-119.71	9.16	3.0%	94/82	116	0.8294	1.499935
193	-90.34	9.18	3.0%	58/48	75	0.8287	1.548728
194	-103.55	9.10	3.0%	83/71	108	0.8276	1.358421
195	-128.32	9.37	3.0%	67/56	92	0.8267	1.922253
196	-73.14	8.88	3.0%	57/49	72	0.8260	1.826622
197	-149.45	9.01	3.0%	69/62	88	0.8251	2.183778
198	-58.66	9.24	4.0%	63/53	81	0.8243	1.589156
199	-80.12	8.50	4.0%	46/36	67	0.8236	1.555174
200	-165.50	8.55	4.0%	81/65	106	0.8226	1.583445
201	-97.28	9.16	4.0%	87/79	115	0.8214	1.769124
202	-114.52	8.81	4.0%	95/74	110	0.8204	1.617505
203	-176.70	9.11	4.0%	47/37	62	0.8197	2.073084
204	-137.61	9.21	4.0%	70/64	85	0.8189	1.502258
205	-163.05	9.11	4.0%	93/79	115	0.8178	1.594409
206	-88.05	8.55	4.0%	52/43	63	0.8171	1.775252
207	-115.82	8.58	4.0%	74/64	90	0.8162	2.004647
208	-127.80	8.13	4.0%	66/56	78	0.8155	1.382540
209	-67.95	8.37	4.0%	50/40	66	0.8148	1.730040
210	-165.89	8.65	4.0%	89/78	119	0.8136	1.627494
211	-186.22	8.72	4.0%	60/52	81	0.8128	1.465239
212	-72.34	8.27	4.0%	52/47	67	0.8122	1.839680
213	-130.03	8.18	4.0%	80/72	104	0.8111	1.542305
214	-94.07	8.43	4.0%	85/74	104	0.8101	1.500368
215	-102.18	8.48	4.0%	53/46	66	0.8095	1.417212
216	-111.15	8.56	4.0%	55/48	69	0.8088	2.249330
217	-139.74	8.20	4.0%	58/49	95	0.8078	1.811956
218	-159.60	8.07	3.0%	84/73	105	0.8068	2.146051
219	-148.50	7.74	3.0%	67/56	93	0.8059	1.868340
220	-114.07	7.77	3.0%	66/50	86	0.8050	2.381290

Continued: Dqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
221	-143.26	8.17	3.0%	71/56	86	0.8042	2.022571
222	-92.65	7.42	3.0%	47/39	66	0.8035	2.189210
223	-88.56	8.05	3.0%	49/42	61	0.8029	1.868906
224	-188.20	8.19	3.0%	66/53	85	0.8021	1.522647
225	-38.81	6.98	4.0%	67/60	86	0.8012	2.334329
226	-68.71	6.57	4.0%	52/43	66	0.8006	1.842790
227	-116.89	6.79	4.0%	53/44	67	0.7999	2.205243
228	-14.31	7.11	5.0%	82/70	115	0.7988	1.685510
229	-162.94	6.83	5.0%	66/57	96	0.7978	1.620507
230	-124.62	6.08	5.0%	67/56	101	0.7968	1.914845
231	-73.59	6.08	5.0%	56/48	69	0.7961	1.837267
232	-71.27	6.02	5.0%	61/48	77	0.7954	1.587263
233	-123.51	5.87	5.0%	50/44	77	0.7946	2.400733
234	-145.55	5.46	5.0%	57/49	67	0.7940	1.631781
235	-40.10	6.41	5.0%	62/53	85	0.7931	1.755935
236	-108.79	5.53	5.0%	72/64	92	0.7922	1.789167
237	-22.37	4.69	5.0%	54/47	72	0.7915	1.762317
238	-141.71	4.75	5.0%	72/64	97	0.7905	1.774301
239	-177.79	4.60	5.0%	99/84	131	0.7892	2.245826
240	-116.77	5.07	5.0%	86/73	112	0.7881	1.759543
241	-139.09	5.23	5.0%	58/50	75	0.7874	1.974718
242	-156.19	4.79	5.0%	60/52	83	0.7866	2.203706
243	-111.15	4.53	5.0%	74/63	97	0.7856	1.737398
244	-129.56	4.77	5.0%	80/66	103	0.7846	1.667024
245	-88.05	4.52	5.0%	52/45	76	0.7838	1.921561
246	-131.30	4.72	5.0%	64/50	84	0.7830	1.928909
247	-65.73	4.94	5.0%	40/33	60	0.7824	1.929502
248	-44.37	4.72	6.0%	72/63	94	0.7815	1.634598
249	-91.99	4.57	6.0%	61/50	85	0.7806	2.350092
250	-110.12	4.46	6.0%	75/63	107	0.7796	1.916054
251	-100.00	4.09	6.0%	49/39	66	0.7789	1.688823
252	-109.86	4.29	6.0%	77/66	97	0.7780	1.922388
253	-74.52	4.30	6.0%	51/45	61	0.7773	1.733022
254	-129.60	3.76	6.0%	86/72	115	0.7762	1.824614
255	-109.38	3.82	6.0%	70/63	95	0.7753	2.196716
256	-117.08	3.98	6.0%	80/74	102	0.7743	1.960264
257	-107.96	3.90	6.0%	67/55	87	0.7734	1.602095
258	-106.20	3.48	6.0%	55/52	73	0.7727	1.613638
259	-102.48	2.81	6.0%	65/60	85	0.7718	2.117504
260	-126.12	2.54	6.0%	75/56	102	0.7708	1.860785
261	-18.85	3.12	7.0%	56/50	69	0.7701	2.449884
262	-17.80	2.76	7.0%	62/53	85	0.7693	2.192970
263	-89.56	2.93	6.0%	50/43	61	0.7687	1.810186
264	-133.07	2.98	6.0%	63/53	84	0.7679	1.967795
265	-120.67	1.70	6.0%	72/57	95	0.7669	1.894046
266	-119.88	1.95	6.0%	80/65	109	0.7658	2.158310
267	-125.49	1.53	6.0%	70/60	87	0.7650	1.904715
268	-55.43	1.60	6.0%	53/46	67	0.7643	1.534912
269	-177.00	1.68	6.0%	68/62	85	0.7635	1.870111
270	-159.18	1.75	6.0%	94/80	126	0.7622	1.998596
271	-80.82	1.60	6.0%	93/75	114	0.7611	2.030462
272	-24.10	1.25	7.0%	57/51	75	0.7604	2.230635
273	-60.46	1.70	6.0%	70/65	98	0.7594	1.773542
274	-131.39	2.50	6.0%	78/72	94	0.7585	1.848874
275	-111.52	1.45	6.0%	55/50	78	0.7577	2.040278
276	-84.92	0.72	6.0%	52/43	69	0.7570	2.631087
277	-105.47	1.22	6.0%	88/73	119	0.7558	2.017855
278	-137.65	1.04	6.0%	58/49	78	0.7551	2.205694
279	-119.34	0.92	6.0%	87/73	113	0.7539	1.793077
280	-66.02	1.05	6.0%	58/52	72	0.7532	1.842956
281	-105.22	0.20	6.0%	61/56	79	0.7524	2.058246
282	-139.41	0.10	6.0%	63/51	81	0.7516	2.045332
283	-87.72	0.02	6.0%	82/69	109	0.7506	1.928713
284	-101.40	0.67	6.0%	90/77	116	0.7494	2.215926
285	-138.52	0.38	6.0%	59/49	80	0.7486	1.896461
286	-95.65	0.90	6.0%	61/58	75	0.7479	1.993398
287	-112.71	-0.18	6.0%	56/47	87	0.7470	1.888194
288	-103.00	-0.94	6.0%	54/42	73	0.7463	1.573785
289	-120.32	0.15	6.0%	62/48	86	0.7454	1.891533
290	-126.25	-0.41	6.0%	55/49	77	0.7447	2.479251
291	-13.08	0.93	7.0%	89/74	119	0.7435	2.656491
292	-113.16	0.39	7.0%	57/47	77	0.7427	1.971083
293	-93.72	-0.33	7.0%	74/67	95	0.7418	1.938459
294	-119.89	-1.27	7.0%	71/59	102	0.7408	2.330393

Continued: Dqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
295	-114.21	-0.93	7.0%	61/51	80	0.7400	1.868612
296	-108.76	-0.58	7.0%	73/63	94	0.7391	1.942222
297	-51.20	-0.62	8.0%	66/55	88	0.7382	2.041443
298	-124.58	-1.72	7.0%	61/43	78	0.7374	2.053473
299	-11.04	-1.63	8.0%	56/45	70	0.7367	2.421329
300	-35.79	-0.83	9.0%	87/73	112	0.7356	1.743524
301	-110.55	-1.19	9.0%	80/75	104	0.7346	1.950846
302	-66.47	-1.51	9.0%	66/59	83	0.7338	1.593293
303	-124.75	-0.83	9.0%	66/53	86	0.7329	1.952725
304	-42.50	-1.80	9.0%	48/40	64	0.7323	2.324720
305	-73.25	-1.98	9.0%	73/64	99	0.7313	1.984730
306	-248.07	-1.98	9.0%	92/79	115	0.7302	2.832942
307	-96.53	-1.99	9.0%	92/81	130	0.7289	2.044392
308	-36.70	-1.52	10.0%	61/48	88	0.7280	2.262197
309	-88.76	-1.73	10.0%	50/45	71	0.7273	2.093510
310	-50.70	-1.33	10.0%	74/63	89	0.7264	2.096354
311	-123.85	-2.55	10.0%	89/77	107	0.7254	1.736894
312	-38.11	-1.84	10.0%	65/53	76	0.7246	1.941114
313	-116.30	-1.71	10.0%	80/71	103	0.7236	2.918505
314	-90.97	-2.74	10.0%	60/50	71	0.7229	2.083903
315	-121.05	-2.32	10.0%	51/42	77	0.7221	2.342954
316	-124.94	-2.80	10.0%	72/64	101	0.7211	2.059118
317	-55.99	-1.98	10.0%	54/43	72	0.7204	2.246438
318	-125.34	-2.49	10.0%	53/44	77	0.7197	1.751741
319	-98.16	-2.20	10.0%	62/53	78	0.7189	2.057590
320	-132.00	-2.81	10.0%	83/70	96	0.7179	2.252908
321	-92.88	-3.04	10.0%	79/69	109	0.7169	2.130932
322	-132.75	-2.05	10.0%	77/69	101	0.7159	2.241798
323	-121.12	-1.73	10.0%	64/56	82	0.7150	1.683468
324	-7.26	-1.95	11.0%	88/75	113	0.7139	2.232975
325	-82.36	-2.49	10.0%	58/53	76	0.7132	2.157781
326	-89.20	-3.31	10.0%	44/40	63	0.7125	2.154182
327	-0.19	-2.76	11.0%	85/67	97	0.7116	2.209197
328	-106.25	-2.98	10.0%	66/60	83	0.7108	2.423050
329	-76.71	-2.51	10.0%	61/52	76	0.7100	1.915567
330	-37.94	-2.25	11.0%	76/67	99	0.7090	2.347589
331	-128.54	-2.70	11.0%	65/53	78	0.7083	1.784946
332	-126.33	-3.57	11.0%	74/63	94	0.7073	1.743576
333	-104.77	-3.69	11.0%	73/65	86	0.7065	2.464583
334	-115.35	-3.45	11.0%	60/50	82	0.7057	2.541074
335	-109.32	-2.95	11.0%	83/71	113	0.7045	2.697170
336	-118.78	-3.39	11.0%	67/53	80	0.7038	2.433858
337	-104.74	-2.91	11.0%	56/50	78	0.7030	2.353596
338	-119.12	-3.97	11.0%	56/48	72	0.7023	1.911754
339	-125.11	-3.49	11.0%	92/82	122	0.7011	2.761507
340	-101.37	-3.31	11.0%	52/47	80	0.7003	2.060145
341	-93.26	-3.33	11.0%	85/70	107	0.6992	2.272704
342	-10.60	-3.71	12.0%	77/69	98	0.6982	2.307490
343	-102.03	-4.33	12.0%	46/39	57	0.6977	2.024607
344	-156.76	-3.33	12.0%	87/74	112	0.6966	2.209063
345	-105.80	-4.12	12.0%	55/47	71	0.6959	2.324233
346	-93.49	-3.49	12.0%	86/70	122	0.6947	2.153958
347	-93.72	-4.24	12.0%	59/50	77	0.6939	2.052843
348	-91.48	-4.41	11.0%	70/62	83	0.6931	1.815138
349	-94.69	-4.50	11.0%	38/32	56	0.6925	3.004305
350	-90.47	-5.12	11.0%	67/55	105	0.6915	2.038075
351	-74.74	-4.88	11.0%	55/49	66	0.6908	2.560126
352	-70.27	-4.52	11.0%	80/63	97	0.6899	2.713474
353	-67.41	-4.81	11.0%	68/60	97	0.6889	2.121194
354	-101.33	-4.29	11.0%	62/56	85	0.6881	3.344830
355	2.88	-4.53	12.0%	55/49	70	0.6874	2.297672
356	-95.35	-4.49	12.0%	43/39	54	0.6868	1.819307
357	-19.34	-4.36	13.0%	72/66	105	0.6858	2.287651
358	-75.30	-4.99	13.0%	57/47	74	0.6851	2.568398
359	-96.85	-5.56	13.0%	52/44	74	0.6843	1.953924
360	-115.84	-4.59	13.0%	72/61	93	0.6834	2.327227
361	-44.33	-5.42	12.0%	59/51	70	0.6827	2.600970
362	-138.62	-4.68	12.0%	67/54	93	0.6818	2.288720
363	-110.92	-5.29	12.0%	86/72	99	0.6808	2.401261
364	-61.54	-6.01	12.0%	64/59	77	0.6801	2.691531
365	-26.08	-5.26	13.0%	90/80	125	0.6788	2.298757
366	-117.51	-5.65	13.0%	88/68	112	0.6777	2.203563
367	-157.56	-5.68	13.0%	76/65	93	0.6768	2.349497
368	-168.98	-5.77	13.0%	79/65	111	0.6757	2.122478

Continued: Dqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
369	-61.73	-5.14	13.0%	55/48	66	0.6750	2.137213
370	-101.14	-5.57	13.0%	69/57	79	0.6743	2.549946
371	-113.69	-6.23	13.0%	79/68	103	0.6732	2.121191
372	-17.16	-5.85	13.0%	85/76	107	0.6722	2.640110
373	-87.04	-6.14	13.0%	54/51	68	0.6715	2.376222
374	-44.31	-6.16	14.0%	73/65	93	0.6706	2.543764
375	-138.66	-6.18	14.0%	78/67	104	0.6695	2.588268
376	-142.42	-6.36	14.0%	81/68	94	0.6686	2.293643
377	-107.88	-6.63	14.0%	56/45	71	0.6679	1.819702
378	-102.65	-6.62	14.0%	82/69	102	0.6669	2.528186
379	-135.78	-6.30	14.0%	102/92	132	0.6656	2.523727
380	-104.15	-6.40	14.0%	71/61	99	0.6646	2.502872
381	-74.15	-6.65	14.0%	79/68	93	0.6637	2.476752
382	-101.10	-7.01	14.0%	63/58	78	0.6629	2.562173
383	-59.41	-6.53	14.0%	74/64	120	0.6617	2.565260
384	-88.37	-7.14	14.0%	86/68	114	0.6606	2.333389
385	-55.81	-6.47	14.0%	71/62	83	0.6598	2.421224
386	-119.45	-6.56	14.0%	97/87	129	0.6585	2.451178
387	-104.06	-7.48	14.0%	68/59	97	0.6575	2.463658
388	-88.78	-7.04	14.0%	58/50	79	0.6568	2.903617
389	-225.69	-7.30	14.0%	105/94	125	0.6555	2.468850
390	-108.51	-7.53	14.0%	63/50	95	0.6546	2.294009
391	-78.40	-7.23	13.0%	95/81	108	0.6535	2.665078
392	-52.77	-7.14	13.0%	59/52	73	0.6528	2.606875
393	-91.88	-6.47	13.0%	112/97	148	0.6513	2.555566
394	-169.27	-7.66	13.0%	71/59	92	0.6504	2.234292
395	-76.62	-6.95	13.0%	65/55	89	0.6495	2.983029
396	-45.65	-7.54	13.0%	60/51	73	0.6488	3.067831
397	-108.80	-6.23	12.0%	86/75	105	0.6478	2.572215
398	-94.41	-6.43	12.0%	54/52	67	0.6471	2.113915
399	-57.73	-6.90	11.0%	70/58	83	0.6463	2.347266
400	-116.53	-7.43	10.0%	66/57	96	0.6453	2.878785
401	-64.06	-7.16	10.0%	66/57	73	0.6446	2.344421
402	-82.71	-6.87	10.0%	80/63	103	0.6436	2.293249
403	-88.45	-7.15	10.0%	46/41	59	0.6430	2.034184
404	-43.51	-7.01	10.0%	66/55	76	0.6423	2.142397
405	-68.23	-6.61	10.0%	60/51	72	0.6416	2.787604
406	-69.70	-7.03	10.0%	56/48	74	0.6408	2.777242
407	-108.84	-6.74	10.0%	54/47	78	0.6400	2.168132
408	-186.94	-6.60	9.0%	99/79	129	0.6388	2.667647
409	-113.75	-6.91	9.0%	97/78	127	0.6375	2.656106
410	-128.93	-7.09	9.0%	92/83	120	0.6363	2.222128
411	-83.97	-7.58	9.0%	86/66	102	0.6353	1.959824
412	-70.45	-6.54	9.0%	77/63	106	0.6343	2.275251
413	-97.23	-6.62	9.0%	77/64	100	0.6333	3.024006
414	-99.56	-6.80	9.0%	64/55	84	0.6324	2.693293
415	-63.07	-7.42	9.0%	64/53	77	0.6317	2.184173
416	-79.77	-8.18	9.0%	69/55	84	0.6308	2.521660
417	-39.87	-6.89	9.0%	85/76	97	0.6299	2.250173
418	-75.80	-6.53	9.0%	60/50	75	0.6291	2.351018
419	-103.09	-6.84	9.0%	91/79	117	0.6280	2.605802
420	-2.62	-7.24	10.0%	98/90	130	0.6267	2.507699
421	-139.94	-7.33	10.0%	85/79	109	0.6256	2.888503
422	-49.98	-6.52	10.0%	52/45	64	0.6250	2.510099
423	-111.81	-6.65	10.0%	76/68	89	0.6241	2.378502
424	-75.11	-7.20	9.0%	52/42	62	0.6235	2.729160
425	-57.07	-7.21	9.0%	93/78	124	0.6223	2.249468
426	-95.80	-7.56	9.0%	60/56	72	0.6216	1.891379
427	-83.27	-7.44	8.0%	85/68	108	0.6205	2.238539
428	-75.26	-6.95	8.0%	86/79	115	0.6193	2.593282
429	-103.54	-7.60	8.0%	63/56	79	0.6186	2.635045
430	-13.14	-6.52	8.0%	62/53	82	0.6178	2.432477
431	-70.46	-7.21	8.0%	64/55	74	0.6170	3.087304
432	-106.51	-7.66	8.0%	85/70	108	0.6159	2.455569
433	-69.22	-7.06	8.0%	58/44	70	0.6153	2.458585
434	-75.87	-6.56	8.0%	88/77	118	0.6141	3.204874
435	-157.57	-7.79	8.0%	100/86	120	0.6129	2.229016
436	-74.99	-7.91	8.0%	74/60	102	0.6119	2.436625
437	-62.67	-7.90	8.0%	92/77	118	0.6107	2.157133
438	-21.12	-7.87	9.0%	54/51	75	0.6100	2.243562
439	-128.16	-8.37	9.0%	104/85	117	0.6088	2.206641
440	-135.96	-7.32	9.0%	57/51	72	0.6081	2.382412
441	-94.82	-7.82	9.0%	92/73	111	0.6070	2.439791
442	-97.15	-8.43	8.0%	82/74	101	0.6060	2.570860

Continued: Dqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
443	-109.84	-8.04	8.0%	107/93	139	0.6046	2.280162
444	-85.37	-7.27	8.0%	62/49	93	0.6037	2.360659
445	-109.28	-8.13	8.0%	67/54	84	0.6029	2.129900
446	-168.06	-6.81	8.0%	104/85	141	0.6015	2.340125
447	-73.60	-7.65	8.0%	94/78	107	0.6004	2.058462
448	-126.81	-6.99	8.0%	90/75	112	0.5993	2.288770
449	-112.46	-8.13	8.0%	51/43	65	0.5987	2.342754
450	-104.32	-8.10	8.0%	92/83	121	0.5975	2.494133
451	-57.74	-7.15	8.0%	55/51	68	0.5968	2.741897
452	15.49	-7.66	9.0%	70/58	79	0.5960	2.302786
453	-61.31	-7.69	9.0%	79/71	94	0.5951	2.284629
454	-73.90	-8.80	9.0%	96/82	116	0.5939	2.642833
455	-37.35	-7.37	9.0%	85/70	99	0.5930	2.485440
456	-128.56	-7.88	9.0%	84/71	108	0.5919	2.182684
457	-94.71	-8.02	8.0%	90/79	106	0.5908	2.413474
458	-75.60	-7.97	8.0%	101/83	127	0.5896	2.497219
459	-71.72	-7.84	8.0%	72/62	90	0.5887	2.596425
460	-69.02	-8.64	8.0%	96/84	118	0.5875	2.169151
461	-103.72	-8.01	8.0%	75/67	99	0.5865	2.357076
462	25.11	-7.43	9.0%	86/78	101	0.5855	2.396995
463	-73.61	-7.28	9.0%	83/72	107	0.5845	2.110120
464	-89.92	-8.13	9.0%	69/56	86	0.5836	1.915256
465	-84.16	-7.81	8.0%	49/42	57	0.5831	2.132327
466	-60.71	-7.93	8.0%	74/63	83	0.5822	2.304075
467	-131.47	-8.28	8.0%	123/111	147	0.5808	2.070986
468	-91.59	-8.10	8.0%	70/61	85	0.5800	2.672718
469	-15.82	-8.61	9.0%	66/51	71	0.5793	2.065389
470	-76.86	-8.44	9.0%	62/52	70	0.5786	2.409805
471	-85.99	-7.99	9.0%	67/53	77	0.5778	2.421055
472	-13.62	-8.20	8.0%	90/77	104	0.5768	2.696404
473	-120.25	-8.96	8.0%	67/52	98	0.5758	2.703084
474	-93.52	-8.36	7.0%	84/78	123	0.5746	2.526147
475	-7.12	-8.50	8.0%	80/63	119	0.5734	2.726379
476	-37.46	-8.79	8.0%	64/56	74	0.5727	2.421207
477	-59.81	-9.24	8.0%	54/45	69	0.5720	2.521936
478	-111.68	-9.15	8.0%	85/74	105	0.5709	2.697672
479	-88.33	-8.46	8.0%	117/97	133	0.5696	2.521690
480	-60.48	-7.87	8.0%	119/100	145	0.5682	2.418006
481	-97.57	-8.51	8.0%	65/54	70	0.5675	2.412708
482	-7.28	-8.67	9.0%	85/71	110	0.5664	2.403064
483	-115.59	-8.62	9.0%	50/40	66	0.5658	2.565601
484	-76.11	-8.43	9.0%	85/72	96	0.5648	2.354622
485	-68.24	-8.37	9.0%	78/64	91	0.5639	2.282912
486	-117.29	-7.94	9.0%	81/68	110	0.5628	2.650545
487	-83.38	-8.09	9.0%	81/72	99	0.5618	2.530859
488	-85.31	-8.23	9.0%	74/60	94	0.5609	2.181522
489	-97.90	-8.15	9.0%	102/91	132	0.5596	2.116793
490	-119.62	-8.65	9.0%	88/76	105	0.5586	2.710799
491	-42.19	-9.00	9.0%	80/67	95	0.5576	2.758430
492	-21.28	-9.06	10.0%	72/59	91	0.5567	2.483609
493	-77.26	-9.01	10.0%	66/54	70	0.5560	1.906712
494	-77.02	-8.23	10.0%	81/71	104	0.5550	2.545696
495	-74.25	-7.82	10.0%	88/71	123	0.5538	2.486060
496	7.52	-7.86	11.0%	64/52	80	0.5530	2.560677
497	-16.34	-8.01	12.0%	78/63	104	0.5520	2.590926
498	-124.95	-7.49	12.0%	91/75	111	0.5509	2.680656
499	-82.96	-8.28	12.0%	94/80	113	0.5497	2.048162
500	-62.86	-8.47	12.0%	83/68	123	0.5485	2.562660
501	-104.88	-8.50	12.0%	56/50	67	0.5479	2.455978
502	-110.80	-7.38	12.0%	73/64	112	0.5467	2.285699
503	-104.97	-8.25	12.0%	101/86	118	0.5456	2.250428
504	-97.57	-8.52	12.0%	98/85	120	0.5444	2.511954
505	-49.08	-8.45	13.0%	66/55	96	0.5434	2.465253
506	-79.82	-8.33	13.0%	81/72	93	0.5425	2.497405
507	-71.54	-8.10	13.0%	87/72	95	0.5416	1.945441
508	-55.67	-7.55	13.0%	62/50	79	0.5408	2.374849
509	-81.90	-8.24	13.0%	94/79	114	0.5397	2.418664
510	-89.39	-7.59	13.0%	67/61	80	0.5389	2.337092
511	-24.51	-7.33	13.0%	80/69	93	0.5380	2.322193
512	-267.70	-7.76	13.0%	118/101	151	0.5365	2.102146
513	-7.51	-7.26	13.0%	109/95	122	0.5353	2.410892
514	-118.83	-6.85	13.0%	79/68	99	0.5343	2.715783
515	-73.54	-6.67	13.0%	75/69	89	0.5334	2.226475
516	-27.63	-6.96	14.0%	95/81	114	0.5323	2.178711

Continued: Dqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
517	-48.40	-7.65	14.0%	74/59	90	0.5314	2.309148
518	-120.53	-7.26	14.0%	65/57	79	0.5306	2.114751
519	-101.68	-7.75	14.0%	91/70	110	0.5295	2.598997
520	-60.39	-7.53	13.0%	63/53	73	0.5288	2.109830
521	-42.08	-7.53	13.0%	80/66	89	0.5279	2.508172
522	-46.06	-7.32	13.0%	98/84	123	0.5267	2.074657
523	-57.13	-7.60	13.0%	75/62	92	0.5258	2.305801
524	8.57	-8.10	14.0%	101/87	131	0.5245	2.333814
525	-135.08	-7.65	14.0%	111/95	141	0.5231	2.192495
526	-62.06	-7.79	14.0%	89/80	118	0.5219	2.712673
527	-93.79	-7.39	14.0%	102/80	127	0.5207	2.268856
528	-97.27	-6.73	14.0%	73/66	93	0.5197	2.710274
529	-83.26	-6.55	14.0%	91/76	108	0.5187	2.518285
530	-144.90	-6.51	13.0%	119/105	139	0.5173	2.374191
531	-11.52	-7.37	14.0%	77/70	91	0.5164	2.438045
532	-70.25	-7.92	14.0%	99/81	126	0.5151	2.268979
533	-54.83	-7.36	14.0%	111/92	142	0.5137	2.271280
534	-89.94	-8.35	14.0%	91/79	113	0.5126	2.496495
535	25.86	-6.96	15.0%	65/57	82	0.5118	2.499490
536	-77.52	-7.59	15.0%	69/59	82	0.5110	2.871064
537	-56.08	-7.10	15.0%	91/81	104	0.5100	2.419352
538	-28.94	-7.18	14.0%	100/87	137	0.5086	2.018821
539	-76.27	-7.64	14.0%	76/64	95	0.5077	2.345099
540	-91.95	-6.88	14.0%	86/78	105	0.5066	2.773739
541	19.35	-6.68	15.0%	77/68	86	0.5058	2.315018
542	-73.22	-7.61	15.0%	70/64	98	0.5048	2.140620
543	-306.71	-7.89	15.0%	106/93	141	0.5034	2.213923
544	-23.04	-7.50	15.0%	92/79	115	0.5023	2.389916
545	-142.22	-8.35	15.0%	76/66	98	0.5013	2.145690
546	-69.30	-7.89	15.0%	171/149	199	0.4993	2.514500
547	-72.75	-8.01	15.0%	75/65	85	0.4985	2.552113
548	-11.40	-7.61	16.0%	85/66	113	0.4974	2.082169
549	-85.24	-7.65	16.0%	55/47	70	0.4967	2.646254
550	-106.98	-7.49	16.0%	76/63	92	0.4958	2.358091
551	-55.04	-7.44	16.0%	120/114	140	0.4944	1.997658
552	-21.50	-7.43	16.0%	73/64	98	0.4934	2.097806
553	-9.66	-7.28	17.0%	77/66	92	0.4925	2.417344
554	-71.31	-8.02	17.0%	82/66	106	0.4914	2.250725
555	-87.79	-6.23	16.0%	52/43	58	0.4909	2.050837
556	-51.19	-7.11	16.0%	66/55	73	0.4901	2.268192
557	2.46	-6.65	17.0%	82/72	97	0.4892	2.341889
558	-54.05	-7.38	17.0%	142/121	169	0.4875	2.160244
559	-73.74	-7.13	17.0%	94/83	136	0.4862	2.387121
560	-45.27	-7.91	17.0%	110/98	134	0.4848	2.631377
561	-81.95	-8.16	17.0%	57/46	65	0.4842	2.708532
562	-87.88	-7.97	16.0%	82/72	105	0.4832	2.319838
563	-53.94	-8.23	17.0%	79/64	102	0.4822	2.361263
564	-94.65	-8.21	17.0%	87/74	108	0.4811	2.409042
565	19.30	-8.25	18.0%	91/78	101	0.4801	2.137862
566	-89.35	-7.62	18.0%	96/85	119	0.4789	2.440790
567	-91.04	-8.25	18.0%	77/65	94	0.4780	2.048914
568	-72.59	-8.32	18.0%	86/72	104	0.4769	1.964140
569	-73.64	-8.14	17.0%	98/80	117	0.4758	2.129681
570	-49.19	-8.47	17.0%	114/97	138	0.4744	2.291566
571	-87.73	-8.05	17.0%	90/75	102	0.4734	2.239203
572	-160.76	-7.52	17.0%	101/89	118	0.4722	2.393065
573	-50.28	-8.22	17.0%	74/63	86	0.4714	2.033550
574	-78.71	-8.43	17.0%	79/67	93	0.4705	2.117790
575	-44.99	-8.32	16.0%	56/45	66	0.4698	2.078488
576	-77.59	-7.96	16.0%	71/60	81	0.4690	2.540833
577	-68.29	-8.06	16.0%	61/53	68	0.4683	2.243906
578	-45.49	-8.86	16.0%	76/66	93	0.4674	2.543055
579	-62.08	-8.81	16.0%	75/62	104	0.4664	2.833652
580	-36.27	-8.59	16.0%	107/91	122	0.4652	2.080521
581	-83.02	-9.31	16.0%	78/65	113	0.4641	2.345103
582	8.21	-8.50	16.0%	81/69	102	0.4631	2.481790
583	-71.60	-9.46	16.0%	101/82	114	0.4619	2.069867
584	-34.60	-8.81	16.0%	75/63	92	0.4610	2.418252
585	-91.93	-9.11	16.0%	64/54	71	0.4603	2.642436
586	-87.06	-9.97	16.0%	104/84	124	0.4591	2.203585
587	-67.73	-8.79	16.0%	82/70	104	0.4581	2.108661
588	-48.51	-9.32	16.0%	145/126	183	0.4562	2.169432
589	-117.87	-8.68	16.0%	104/85	134	0.4549	2.013673
590	-56.32	-9.21	16.0%	86/74	103	0.4539	2.601542

Continued: Dqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
591	-69.31	-8.06	16.0%	132/108	147	0.4524	2.231632
592	-10.46	-9.25	16.0%	100/83	121	0.4512	2.074312
593	-113.61	-8.73	16.0%	87/73	105	0.4502	2.053865
594	-66.34	-8.56	16.0%	91/78	126	0.4490	1.909341
595	-145.73	-8.29	16.0%	109/91	132	0.4476	2.207506
596	-51.13	-8.86	15.0%	81/72	108	0.4466	2.333156
597	-109.11	-8.13	14.0%	210/174	229	0.4443	2.317612
598	-73.73	-8.63	14.0%	111/93	139	0.4429	2.259448
599	-34.82	-7.91	14.0%	72/64	81	0.4421	2.338253
600	-63.88	-8.72	14.0%	66/53	75	0.4414	2.298889
601	-48.30	-8.52	14.0%	74/61	87	0.4405	2.016996
602	-13.21	-8.66	14.0%	77/69	89	0.4397	2.585195
603	-70.90	-8.57	14.0%	90/72	113	0.4385	2.016218
604	-54.63	-8.62	14.0%	95/83	113	0.4374	1.975721
605	-11.33	-8.95	13.0%	78/68	89	0.4365	1.996933
606	8.66	-8.10	13.0%	116/101	133	0.4352	2.023362
607	-39.84	-8.86	13.0%	188/161	225	0.4330	2.137021
608	-59.03	-8.65	13.0%	208/173	235	0.4307	2.316812
609	-50.52	-8.29	13.0%	106/91	129	0.4294	2.327556
610	-45.26	-8.28	13.0%	104/88	125	0.4281	1.907753
611	-83.23	-8.23	13.0%	77/67	87	0.4273	2.297914
612	18.01	-8.21	14.0%	85/65	101	0.4263	1.796926
613	-60.62	-7.60	14.0%	72/64	84	0.4255	2.026040
614	-21.81	-8.67	15.0%	81/64	115	0.4243	2.443171
615	-76.30	-8.19	15.0%	101/81	119	0.4231	2.645254
616	-32.08	-7.87	14.0%	123/110	142	0.4217	1.903226
617	2.80	-7.24	14.0%	105/87	113	0.4206	1.967727
618	-59.28	-7.79	14.0%	79/62	90	0.4197	2.653889
619	-73.29	-7.91	14.0%	87/72	103	0.4187	1.936774
620	-75.66	-7.92	14.0%	103/87	122	0.4175	2.033319
621	-42.25	-7.37	14.0%	86/79	97	0.4165	2.010066
622	-59.75	-7.13	14.0%	96/76	129	0.4153	2.419869
623	-42.36	-7.83	14.0%	153/130	188	0.4134	2.334609
624	-52.96	-7.50	13.0%	162/143	178	0.4116	2.316606
625	-30.70	-7.52	13.0%	119/95	147	0.4102	2.427752
626	-117.74	-6.90	13.0%	101/83	126	0.4089	2.243680
627	-108.18	-7.53	13.0%	146/122	168	0.4073	2.483319
628	-53.30	-6.26	13.0%	79/63	95	0.4063	2.208423
629	30.03	-8.29	14.0%	80/70	95	0.4054	2.168643
630	-44.48	-7.59	14.0%	114/99	132	0.4041	2.176357
631	-118.45	-7.54	13.0%	96/74	105	0.4030	2.434165
632	-108.76	-7.64	13.0%	89/77	103	0.4020	1.943781
633	-58.83	-7.73	13.0%	104/86	117	0.4009	2.042094
634	-42.07	-8.09	13.0%	113/91	137	0.3995	2.387232
635	3.75	-7.99	13.0%	88/76	123	0.3983	2.293650
636	-52.73	-8.17	13.0%	108/92	134	0.3970	2.495462
637	-38.42	-6.49	13.0%	131/110	174	0.3952	1.991947
638	-186.90	-7.64	13.0%	402/351	453	0.3908	2.106445
639	-24.53	-6.65	13.0%	123/104	145	0.3893	2.155747
640	-21.36	-7.07	14.0%	79/68	90	0.3884	2.634768
641	17.84	-7.84	14.0%	93/78	124	0.3872	2.013506
642	-80.62	-7.30	14.0%	84/66	97	0.3862	2.392860
643	-42.70	-7.31	14.0%	112/95	126	0.3850	2.120487
644	-43.50	-6.54	14.0%	92/75	100	0.3840	2.210223
645	-43.96	-7.35	14.0%	143/125	172	0.3823	2.021225
646	-43.59	-8.32	14.0%	83/73	92	0.3814	2.160152
647	-69.05	-7.48	14.0%	138/122	167	0.3797	2.032001
648	-70.69	-6.93	13.0%	81/70	88	0.3789	2.212303
649	-163.48	-7.01	13.0%	189/163	217	0.3767	2.301119
650	-15.40	-6.74	13.0%	93/77	99	0.3757	2.079361
651	-37.27	-6.65	13.0%	101/81	120	0.3745	2.295786
652	-26.76	-6.72	12.0%	126/107	156	0.3730	1.865452
653	26.88	-6.06	12.0%	166/137	192	0.3711	1.916417
654	-26.12	-6.24	12.0%	90/80	105	0.3701	2.148350
655	-55.37	-6.95	12.0%	68/61	84	0.3692	2.276608
656	-83.06	-6.82	12.0%	140/119	155	0.3677	2.255286
657	-45.27	-6.37	11.0%	98/75	116	0.3665	1.995249
658	-41.83	-6.02	11.0%	94/79	118	0.3654	2.069674
659	-18.32	-5.75	11.0%	81/64	85	0.3645	2.330359
660	-99.53	-6.80	11.0%	190/163	216	0.3624	2.249133
661	-35.83	-6.12	12.0%	120/92	129	0.3611	1.974384
662	-281.56	-5.49	12.0%	889/760	1000	0.3512	2.212093
663	-74.84	-4.93	11.0%	113/101	145	0.3498	2.068402
664	13.28	-5.17	12.0%	71/64	95	0.3488	1.949820

Continued: Dqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
665	-39.77	-5.19	12.0%	357/300	409	0.3448	2.104585
666	-54.37	-5.49	12.0%	92/78	120	0.3436	2.417081
667	-10.52	-4.46	12.0%	80/68	92	0.3427	2.156776
668	-214.57	-4.31	12.0%	487/415	577	0.3370	2.100184
669	-56.15	-3.68	12.0%	163/146	198	0.3350	2.170922
670	-355.22	-3.09	12.0%	876/758	1000	0.3251	2.030768
671	-38.59	-2.82	12.0%	217/186	236	0.3228	2.291488
672	-289.35	-3.62	12.0%	760/630	863	0.3142	2.148399
673	-11.89	-3.98	12.0%	98/84	106	0.3132	2.075092
674	-32.68	-1.98	12.0%	138/116	151	0.3117	2.027457
675	-56.54	-2.54	12.0%	129/105	151	0.3102	2.170231
676	23.17	-2.60	13.0%	240/212	264	0.3076	2.100467
677	-220.60	-2.81	13.0%	898/772	1000	0.2977	2.134198
678	-251.06	-2.08	13.0%	883/761	1000	0.2878	2.145420
679	-19.55	-0.48	13.0%	94/85	100	0.2868	2.365779
680	-56.86	-0.45	13.0%	159/137	173	0.2851	2.316653
681	-52.09	-0.42	13.0%	82/66	95	0.2842	2.281172
682	-65.02	-0.76	12.0%	101/81	121	0.2830	2.328117
683	-195.45	1.09	12.0%	862/728	1000	0.2731	2.039750
684	30.89	1.71	13.0%	126/109	152	0.2715	1.905458
685	-63.12	2.26	13.0%	136/122	159	0.2700	2.376061
686	-19.75	2.37	13.0%	107/84	131	0.2687	1.821338
687	-34.24	1.55	13.0%	160/130	175	0.2669	1.826779
688	-278.82	3.10	13.0%	788/668	1000	0.2570	2.073747
689	47.91	2.75	14.0%	98/84	133	0.2557	2.058165
690	-77.00	3.32	14.0%	139/111	165	0.2541	1.843737
691	-19.06	2.59	14.0%	170/146	214	0.2520	2.010382
692	-27.10	2.43	13.0%	113/96	139	0.2506	2.590075
693	-41.60	2.47	13.0%	83/66	90	0.2497	2.231731
694	-32.85	2.84	13.0%	182/158	203	0.2477	2.411900
695	-129.29	2.89	13.0%	682/570	1000	0.2378	2.064959
696	-267.62	3.11	13.0%	752/638	1000	0.2279	2.159448
697	142.74	4.58	14.0%	394/334	550	0.2225	2.294904
698	-238.12	5.20	14.0%	791/674	1000	0.2126	2.170319
699	-10.56	4.61	14.0%	117/102	136	0.2112	2.438938
700	-150.44	6.38	14.0%	797/687	1000	0.2013	2.165641
701	-349.20	5.11	14.0%	596/497	688	0.1945	2.312673
702	-265.05	6.98	14.0%	827/722	1000	0.1846	2.125260
703	-229.42	5.97	14.0%	588/511	712	0.1775	2.138474
704	-318.09	5.59	14.0%	787/690	1000	0.1676	2.157092
705	-220.84	4.10	14.0%	802/669	1000	0.1577	2.179261
706	-152.97	5.72	14.0%	689/578	1000	0.1478	2.059767
707	-352.30	6.14	14.0%	865/709	1000	0.1379	2.020059
708	-199.27	6.17	14.0%	332/271	386	0.1341	2.141918
709	-377.70	6.83	14.0%	847/727	1000	0.1242	2.161853
710	-278.28	5.29	14.0%	836/717	1000	0.1143	1.990823
711	-391.62	6.27	14.0%	674/574	794	0.1065	1.948091
712	-343.55	6.42	13.0%	805/677	1000	0.0966	2.003269
713	-265.80	7.82	13.0%	810/681	1000	0.0867	2.071103
714	-389.61	8.81	12.0%	812/674	1000	0.0768	2.044056
715	-379.04	8.22	12.0%	786/675	1000	0.0669	2.103738
716	-390.55	8.12	12.0%	667/581	896	0.0580	2.024966
717	-321.72	8.16	12.0%	762/655	1000	0.0481	2.015002
718	-337.83	9.60	12.0%	766/656	1000	0.0382	2.024084
719	-333.28	10.50	12.0%	747/637	1000	0.0283	1.899911
720	-289.75	10.74	12.0%	746/629	1000	0.0184	1.943522
721	-285.18	11.29	12.0%	735/637	1000	0.0100	1.955590
722	-275.17	11.23	12.0%	724/625	1000	0.0100	1.874177
723	-254.40	10.93	12.0%	753/629	1000	0.0100	1.779097
724	-241.38	11.92	12.0%	741/646	1000	0.0100	1.776037
725	-462.16	12.88	12.0%	706/614	883	0.0100	1.711945
726	-259.01	13.35	12.0%	739/633	1000	0.0100	1.704946
727	-257.28	11.78	12.0%	370/309	464	0.0100	1.629807
728	-270.76	11.61	12.0%	807/684	1000	0.0100	1.694345
729	-217.27	13.76	11.0%	764/646	1000	0.0100	1.570798
730	-282.25	14.56	11.0%	820/685	1000	0.0100	1.583399
731	-262.90	12.10	11.0%	778/646	1000	0.0100	1.563304
732	-277.81	12.32	11.0%	774/655	1000	0.0100	1.572621
733	-247.61	13.28	11.0%	825/690	1000	0.0100	1.556724
734	-250.66	13.95	11.0%	788/665	1000	0.0100	1.502125
735	-269.58	15.11	10.0%	787/685	1000	0.0100	1.487064
736	-241.12	14.56	10.0%	757/646	1000	0.0100	1.490412
737	-134.98	14.30	10.0%	308/266	390	0.0100	1.548411
738	-252.56	14.99	10.0%	768/640	1000	0.0100	1.531091

Continued: Dqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
739	-242.95	15.88	10.0%	786/673	1000	0.0100	1.516415
740	-281.51	15.68	9.0%	789/681	1000	0.0100	1.507604
741	-261.06	14.72	8.0%	805/669	1000	0.0100	1.522440
742	-274.81	15.86	8.0%	842/714	1000	0.0100	1.452034
743	-297.24	15.00	8.0%	824/708	1000	0.0100	1.507917
744	-247.90	15.13	8.0%	762/638	1000	0.0100	1.470607
745	-326.37	14.99	8.0%	843/705	1000	0.0100	1.469853
746	-210.71	15.72	8.0%	795/660	1000	0.0100	1.425258
747	-43.13	14.61	8.0%	569/496	676	0.0100	1.380574
748	-233.74	13.52	8.0%	828/697	1000	0.0100	1.389422
749	-282.89	12.59	8.0%	757/656	1000	0.0100	1.337918
750	283.77	12.36	9.0%	208/183	281	0.0100	1.409272
751	-222.86	12.75	9.0%	780/663	1000	0.0100	1.329391
752	-293.07	13.11	9.0%	783/661	1000	0.0100	1.298325
753	13.51	12.92	8.0%	140/120	186	0.0100	1.271506
754	-261.42	12.30	8.0%	845/717	1000	0.0100	1.250849
755	229.41	12.70	9.0%	295/238	417	0.0100	1.155829
756	-226.25	11.74	9.0%	771/675	1000	0.0100	1.211849
757	-230.21	11.85	9.0%	777/661	1000	0.0100	1.245424
758	-215.26	12.22	9.0%	767/657	1000	0.0100	1.233978
759	-240.95	14.22	9.0%	759/653	1000	0.0100	1.225716
760	-243.19	12.95	9.0%	849/717	1000	0.0100	1.210360
761	280.77	13.03	9.0%	218/179	352	0.0100	1.120081
762	-224.23	11.91	9.0%	803/683	1000	0.0100	1.215089
763	-273.66	11.28	9.0%	766/654	1000	0.0100	1.170927
764	-238.39	11.76	8.0%	732/623	1000	0.0100	1.154907
765	-255.55	12.92	7.0%	755/651	1000	0.0100	1.150467
766	-272.28	13.06	7.0%	813/687	1000	0.0100	1.134587
767	294.46	11.73	8.0%	187/160	315	0.0100	1.110565
768	-255.80	12.40	8.0%	782/659	1000	0.0100	1.150458
769	-268.51	11.04	8.0%	731/616	1000	0.0100	1.127829
770	-249.03	12.61	8.0%	791/667	1000	0.0100	1.091610
771	-222.22	15.13	8.0%	796/676	1000	0.0100	1.075483
772	-265.43	14.98	8.0%	739/647	1000	0.0100	1.090671
773	-252.85	16.34	8.0%	780/652	1000	0.0100	1.121778
774	110.02	18.17	9.0%	534/460	782	0.0100	1.091888
775	4.69	16.71	10.0%	703/617	978	0.0100	1.010688
776	141.02	19.22	10.0%	450/383	623	0.0100	0.971257
777	-235.72	20.33	10.0%	774/653	1000	0.0100	0.990682
778	113.26	22.22	11.0%	482/410	668	0.0100	0.975349
779	89.56	22.54	12.0%	483/413	692	0.0100	0.973636
780	231.58	23.56	13.0%	246/199	328	0.0100	0.950733
781	-278.04	25.20	13.0%	768/649	1000	0.0100	1.005221
782	101.34	26.33	14.0%	484/408	685	0.0100	0.923669
783	125.52	26.72	15.0%	449/378	649	0.0100	0.996846
784	-252.67	28.50	14.0%	718/635	1000	0.0100	0.897850
785	-225.49	31.69	14.0%	758/648	1000	0.0100	0.914854
786	-244.37	31.20	14.0%	748/645	1000	0.0100	0.846199
787	302.30	33.03	15.0%	145/126	220	0.0100	0.825561
788	-249.30	33.05	15.0%	706/626	1000	0.0100	0.871418
789	231.18	33.47	15.0%	206/176	321	0.0100	0.921703
790	-204.03	35.67	15.0%	747/644	1000	0.0100	0.841834
791	-163.08	37.69	15.0%	699/586	1000	0.0100	0.817303
792	-174.26	38.42	15.0%	720/638	1000	0.0100	0.797543
793	252.80	41.54	16.0%	282/237	552	0.0100	0.815084
794	118.24	41.12	17.0%	422/355	605	0.0100	0.792268
795	-349.66	41.86	17.0%	111/84	112	0.0100	0.740762
796	283.12	43.46	18.0%	154/134	205	0.0100	0.822326
797	230.59	44.31	18.0%	239/206	350	0.0100	0.828459
798	-96.52	44.22	18.0%	292/264	395	0.0100	0.848400
799	-114.75	48.13	18.0%	762/666	1000	0.0100	0.779406
800	218.86	48.56	19.0%	269/227	366	0.0100	0.757581

Table 4: Complete per-iteration training output - Ddqn Modified (800 episodes).

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
1	-147.06	0.24	0.0%	50/40	70	0.9993	nan
2	-359.38	0.24	0.0%	67/61	93	0.9984	nan
3	-158.89	0.24	0.0%	102/86	122	0.9972	nan
4	-141.98	0.24	0.0%	67/58	85	0.9963	nan
5	-173.75	0.24	0.0%	84/72	117	0.9952	nan
6	-329.77	0.24	0.0%	78/67	104	0.9941	nan
7	-327.33	0.24	0.0%	57/47	79	0.9934	nan
8	-179.77	0.24	0.0%	58/47	70	0.9927	nan
9	-351.41	0.24	0.0%	79/63	108	0.9916	nan
10	-298.78	0.24	0.0%	90/80	118	0.9904	nan
11	-78.47	-0.37	0.0%	51/43	68	0.9898	2.386483
12	-159.12	-0.73	0.0%	58/45	78	0.9890	2.190137
13	-116.79	-0.70	0.0%	74/67	96	0.9880	2.108651
14	-90.14	-0.43	0.0%	63/49	84	0.9872	1.892458
15	-168.22	-0.04	0.0%	80/66	115	0.9861	1.636603
16	-243.62	-0.08	0.0%	54/41	74	0.9853	1.637466
17	-280.66	-0.16	0.0%	77/61	99	0.9844	1.787955
18	-175.90	-0.02	0.0%	76/66	92	0.9834	1.366507
19	-160.71	-0.10	0.0%	69/62	93	0.9825	1.552153
20	-33.68	-0.14	5.0%	51/44	68	0.9819	1.523708
21	-159.27	-0.18	4.8%	48/33	64	0.9812	1.428822
22	-153.45	0.18	4.5%	68/55	87	0.9804	1.356763
23	-175.76	-0.30	4.3%	54/49	68	0.9797	1.635212
24	-132.43	0.00	4.2%	54/46	75	0.9789	1.258110
25	-120.53	0.20	4.0%	63/58	78	0.9782	1.202664
26	-93.94	0.11	3.8%	49/38	65	0.9775	1.224705
27	-295.67	0.18	3.7%	62/50	76	0.9768	1.157043
28	-355.29	0.13	3.6%	81/71	105	0.9757	1.289707
29	-108.34	0.60	3.4%	48/41	63	0.9751	2.039138
30	-192.05	0.29	3.3%	49/43	74	0.9744	1.637235
31	-336.50	0.84	3.2%	104/96	137	0.9730	1.282972
32	-135.93	0.57	3.1%	69/64	89	0.9721	1.419130
33	-144.84	0.71	3.0%	99/88	126	0.9709	1.328042
34	-188.46	0.35	2.9%	88/76	114	0.9698	1.622996
35	-33.14	0.25	2.9%	74/62	104	0.9687	1.409984
36	-79.37	0.42	2.8%	108/101	143	0.9673	1.437567
37	-118.32	0.63	2.7%	56/43	72	0.9666	1.540107
38	-180.27	0.42	2.6%	47/34	58	0.9660	1.679823
39	-111.14	1.17	2.6%	60/55	76	0.9653	1.555382
40	-171.94	0.83	2.5%	93/74	112	0.9642	1.663554
41	-122.42	0.96	2.4%	88/76	119	0.9630	1.392620
42	-135.08	0.98	2.4%	64/56	81	0.9622	1.438564
43	-122.99	0.71	2.3%	74/63	102	0.9612	1.748344
44	-188.73	0.67	2.3%	79/72	108	0.9601	1.445632
45	-19.38	0.97	4.4%	56/49	72	0.9594	1.604415
46	-150.89	1.00	4.3%	74/62	94	0.9585	1.471473
47	-139.13	0.55	4.3%	72/65	94	0.9575	1.416374
48	-477.24	0.94	4.2%	83/67	104	0.9565	1.389513
49	-244.74	0.83	4.1%	105/87	126	0.9553	1.377756
50	-116.23	1.23	4.0%	50/44	66	0.9546	1.647607
51	-149.50	1.31	3.9%	64/57	88	0.9537	1.351277
52	-370.78	1.63	3.8%	99/84	122	0.9525	1.480450
53	-194.47	1.42	3.8%	79/66	107	0.9515	1.238991
54	-128.05	1.30	3.7%	41/33	56	0.9509	1.682211
55	-113.27	1.02	3.6%	56/50	79	0.9501	1.474609
56	-137.83	1.47	3.6%	77/74	98	0.9492	1.430941
57	-68.91	1.48	3.5%	51/42	66	0.9485	1.502843
58	-136.61	1.67	3.4%	80/67	109	0.9474	1.694837
59	-156.19	1.34	3.4%	46/38	62	0.9468	1.552294
60	-164.89	1.40	3.3%	69/60	96	0.9459	1.267330
61	-118.96	1.40	3.3%	54/42	69	0.9452	1.568290
62	-141.97	1.64	3.2%	46/38	67	0.9445	1.489827
63	-292.10	1.36	3.2%	81/70	112	0.9434	1.364502
64	-141.48	1.57	3.1%	90/82	121	0.9422	1.401506
65	-149.22	1.54	3.1%	65/55	90	0.9413	1.500160
66	-142.79	2.73	3.0%	56/45	77	0.9406	1.406897
67	-164.01	2.26	3.0%	79/68	105	0.9395	1.487290
68	-211.88	2.48	2.9%	86/77	109	0.9384	1.447779
69	-156.62	2.33	2.9%	67/57	90	0.9376	1.395930
70	-252.11	2.77	2.9%	65/58	94	0.9366	1.319399
71	-126.05	2.07	2.8%	72/65	98	0.9356	1.279171
72	-82.59	2.47	2.8%	101/85	127	0.9344	1.431185

Continued: Ddqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
73	-123.53	2.69	2.7%	48/40	70	0.9337	1.417930
74	-244.55	2.82	2.7%	59/48	75	0.9330	1.472802
75	-113.60	2.77	2.7%	41/34	63	0.9323	1.522064
76	-125.08	2.73	2.6%	64/52	83	0.9315	1.256072
77	-134.89	2.23	2.6%	64/51	79	0.9307	1.350415
78	-114.20	3.41	2.6%	60/54	83	0.9299	1.532914
79	-111.96	3.06	2.5%	45/38	69	0.9292	1.401979
80	-38.14	3.04	3.8%	62/53	75	0.9285	1.410024
81	-99.86	3.42	3.7%	46/35	64	0.9278	1.368483
82	-150.59	3.44	3.7%	83/62	103	0.9268	1.484184
83	-130.46	3.30	3.6%	60/53	78	0.9261	1.294035
84	-109.02	4.11	3.6%	55/47	70	0.9254	1.841997
85	-155.14	3.85	3.5%	70/58	95	0.9244	1.523407
86	-122.62	3.84	3.5%	72/63	89	0.9235	1.680310
87	-88.16	4.11	3.4%	84/67	98	0.9226	1.268978
88	-121.49	4.06	3.4%	49/44	62	0.9220	1.432478
89	-162.88	3.77	3.4%	47/42	60	0.9214	1.635413
90	-234.93	5.17	3.3%	74/63	89	0.9205	1.454705
91	-156.30	4.61	3.3%	89/74	111	0.9194	1.369547
92	-118.90	4.75	3.3%	57/50	75	0.9186	1.651048
93	-192.76	4.98	3.2%	80/68	101	0.9176	1.418183
94	-137.11	4.95	3.2%	98/89	125	0.9164	1.570037
95	-157.66	5.36	3.2%	70/63	96	0.9155	1.596823
96	-122.84	4.99	3.1%	84/76	121	0.9143	1.740050
97	-329.45	5.38	3.1%	89/78	102	0.9132	1.292990
98	-104.62	5.45	3.1%	52/45	68	0.9126	1.278991
99	-98.92	5.13	3.0%	52/47	67	0.9119	1.437163
100	-131.15	5.61	3.0%	85/74	115	0.9108	1.539776
101	-89.89	6.11	3.0%	52/46	65	0.9101	1.674902
102	-192.47	6.15	3.0%	74/66	92	0.9092	1.449114
103	-181.96	6.03	3.0%	81/69	101	0.9082	1.412471
104	-90.94	5.76	3.0%	49/45	63	0.9076	1.707469
105	-222.74	6.78	3.0%	189/158	247	0.9051	1.741155
106	-139.61	7.21	3.0%	87/76	110	0.9041	1.434954
107	-42.84	6.66	3.0%	62/54	77	0.9033	1.432893
108	-47.95	6.36	4.0%	70/61	85	0.9025	1.686290
109	-195.82	6.45	4.0%	102/80	131	0.9012	1.610831
110	-192.93	7.43	4.0%	70/58	85	0.9003	1.678899
111	-82.56	7.39	4.0%	41/36	60	0.8997	1.546077
112	-252.61	7.73	4.0%	99/79	117	0.8986	1.581710
113	-126.41	7.25	4.0%	70/61	82	0.8978	1.506131
114	-5.12	6.98	4.0%	102/87	118	0.8966	1.416142
115	-133.71	8.01	4.0%	76/63	99	0.8956	1.641036
116	-115.56	8.23	4.0%	95/83	121	0.8944	1.775353
117	-120.39	7.87	4.0%	58/51	81	0.8936	1.640497
118	-144.10	7.62	4.0%	51/43	63	0.8930	1.404265
119	-156.34	8.01	4.0%	69/63	87	0.8921	1.615148
120	-115.73	8.44	3.0%	98/85	116	0.8910	1.352230
121	-90.91	8.65	3.0%	53/46	63	0.8903	1.565654
122	-26.29	8.54	3.0%	114/95	146	0.8889	1.483326
123	-98.83	8.75	3.0%	45/41	64	0.8883	1.677827
124	-141.38	8.61	3.0%	51/42	65	0.8876	1.446950
125	-99.61	8.44	3.0%	77/64	98	0.8867	1.572127
126	-131.89	9.38	3.0%	86/73	115	0.8855	1.565691
127	-144.22	9.46	3.0%	66/52	89	0.8846	1.735563
128	-136.63	9.93	3.0%	83/69	93	0.8837	1.688920
129	-141.34	9.80	3.0%	76/66	102	0.8827	1.649844
130	-82.73	9.89	4.0%	61/52	86	0.8819	1.726284
131	-106.03	10.17	4.0%	83/76	101	0.8809	1.637935
132	-134.98	9.93	4.0%	72/63	93	0.8799	1.275780
133	-102.01	10.14	4.0%	50/44	65	0.8793	1.306813
134	-24.89	9.92	5.0%	86/70	103	0.8783	1.339044
135	-131.58	10.07	5.0%	75/65	94	0.8773	1.576257
136	-18.48	10.29	6.0%	53/41	70	0.8766	1.268129
137	-205.93	10.52	6.0%	87/72	112	0.8755	1.615869
138	-127.08	10.26	6.0%	82/72	102	0.8745	1.591118
139	-127.53	10.70	6.0%	50/44	62	0.8739	1.496724
140	-67.32	10.14	6.0%	94/79	115	0.8728	1.424735
141	-80.70	10.45	6.0%	64/52	88	0.8719	1.553341
142	-191.75	10.66	6.0%	106/88	133	0.8706	1.456855
143	-125.77	10.70	6.0%	66/57	88	0.8697	1.467778
144	-141.11	10.31	6.0%	53/47	69	0.8690	1.367368
145	-129.10	10.43	5.0%	79/67	105	0.8680	1.552796
146	-117.71	10.69	5.0%	55/43	76	0.8672	1.371270

Continued: Ddqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
147	-116.53	10.50	5.0%	59/49	76	0.8665	1.269878
148	-95.96	10.70	5.0%	99/88	129	0.8652	1.509813
149	-89.75	11.05	5.0%	50/42	63	0.8646	1.443166
150	-3.34	10.63	6.0%	52/46	70	0.8639	1.450981
151	-117.29	11.01	6.0%	57/44	79	0.8631	1.746479
152	-111.21	10.86	6.0%	58/45	80	0.8623	1.359541
153	-129.73	11.36	6.0%	76/62	100	0.8613	1.652116
154	-128.91	11.42	6.0%	76/62	96	0.8604	1.336846
155	-69.57	10.33	6.0%	49/44	65	0.8597	1.427486
156	-142.92	10.55	6.0%	68/61	99	0.8588	1.678102
157	-148.70	11.13	6.0%	51/42	58	0.8582	1.218731
158	-75.27	11.11	6.0%	43/38	63	0.8576	1.529061
159	-66.74	11.36	6.0%	51/39	65	0.8569	1.792679
160	-89.30	10.37	6.0%	47/42	67	0.8563	1.485745
161	-125.99	10.89	6.0%	72/66	85	0.8554	1.555554
162	-136.18	10.86	6.0%	78/72	100	0.8544	1.680652
163	-133.11	10.81	6.0%	51/46	74	0.8537	1.755086
164	-67.10	10.76	6.0%	58/48	71	0.8530	1.993776
165	-124.29	10.97	6.0%	68/62	87	0.8521	1.079844
166	-32.05	10.75	7.0%	88/75	118	0.8510	1.433543
167	-117.38	10.79	7.0%	92/79	107	0.8499	1.702143
168	-132.02	11.26	7.0%	55/46	71	0.8492	1.625924
169	-97.83	11.47	7.0%	78/63	102	0.8482	1.387736
170	-121.96	10.41	7.0%	61/48	78	0.8474	1.212189
171	-107.72	10.80	7.0%	73/63	102	0.8464	1.842333
172	-115.52	11.17	7.0%	56/48	71	0.8457	1.449686
173	-12.58	10.76	7.0%	63/53	80	0.8449	1.412804
174	-102.68	10.77	7.0%	61/57	76	0.8442	1.344860
175	-394.33	11.37	7.0%	78/72	105	0.8431	1.245979
176	-133.48	11.10	7.0%	62/56	86	0.8423	1.370241
177	-93.64	11.49	7.0%	73/66	92	0.8414	1.489331
178	-139.51	11.22	7.0%	52/42	66	0.8407	1.580868
179	-136.60	11.55	7.0%	71/53	97	0.8397	1.325594
180	-132.83	11.50	6.0%	79/64	99	0.8388	1.417850
181	-173.78	11.85	6.0%	70/62	90	0.8379	1.349030
182	-279.45	11.32	6.0%	77/66	100	0.8369	1.375255
183	-143.08	11.86	6.0%	74/66	93	0.8360	1.483475
184	-117.20	11.38	6.0%	48/39	70	0.8353	1.412628
185	-131.84	10.82	6.0%	87/73	106	0.8342	1.252107
186	-90.69	10.99	6.0%	103/92	127	0.8330	1.307405
187	-123.88	10.84	6.0%	74/63	101	0.8320	1.416536
188	-147.75	10.36	6.0%	53/43	78	0.8312	1.304858
189	-99.55	10.42	6.0%	80/78	102	0.8302	1.390492
190	-178.08	10.72	6.0%	58/51	68	0.8295	1.368212
191	-274.04	10.81	6.0%	69/63	88	0.8286	1.272274
192	-180.70	10.33	6.0%	94/82	115	0.8275	1.306594
193	-108.46	10.28	6.0%	55/45	79	0.8267	1.408997
194	-142.84	10.77	6.0%	113/94	142	0.8253	1.598896
195	-126.38	10.61	6.0%	72/61	91	0.8244	1.455205
196	-75.24	10.08	6.0%	48/41	65	0.8238	1.426783
197	-110.52	9.79	6.0%	64/57	87	0.8229	1.426670
198	-92.43	10.68	6.0%	66/55	81	0.8221	1.393502
199	-15.33	9.92	7.0%	50/39	69	0.8214	1.446979
200	-175.29	10.50	7.0%	92/75	106	0.8204	1.349295
201	-104.24	10.43	7.0%	94/83	115	0.8192	1.478711
202	-189.64	10.01	7.0%	65/48	86	0.8184	1.489841
203	-110.18	10.01	7.0%	54/44	67	0.8177	1.248763
204	-112.10	10.19	7.0%	63/57	78	0.8169	1.682364
205	-122.90	9.28	7.0%	112/93	135	0.8156	1.539801
206	15.88	9.92	7.0%	56/47	69	0.8149	1.439841
207	-94.73	9.92	7.0%	77/67	100	0.8139	1.564357
208	-103.72	10.09	6.0%	51/42	70	0.8132	1.350598
209	-81.84	9.94	6.0%	50/40	65	0.8126	1.298051
210	-114.22	9.12	6.0%	101/87	124	0.8114	1.562407
211	-96.80	9.80	6.0%	76/66	95	0.8104	1.470561
212	-84.09	9.16	6.0%	58/51	65	0.8098	1.517450
213	-132.95	9.94	6.0%	69/61	90	0.8089	1.723398
214	-145.20	9.83	6.0%	54/46	84	0.8081	1.551081
215	-43.28	9.82	6.0%	53/46	68	0.8074	1.823296
216	-123.47	8.72	6.0%	59/50	76	0.8066	1.700435
217	-89.41	8.92	6.0%	89/75	114	0.8055	1.898866
218	-133.98	8.60	6.0%	79/70	95	0.8046	1.929999
219	-147.79	8.15	6.0%	65/56	84	0.8037	1.955397
220	-119.67	9.02	6.0%	69/53	85	0.8029	1.696192

Continued: Ddqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
221	-157.43	9.12	6.0%	67/53	88	0.8020	1.235788
222	-90.34	8.36	6.0%	46/38	59	0.8014	2.064327
223	-81.41	8.50	6.0%	49/42	61	0.8008	2.054167
224	-128.26	9.37	6.0%	65/52	87	0.8000	1.752974
225	-107.04	8.54	6.0%	75/68	97	0.7990	1.832046
226	-77.53	8.28	6.0%	46/38	68	0.7983	1.689542
227	-116.36	8.38	6.0%	50/41	68	0.7977	1.636362
228	-115.13	8.03	6.0%	66/57	98	0.7967	1.877510
229	-119.83	8.42	6.0%	77/67	102	0.7957	1.663348
230	-165.30	8.30	5.0%	108/91	130	0.7944	2.068948
231	38.00	8.23	5.0%	48/42	67	0.7937	1.663620
232	-91.59	8.56	5.0%	63/50	83	0.7929	1.896833
233	-119.44	9.13	5.0%	64/56	82	0.7921	1.783605
234	2.87	8.62	4.0%	59/51	81	0.7913	1.785562
235	-97.05	8.83	4.0%	75/65	99	0.7903	1.649347
236	-119.54	9.09	3.0%	68/60	81	0.7895	1.928924
237	-78.49	8.98	3.0%	64/54	80	0.7887	1.690285
238	-115.67	8.73	3.0%	97/87	118	0.7876	1.955161
239	-186.17	7.86	3.0%	87/74	120	0.7864	1.873306
240	-46.94	8.58	4.0%	90/77	112	0.7853	1.664483
241	-99.21	8.28	4.0%	56/48	69	0.7846	1.571158
242	-134.37	8.56	4.0%	59/52	84	0.7837	2.034220
243	-144.97	8.67	4.0%	66/56	90	0.7829	1.748567
244	-134.93	8.74	4.0%	81/67	110	0.7818	1.617245
245	-2.40	8.45	5.0%	64/57	76	0.7810	1.964311
246	-102.67	8.70	5.0%	61/47	79	0.7802	1.924018
247	-75.14	8.49	5.0%	52/44	65	0.7796	2.225628
248	-110.41	8.54	5.0%	65/57	88	0.7787	1.805343
249	-139.39	8.78	5.0%	70/57	85	0.7779	1.739708
250	-101.74	7.19	4.0%	69/59	99	0.7769	1.706151
251	-102.34	7.67	4.0%	60/48	76	0.7761	1.725585
252	-125.25	8.39	4.0%	60/51	78	0.7754	2.201456
253	-69.37	7.72	4.0%	48/43	56	0.7748	1.786990
254	-138.20	7.94	4.0%	88/73	113	0.7737	1.781611
255	-126.95	8.38	4.0%	86/75	107	0.7726	1.643079
256	-115.81	7.26	4.0%	82/76	108	0.7716	1.853297
257	-167.92	7.36	4.0%	84/70	110	0.7705	1.988082
258	-147.94	7.33	4.0%	73/68	95	0.7695	1.960925
259	-107.43	7.37	4.0%	67/62	84	0.7687	1.752842
260	-179.20	6.80	4.0%	72/54	89	0.7678	1.822595
261	-22.89	6.27	5.0%	46/40	65	0.7672	1.966515
262	-138.06	6.28	5.0%	57/48	81	0.7664	1.864911
263	-2.00	6.08	5.0%	59/50	75	0.7656	2.004632
264	-137.72	6.37	5.0%	74/63	95	0.7647	1.756917
265	-118.17	6.21	5.0%	82/66	103	0.7637	1.849854
266	-86.35	6.77	4.0%	88/72	112	0.7626	1.797844
267	-125.65	5.42	4.0%	81/69	104	0.7615	1.934352
268	-72.95	5.44	4.0%	56/49	70	0.7608	1.936458
269	-116.54	5.39	4.0%	90/78	112	0.7597	1.737569
270	-189.14	6.04	4.0%	85/72	119	0.7586	1.785241
271	-138.95	5.34	4.0%	74/60	101	0.7576	1.715611
272	-94.49	4.89	4.0%	60/54	79	0.7568	1.803476
273	-137.23	4.82	4.0%	79/73	99	0.7558	1.852944
274	-127.28	5.12	4.0%	88/81	114	0.7547	1.783857
275	-103.64	4.77	4.0%	68/61	86	0.7538	1.869400
276	-84.19	5.37	4.0%	54/45	70	0.7531	1.662562
277	-114.72	4.27	4.0%	93/76	109	0.7520	2.080872
278	-188.09	4.16	4.0%	62/52	78	0.7513	2.022087
279	-216.23	4.39	4.0%	93/78	111	0.7502	1.984946
280	-72.05	4.28	4.0%	57/51	66	0.7495	2.276519
281	-99.37	4.03	4.0%	63/58	85	0.7487	1.980543
282	-92.61	3.88	4.0%	60/48	78	0.7479	2.019114
283	-136.17	4.04	4.0%	55/46	86	0.7471	1.793419
284	-91.00	3.97	4.0%	89/76	123	0.7458	1.801672
285	-106.81	3.79	4.0%	48/40	73	0.7451	2.003878
286	-94.01	4.23	4.0%	53/51	71	0.7444	2.007622
287	-87.44	4.35	4.0%	87/74	107	0.7434	2.280867
288	-92.22	4.70	4.0%	52/40	71	0.7426	1.944921
289	-107.00	3.74	4.0%	74/58	91	0.7417	2.011357
290	-107.43	3.22	4.0%	56/50	83	0.7409	1.932008
291	-79.33	2.84	4.0%	84/70	110	0.7398	2.243563
292	-109.77	3.49	4.0%	67/57	80	0.7390	1.876652
293	-124.67	3.75	4.0%	63/58	84	0.7382	2.113985
294	-118.38	2.75	4.0%	78/66	98	0.7372	1.956947

Continued: Ddqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
295	-143.76	2.49	4.0%	69/58	88	0.7364	2.052511
296	-106.21	2.69	4.0%	73/63	95	0.7354	1.692481
297	-115.73	3.07	4.0%	66/55	81	0.7346	1.771754
298	-117.34	2.76	4.0%	54/37	72	0.7339	1.881097
299	-89.59	2.47	3.0%	53/42	70	0.7332	1.947227
300	-2.19	1.90	4.0%	74/63	109	0.7321	1.941613
301	-106.13	1.79	4.0%	75/70	98	0.7312	2.412922
302	-74.44	2.23	4.0%	55/48	71	0.7305	2.603833
303	-75.15	2.41	4.0%	72/59	101	0.7295	1.972003
304	-25.73	2.35	4.0%	56/47	69	0.7288	1.903994
305	-204.88	2.49	4.0%	76/67	103	0.7278	1.866876
306	-108.63	1.27	4.0%	67/55	88	0.7269	1.947352
307	-90.68	1.30	4.0%	88/77	117	0.7257	2.050842
308	-110.86	1.50	4.0%	68/55	83	0.7249	1.638188
309	-20.39	1.81	5.0%	60/55	82	0.7241	1.972457
310	-89.00	1.35	5.0%	66/55	80	0.7233	2.685224
311	-124.48	0.90	5.0%	78/68	109	0.7222	2.140365
312	-94.62	0.83	5.0%	52/42	71	0.7215	2.299327
313	-108.63	0.91	5.0%	60/53	80	0.7207	2.041754
314	-47.18	0.64	6.0%	39/32	62	0.7201	1.818984
315	-131.95	0.69	6.0%	62/51	74	0.7194	2.056831
316	-169.51	0.98	6.0%	75/66	100	0.7184	2.154809
317	-111.08	0.54	6.0%	68/56	76	0.7177	2.236956
318	-35.40	0.64	7.0%	60/50	84	0.7168	1.836063
319	-109.95	0.56	7.0%	66/57	81	0.7160	2.195182
320	-110.04	0.49	7.0%	70/61	88	0.7151	1.553355
321	-78.21	1.49	7.0%	101/87	130	0.7139	1.923665
322	-139.79	0.40	7.0%	72/65	101	0.7129	1.860809
323	-104.23	1.16	7.0%	65/56	80	0.7121	2.028235
324	-109.97	0.30	7.0%	85/74	99	0.7111	2.227439
325	-103.88	0.22	7.0%	57/53	67	0.7104	2.265546
326	-69.69	0.04	7.0%	59/53	69	0.7097	1.869105
327	-81.16	0.30	7.0%	67/52	79	0.7090	2.068484
328	-116.51	-0.47	7.0%	69/63	91	0.7081	1.801446
329	-145.51	-0.20	7.0%	63/54	73	0.7073	1.993966
330	-75.60	-0.10	7.0%	79/70	95	0.7064	2.390815
331	-104.43	-0.20	7.0%	53/45	72	0.7057	2.110758
332	-128.38	0.49	7.0%	79/67	96	0.7047	2.461856
333	-122.19	0.23	7.0%	86/73	115	0.7036	2.030703
334	-78.18	-1.01	7.0%	69/58	94	0.7027	2.132227
335	-116.14	-0.37	7.0%	77/66	106	0.7016	2.076934
336	-64.61	-0.78	7.0%	62/49	77	0.7009	2.772930
337	-89.84	-1.04	7.0%	57/51	76	0.7001	1.817834
338	-107.23	-0.35	7.0%	54/46	66	0.6994	2.198281
339	-17.44	-0.47	7.0%	104/93	133	0.6981	2.116340
340	-138.65	-0.52	6.0%	61/54	76	0.6974	1.962794
341	-117.39	-0.94	6.0%	79/66	92	0.6965	2.341204
342	-112.57	-1.49	6.0%	64/57	86	0.6956	2.170988
343	-42.92	-1.30	6.0%	51/43	67	0.6950	2.078996
344	-99.38	-1.25	6.0%	85/72	120	0.6938	1.965393
345	-105.35	-1.83	5.0%	66/56	83	0.6929	2.346100
346	-57.52	-2.07	5.0%	75/60	110	0.6919	2.046996
347	-135.32	-1.12	5.0%	66/57	82	0.6910	2.296311
348	-44.66	-1.53	5.0%	61/55	77	0.6903	2.434074
349	-79.04	-0.98	5.0%	58/48	68	0.6896	2.314190
350	-68.99	-1.09	5.0%	78/64	121	0.6884	2.305848
351	-100.96	-1.63	5.0%	52/47	65	0.6878	2.314345
352	-97.71	-0.79	5.0%	64/52	82	0.6870	2.064232
353	-68.87	-1.92	5.0%	88/77	119	0.6858	2.198974
354	-90.08	-1.09	5.0%	63/57	83	0.6850	2.267534
355	-86.21	-1.76	5.0%	53/47	75	0.6842	1.864667
356	-102.30	-1.02	5.0%	53/48	70	0.6835	1.983293
357	-92.14	-2.91	5.0%	84/75	104	0.6825	2.176815
358	-76.95	-2.77	5.0%	48/38	64	0.6819	1.813678
359	-102.86	-2.34	5.0%	68/59	78	0.6811	2.444903
360	-98.95	-2.35	5.0%	69/58	84	0.6802	2.233517
361	-90.80	-2.07	4.0%	46/39	62	0.6796	1.830283
362	-48.57	-2.36	5.0%	78/64	95	0.6787	2.303985
363	-103.55	-2.70	5.0%	77/65	96	0.6777	1.957133
364	-84.43	-2.05	5.0%	63/59	74	0.6770	2.394387
365	-101.46	-1.47	5.0%	79/70	113	0.6759	2.097140
366	-101.88	-2.40	5.0%	76/58	96	0.6749	2.115573
367	-148.25	-2.49	5.0%	71/60	90	0.6741	2.461953
368	-155.47	-2.97	5.0%	82/68	98	0.6731	1.969755

Continued: Ddqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
369	-83.26	-2.64	5.0%	51/45	58	0.6725	2.322549
370	-84.45	-2.15	5.0%	56/46	76	0.6718	2.283960
371	-28.30	-1.73	6.0%	81/70	98	0.6708	1.749963
372	-131.63	-2.17	6.0%	75/67	100	0.6698	1.984048
373	-197.26	-1.79	6.0%	85/78	105	0.6688	2.026279
374	-117.09	-2.15	6.0%	59/55	79	0.6680	1.923406
375	-125.53	-3.29	6.0%	80/69	112	0.6669	2.291135
376	-114.79	-2.98	6.0%	63/53	84	0.6660	2.070814
377	-31.79	-2.95	6.0%	72/60	81	0.6652	2.224700
378	-93.59	-3.04	6.0%	74/61	91	0.6643	2.318149
379	-131.37	-4.14	6.0%	88/80	115	0.6632	2.177648
380	-86.16	-3.52	6.0%	72/62	105	0.6622	2.151074
381	-77.35	-3.73	6.0%	74/63	91	0.6613	1.890655
382	-76.88	-3.27	6.0%	75/69	92	0.6603	2.260427
383	-88.34	-3.58	6.0%	101/88	129	0.6591	1.971006
384	-112.54	-3.69	6.0%	78/61	112	0.6580	2.182478
385	-111.71	-3.95	6.0%	50/44	69	0.6573	2.263710
386	-118.47	-3.98	6.0%	88/78	117	0.6561	2.162028
387	-83.74	-3.93	6.0%	74/64	101	0.6551	2.213275
388	-100.95	-3.81	6.0%	66/58	86	0.6543	2.282869
389	-202.89	-3.46	6.0%	109/98	136	0.6529	2.210840
390	-87.40	-3.68	6.0%	97/82	123	0.6517	2.131545
391	-151.41	-3.88	6.0%	54/48	80	0.6509	2.351941
392	-41.14	-3.10	6.0%	61/54	75	0.6502	2.079045
393	-82.63	-3.37	6.0%	88/76	126	0.6489	2.302276
394	-49.82	-3.41	6.0%	93/79	113	0.6478	1.895729
395	-129.58	-4.39	6.0%	54/46	76	0.6470	2.021301
396	-23.40	-4.98	6.0%	53/47	70	0.6464	1.753484
397	-92.00	-4.04	6.0%	89/78	117	0.6452	2.411097
398	-57.22	-4.55	6.0%	70/63	77	0.6444	1.828591
399	-66.52	-3.84	6.0%	67/55	78	0.6437	2.070732
400	-71.17	-4.38	5.0%	64/55	88	0.6428	1.887840
401	-74.73	-4.90	5.0%	49/44	56	0.6422	1.999833
402	-70.83	-4.90	5.0%	85/68	116	0.6411	2.337375
403	-74.95	-5.04	5.0%	59/51	72	0.6404	2.077291
404	16.25	-3.80	6.0%	52/44	68	0.6397	2.051040
405	-65.50	-4.72	6.0%	54/47	70	0.6390	2.476810
406	-82.55	-4.00	6.0%	56/48	79	0.6382	2.006606
407	-107.84	-4.35	6.0%	66/55	86	0.6374	1.933678
408	-184.69	-4.06	6.0%	96/77	127	0.6361	1.899450
409	-149.16	-3.81	5.0%	90/73	128	0.6348	1.859974
410	-125.76	-4.54	5.0%	84/75	105	0.6338	2.290539
411	-127.37	-4.37	5.0%	72/55	92	0.6329	2.290295
412	-92.10	-5.39	5.0%	75/61	110	0.6318	1.871263
413	-78.28	-4.95	5.0%	76/63	107	0.6307	2.141004
414	-94.12	-5.17	4.0%	74/63	89	0.6299	1.956309
415	-131.90	-5.80	4.0%	51/42	61	0.6293	2.158304
416	-79.18	-5.10	4.0%	65/52	80	0.6285	1.950994
417	-67.43	-4.63	4.0%	65/58	77	0.6277	2.547229
418	-66.07	-4.67	3.0%	70/59	84	0.6269	2.061347
419	-117.54	-3.94	3.0%	71/61	101	0.6259	2.295975
420	-94.91	-4.57	3.0%	82/76	115	0.6247	2.046820
421	-85.85	-4.23	3.0%	72/67	98	0.6238	1.930060
422	-50.36	-5.07	3.0%	66/59	81	0.6230	2.081075
423	-57.41	-4.44	3.0%	61/55	73	0.6222	2.047828
424	-132.22	-4.18	3.0%	47/38	57	0.6217	2.136458
425	-58.26	-4.75	3.0%	91/76	113	0.6206	1.884422
426	-98.06	-4.74	3.0%	51/48	72	0.6198	1.887085
427	-104.04	-4.65	3.0%	72/57	99	0.6189	2.142026
428	-70.85	-3.81	3.0%	85/78	113	0.6178	1.914412
429	-67.11	-4.32	3.0%	74/66	96	0.6168	1.908115
430	-27.28	-4.69	3.0%	77/66	97	0.6158	1.681516
431	-36.48	-5.32	3.0%	53/47	66	0.6152	2.045372
432	-109.75	-5.98	3.0%	90/75	116	0.6140	2.155313
433	-78.94	-4.58	3.0%	52/38	67	0.6134	1.971705
434	-9.96	-4.28	4.0%	80/70	113	0.6123	2.029550
435	-188.69	-5.00	4.0%	113/98	136	0.6109	1.807457
436	-89.67	-5.30	4.0%	66/53	104	0.6099	1.992192
437	-18.84	-5.17	5.0%	88/73	117	0.6087	1.992053
438	-96.23	-5.26	5.0%	59/54	72	0.6080	2.099594
439	-104.88	-4.93	5.0%	103/84	118	0.6068	2.144098
440	-55.18	-5.62	5.0%	81/73	95	0.6059	2.241344
441	-107.25	-5.72	5.0%	69/54	101	0.6049	1.972026
442	-93.49	-5.55	5.0%	86/78	123	0.6037	2.002541

Continued: Ddqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
443	-82.81	-5.79	5.0%		84/76	108	0.6026 2.008387
444	-29.03	-5.26	6.0%		83/68	105	0.6016 1.982757
445	-119.08	-5.45	6.0%		68/55	76	0.6008 1.942972
446	-222.18	-4.82	6.0%		93/76	139	0.5994 1.832827
447	-83.73	-4.94	6.0%		56/48	72	0.5987 1.981948
448	-121.61	-4.81	6.0%		81/67	104	0.5977 1.839310
449	-91.92	-5.30	6.0%		57/48	70	0.5970 1.925585
450	-147.51	-5.20	6.0%		88/80	107	0.5960 2.013807
451	-63.26	-5.31	6.0%		67/58	80	0.5952 1.762592
452	23.19	-4.90	7.0%		64/53	71	0.5945 2.118002
453	-85.01	-5.34	7.0%		73/65	81	0.5937 2.089407
454	-115.99	-5.72	7.0%		91/77	107	0.5926 1.987391
455	-119.39	-5.80	7.0%		66/54	89	0.5917 2.202953
456	-87.67	-5.12	7.0%		80/69	104	0.5907 1.981262
457	-94.28	-4.84	7.0%		98/86	128	0.5894 2.300646
458	-101.17	-5.80	7.0%		77/64	114	0.5883 2.143130
459	-106.20	-7.07	7.0%		61/54	86	0.5874 1.911042
460	-81.54	-5.94	7.0%		57/50	75	0.5867 1.962188
461	-402.49	-6.05	7.0%	520/455	624	0.5805 1.843069	
462	-16.17	-6.67	7.0%		61/57	81	0.5797 1.927949
463	-101.93	-6.19	7.0%		95/83	104	0.5787 1.944703
464	-88.30	-6.29	7.0%		64/51	77	0.5779 2.025199
465	-34.01	-6.95	7.0%		57/49	66	0.5773 2.220054
466	-66.87	-6.28	7.0%		72/61	81	0.5765 2.190353
467	-193.56	-5.70	7.0%		84/78	111	0.5754 2.020570
468	-96.00	-6.31	7.0%		64/56	82	0.5746 1.987890
469	-33.90	-6.25	7.0%		72/57	90	0.5737 1.901309
470	-67.17	-6.71	7.0%		56/46	70	0.5730 2.467051
471	-58.00	-6.13	6.0%		77/60	92	0.5721 2.004248
472	20.63	-5.26	7.0%		68/61	87	0.5712 1.972136
473	-63.28	-6.21	7.0%	105/90	128	0.5699 1.964601	
474	-53.67	-6.75	7.0%		86/80	129	0.5687 2.004130
475	-92.56	-7.01	7.0%		73/56	104	0.5676 2.207092
476	-61.33	-6.61	7.0%		79/69	90	0.5667 1.862781
477	-35.36	-6.77	7.0%		57/48	74	0.5660 2.164878
478	-117.58	-6.40	7.0%		71/60	84	0.5652 2.005980
479	-91.93	-7.02	7.0%		83/73	118	0.5640 1.841003
480	-92.02	-7.66	7.0%		89/76	122	0.5628 1.968562
481	-81.16	-6.72	7.0%		68/57	76	0.5620 2.216176
482	-106.47	-7.01	7.0%		85/71	116	0.5609 1.768411
483	-78.62	-6.53	7.0%		54/44	64	0.5603 2.065504
484	-71.94	-6.87	7.0%	131/113	160	0.5587 2.061530	
485	-68.51	-7.44	7.0%		79/65	94	0.5577 2.072954
486	-112.30	-8.10	7.0%		88/73	106	0.5567 1.892490
487	-19.62	-7.46	7.0%		104/92	122	0.5555 1.892545
488	-92.77	-8.07	7.0%		70/57	90	0.5546 2.200242
489	-110.05	-8.33	7.0%		82/72	118	0.5534 1.865310
490	-194.53	-7.18	7.0%		100/88	118	0.5523 2.186059
491	-78.04	-7.20	7.0%		75/62	101	0.5513 1.893303
492	-118.77	-8.30	7.0%		75/62	91	0.5504 1.790737
493	-78.76	-7.43	7.0%		61/50	69	0.5497 1.796395
494	-71.45	-8.18	7.0%		74/66	98	0.5487 2.143378
495	-74.58	-8.52	7.0%		87/71	111	0.5476 2.352974
496	-90.49	-7.90	7.0%		65/52	89	0.5467 1.875413
497	25.29	-8.42	8.0%		83/67	114	0.5456 1.765791
498	-86.08	-8.00	8.0%	120/100	140	0.5442 2.042550	
499	-94.73	-8.88	8.0%		84/70	110	0.5431 2.028375
500	-97.32	-8.89	8.0%		99/82	131	0.5418 2.052579
501	-60.01	-8.24	8.0%		66/57	78	0.5411 2.033611
502	-81.12	-8.25	8.0%	118/102	148	0.5396 2.024400	
503	-111.50	-9.24	8.0%		94/81	113	0.5385 1.946109
504	-133.67	-8.11	7.0%		80/67	112	0.5374 2.171576
505	-84.90	-9.05	7.0%		83/69	105	0.5363 1.768968
506	-48.23	-8.54	7.0%		86/77	99	0.5353 1.791080
507	-51.94	-8.80	7.0%		64/56	85	0.5345 2.138271
508	-76.20	-8.44	7.0%		69/56	78	0.5337 2.056788
509	-13.65	-8.59	7.0%		86/73	109	0.5327 2.011121
510	17.52	-8.64	8.0%		76/66	90	0.5318 1.987797
511	-53.65	-8.80	8.0%		65/57	81	0.5310 1.858649
512	-253.03	-8.82	8.0%	122/105	140	0.5296 1.972076	
513	-81.87	-7.07	8.0%		55/46	70	0.5289 1.836908
514	-82.53	-8.06	8.0%		84/73	101	0.5279 2.055508
515	-48.15	-9.16	8.0%		110/99	131	0.5266 2.086203
516	-88.77	-8.61	8.0%		78/68	113	0.5255 1.958263

Continued: Ddqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
517	10.41	-7.92	9.0%	72/57	83	0.5246	1.785639
518	-51.02	-9.22	9.0%	75/67	94	0.5237	1.932735
519	-21.70	-8.13	10.0%	69/55	109	0.5226	1.958850
520	-37.04	-8.00	10.0%	72/61	86	0.5218	2.134314
521	-87.93	-8.04	10.0%	64/51	75	0.5210	1.914689
522	-118.65	-8.27	10.0%	87/74	110	0.5199	1.908932
523	-75.15	-8.77	10.0%	65/52	82	0.5191	2.089218
524	-87.08	-8.27	10.0%	80/69	107	0.5181	2.044981
525	-142.50	-8.49	10.0%	102/88	138	0.5167	2.110900
526	-13.41	-9.44	11.0%	90/81	115	0.5156	1.963175
527	-104.05	-10.45	11.0%	102/80	131	0.5143	1.807563
528	-140.82	-9.99	11.0%	92/83	109	0.5132	2.260762
529	-86.80	-9.44	11.0%	85/70	96	0.5122	2.042048
530	-77.80	-9.62	11.0%	60/54	74	0.5115	1.889919
531	23.92	-9.96	12.0%	64/59	82	0.5107	1.574592
532	-48.26	-10.84	12.0%	78/66	113	0.5096	2.059140
533	11.40	-9.01	12.0%	128/107	159	0.5080	1.830436
534	-66.79	-8.86	11.0%	97/85	142	0.5066	2.041982
535	50.74	-9.47	12.0%	80/70	95	0.5057	2.040575
536	-52.26	-9.91	12.0%	77/66	96	0.5047	2.102263
537	-206.66	-9.07	11.0%	99/89	110	0.5036	2.114414
538	-26.65	-9.05	11.0%	102/88	132	0.5023	2.178071
539	-133.93	-9.34	11.0%	87/73	100	0.5013	1.883591
540	-26.67	-9.45	11.0%	99/89	122	0.5001	2.331480
541	-47.39	-9.36	11.0%	65/56	83	0.4993	1.975445
542	-120.60	-10.01	11.0%	140/124	156	0.4978	1.981160
543	-144.01	-10.26	11.0%	100/87	135	0.4964	2.119526
544	-71.22	-10.30	10.0%	65/54	81	0.4956	2.172306
545	-221.38	-10.49	10.0%	104/90	111	0.4945	1.955705
546	-45.38	-10.30	10.0%	82/73	112	0.4934	1.825247
547	-16.10	-10.80	10.0%	85/72	99	0.4924	2.200347
548	-59.89	-10.13	10.0%	111/91	137	0.4911	2.224359
549	10.97	-10.44	11.0%	61/53	69	0.4904	2.171292
550	-105.14	-9.55	11.0%	76/63	82	0.4896	1.866578
551	-293.89	-9.79	11.0%	110/104	130	0.4883	1.886668
552	-84.11	-9.24	10.0%	77/67	96	0.4873	2.246785
553	-102.78	-9.00	10.0%	76/65	102	0.4863	2.168410
554	-13.36	-9.77	11.0%	76/62	102	0.4853	2.187294
555	-85.17	-9.37	11.0%	65/53	76	0.4846	1.856470
556	-49.09	-10.11	11.0%	54/47	66	0.4839	2.281491
557	-46.95	-10.42	11.0%	62/54	88	0.4830	2.266634
558	-113.98	-10.20	11.0%	90/77	116	0.4819	2.032989
559	-66.25	-10.30	11.0%	86/76	127	0.4806	2.196885
560	1.97	-9.79	12.0%	75/67	103	0.4796	2.154144
561	-68.86	-10.15	12.0%	62/51	70	0.4789	1.972199
562	-114.61	-11.02	11.0%	85/74	101	0.4779	2.058778
563	-116.21	-9.34	11.0%	63/51	90	0.4770	1.729391
564	-96.39	-10.05	11.0%	71/60	94	0.4761	2.062476
565	-75.21	-8.88	11.0%	72/60	88	0.4752	2.050280
566	-13.10	-8.77	11.0%	103/91	141	0.4738	2.070684
567	-137.24	-9.43	11.0%	90/77	100	0.4728	2.437231
568	-259.33	-10.13	11.0%	112/96	121	0.4716	2.070155
569	-97.65	-10.55	11.0%	104/86	119	0.4705	1.700456
570	5.57	-10.35	12.0%	68/56	98	0.4695	2.215971
571	-99.59	-10.78	12.0%	52/44	59	0.4689	2.004205
572	-111.33	-8.81	11.0%	91/81	109	0.4678	2.044138
573	-73.71	-9.67	11.0%	66/55	92	0.4669	2.452548
574	-49.71	-9.50	11.0%	73/62	88	0.4661	2.177243
575	-45.05	-10.53	11.0%	76/61	90	0.4652	1.919716
576	21.79	-10.99	12.0%	53/47	75	0.4644	2.200023
577	-59.99	-9.99	12.0%	73/63	87	0.4636	1.784889
578	-42.78	-11.46	12.0%	92/80	106	0.4625	2.188487
579	-54.86	-11.73	12.0%	89/73	116	0.4614	1.868557
580	-107.91	-10.48	12.0%	65/55	72	0.4606	2.342958
581	-130.58	-10.49	12.0%	101/86	131	0.4594	2.316678
582	-43.15	-11.14	12.0%	88/74	105	0.4583	1.797235
583	-195.35	-11.35	12.0%	234/197	270	0.4556	1.844478
584	-57.29	-11.75	12.0%	72/61	90	0.4547	2.022992
585	-32.76	-10.65	12.0%	73/61	78	0.4540	2.005082
586	-59.74	-10.18	12.0%	117/91	139	0.4526	2.032171
587	-17.24	-10.26	13.0%	63/52	84	0.4518	1.782082
588	-88.34	-11.53	13.0%	73/65	119	0.4506	1.960669
589	-63.04	-11.65	13.0%	176/150	203	0.4486	2.334276
590	-77.64	-12.04	13.0%	77/66	99	0.4476	2.317746

Continued: Ddqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
591	-121.90	-11.67	13.0%	92/77	110	0.4465	1.895738
592	-70.58	-11.62	13.0%	94/78	130	0.4452	2.118468
593	-72.95	-11.87	13.0%	80/67	92	0.4443	2.254300
594	19.38	-10.79	14.0%	102/88	138	0.4429	2.009324
595	-72.63	-11.53	14.0%	132/110	169	0.4413	2.236106
596	-77.89	-11.44	14.0%	68/60	110	0.4402	2.348891
597	-125.09	-11.31	13.0%	92/81	106	0.4391	2.279626
598	24.25	-11.40	14.0%	88/71	120	0.4379	2.456524
599	-33.78	-12.71	14.0%	56/50	63	0.4373	1.981236
600	-58.30	-12.76	14.0%	67/54	77	0.4366	2.118846
601	-84.83	-12.10	14.0%	69/57	80	0.4358	2.049110
602	-42.22	-12.28	14.0%	66/59	81	0.4350	2.630822
603	-45.10	-12.02	14.0%	92/74	113	0.4338	2.169318
604	-40.31	-11.64	14.0%	98/86	114	0.4327	2.228674
605	-29.35	-12.13	14.0%	74/64	91	0.4318	2.105539
606	37.71	-11.78	14.0%	91/80	107	0.4308	2.099821
607	-109.42	-12.22	14.0%	90/76	116	0.4296	2.080977
608	-31.30	-11.30	14.0%	158/134	177	0.4279	2.395547
609	-38.98	-11.53	14.0%	160/139	194	0.4259	2.201729
610	4.18	-11.40	14.0%	91/76	114	0.4248	2.247584
611	-107.20	-11.00	14.0%	80/70	89	0.4239	2.274539
612	16.93	-12.28	15.0%	63/47	89	0.4230	2.530172
613	-114.43	-11.03	15.0%	93/83	105	0.4220	2.238513
614	-7.51	-11.62	16.0%	94/75	119	0.4208	1.943948
615	-104.61	-12.43	16.0%	72/56	87	0.4200	2.308477
616	-51.77	-12.82	16.0%	94/83	109	0.4189	1.902289
617	-75.90	-12.49	15.0%	59/47	66	0.4182	2.487051
618	-30.35	-12.84	16.0%	57/43	79	0.4175	2.085419
619	13.37	-12.28	16.0%	64/54	82	0.4166	2.253555
620	1.53	-12.66	17.0%	110/92	136	0.4153	1.945921
621	-55.04	-11.62	17.0%	145/130	166	0.4137	2.184211
622	-22.47	-12.36	17.0%	129/103	159	0.4121	2.052338
623	-50.01	-11.87	17.0%	146/124	185	0.4102	2.126616
624	-58.20	-11.37	17.0%	105/95	116	0.4091	2.213464
625	-66.96	-12.52	17.0%	121/96	135	0.4078	2.268354
626	-84.23	-11.99	16.0%	92/75	121	0.4066	2.074313
627	-52.64	-11.26	16.0%	81/65	101	0.4056	2.249748
628	-89.63	-10.85	16.0%	113/90	127	0.4043	2.204558
629	4.61	-12.85	17.0%	71/63	85	0.4035	2.504845
630	-40.24	-12.02	17.0%	114/99	133	0.4021	2.100830
631	-77.59	-10.54	16.0%	143/114	171	0.4005	2.224635
632	-209.72	-11.96	16.0%	119/104	140	0.3991	2.493793
633	1.57	-10.91	17.0%	78/63	92	0.3982	2.120494
634	-76.06	-11.25	17.0%	74/57	102	0.3971	3.011856
635	-1.29	-11.69	16.0%	100/84	134	0.3958	2.139691
636	-25.29	-9.68	16.0%	68/61	106	0.3948	2.183194
637	20.81	-10.64	17.0%	77/67	121	0.3936	2.518503
638	-74.16	-10.43	17.0%	87/73	101	0.3926	1.907487
639	39.93	-11.44	18.0%	86/73	111	0.3915	1.846917
640	-73.79	-10.37	18.0%	70/60	92	0.3906	2.024549
641	-9.76	-10.85	18.0%	111/95	151	0.3891	2.218831
642	-70.95	-9.81	18.0%	89/71	103	0.3881	2.312172
643	-68.04	-10.44	18.0%	56/46	74	0.3873	1.609975
644	-97.91	-10.37	18.0%	73/56	81	0.3865	2.124223
645	-301.07	-12.16	18.0%	716/623	817	0.3784	2.145035
646	-88.45	-9.96	19.0%	351/308	386	0.3746	2.290902
647	-27.30	-10.48	19.0%	87/77	125	0.3734	2.248221
648	53.69	-10.22	20.0%	85/74	92	0.3725	1.854869
649	-24.82	-11.65	19.0%	95/84	115	0.3713	2.124249
650	-68.04	-11.77	19.0%	85/70	94	0.3704	1.794026
651	-71.59	-11.38	19.0%	55/43	101	0.3694	2.192416
652	-61.36	-11.15	19.0%	89/74	110	0.3683	2.189765
653	-94.54	-11.23	19.0%	134/109	163	0.3667	2.121738
654	-65.46	-12.09	18.0%	77/68	87	0.3658	2.263966
655	-33.09	-12.94	18.0%	91/80	101	0.3648	2.284419
656	-96.77	-12.14	18.0%	123/105	142	0.3634	2.382323
657	-50.14	-12.30	18.0%	72/55	82	0.3626	2.402558
658	-13.04	-12.41	18.0%	110/91	140	0.3612	2.089180
659	-52.75	-12.51	18.0%	69/53	81	0.3604	2.120988
660	-5.38	-13.89	17.0%	132/113	154	0.3589	2.041734
661	-97.67	-13.15	17.0%	71/57	85	0.3581	1.921940
662	-316.55	-12.75	17.0%	836/715	1000	0.3482	2.190065
663	-54.86	-12.69	17.0%	124/111	158	0.3466	2.074451
664	-41.10	-12.60	17.0%	155/137	171	0.3449	2.407491

Continued: Ddqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
665	-35.87	-13.33	17.0%	94/81	122	0.3437	2.584950
666	-45.47	-12.74	17.0%	109/92	142	0.3423	2.277532
667	-30.35	-12.46	17.0%	71/61	78	0.3415	2.280312
668	59.94	-11.83	18.0%	96/81	133	0.3402	2.100520
669	-19.41	-12.33	18.0%	109/97	141	0.3388	2.281851
670	35.69	-12.39	18.0%	85/73	96	0.3378	2.395418
671	-49.44	-12.61	18.0%	69/63	84	0.3370	2.437915
672	-22.60	-11.01	18.0%	123/104	151	0.3355	2.506455
673	-17.79	-11.80	18.0%	71/60	81	0.3347	2.316325
674	-313.90	-11.31	18.0%	505/419	570	0.3291	2.283250
675	-39.23	-9.98	18.0%	138/113	163	0.3275	2.249028
676	-49.82	-11.65	17.0%	105/94	124	0.3262	2.022367
677	25.93	-10.78	17.0%	180/156	218	0.3241	2.011047
678	20.56	-10.97	18.0%	83/71	110	0.3230	2.201193
679	-27.12	-9.70	18.0%	97/87	105	0.3219	2.382134
680	-21.21	-11.33	18.0%	113/97	124	0.3207	2.300909
681	-214.04	-10.45	18.0%	151/120	171	0.3190	2.219344
682	-259.20	-10.43	18.0%	917/763	1000	0.3091	2.279247
683	-14.79	-10.69	18.0%	152/135	168	0.3075	2.317161
684	-343.41	-9.98	18.0%	921/777	1000	0.2976	2.457500
685	-38.65	-9.52	18.0%	107/95	137	0.2962	2.511616
686	-249.04	-7.43	18.0%	612/510	718	0.2891	2.424081
687	-47.96	-9.05	17.0%	128/106	151	0.2876	2.268672
688	-302.34	-8.39	17.0%	875/744	1000	0.2777	2.381336
689	0.22	-10.80	17.0%	110/94	143	0.2763	2.243657
690	-49.72	-9.66	17.0%	96/79	119	0.2751	2.429678
691	18.00	-9.31	17.0%	90/77	134	0.2738	2.380144
692	-12.14	-10.27	17.0%	97/81	123	0.2726	2.344593
693	-74.15	-11.20	17.0%	77/61	86	0.2717	2.167215
694	-15.78	-9.88	16.0%	104/91	130	0.2704	2.259035
695	-14.90	-11.49	16.0%	88/74	102	0.2694	2.355068
696	-29.04	-10.06	16.0%	151/129	184	0.2676	2.513778
697	23.81	-10.24	17.0%	80/67	110	0.2665	2.359177
698	-282.48	-10.81	16.0%	873/742	1000	0.2566	2.439647
699	14.63	-9.64	17.0%	99/86	119	0.2554	2.132680
700	-39.19	-10.85	17.0%	130/104	150	0.2539	2.354987
701	-243.89	-10.58	17.0%	849/708	1000	0.2440	2.651805
702	-10.00	-11.16	17.0%	114/103	130	0.2428	2.365660
703	-35.93	-10.75	17.0%	117/103	147	0.2413	2.356950
704	-254.85	-12.91	17.0%	807/709	1000	0.2314	2.510911
705	44.42	-11.30	18.0%	84/66	126	0.2302	1.988246
706	-36.29	-12.07	18.0%	100/82	110	0.2291	2.475999
707	43.25	-11.72	19.0%	116/95	130	0.2278	2.371140
708	-139.76	-11.83	19.0%	722/604	1000	0.2179	2.490935
709	19.30	-11.07	19.0%	97/83	105	0.2168	2.237694
710	-168.70	-11.59	18.0%	142/120	187	0.2150	2.437744
711	-271.42	-10.41	18.0%	771/656	1000	0.2051	2.483968
712	-253.06	-10.66	17.0%	819/689	1000	0.1952	2.452659
713	-408.92	-11.05	17.0%	641/535	737	0.1879	2.585576
714	-228.84	-12.02	16.0%	767/639	1000	0.1780	2.467988
715	-297.93	-11.84	16.0%	818/701	1000	0.1681	2.509930
716	-284.40	-12.42	16.0%	827/727	1000	0.1582	2.411014
717	-291.14	-12.66	16.0%	830/711	1000	0.1483	2.354611
718	-350.49	-12.90	15.0%	816/701	1000	0.1384	2.376637
719	-487.55	-13.18	14.0%	732/624	863	0.1298	2.344217
720	-349.35	-14.02	13.0%	540/456	651	0.1234	2.440646
721	-262.74	-13.98	13.0%	351/301	441	0.1190	2.347339
722	-207.20	-14.37	13.0%	196/170	256	0.1165	2.364700
723	6.44	-14.10	13.0%	106/86	115	0.1154	2.188273
724	-281.06	-13.05	13.0%	379/332	450	0.1109	2.328563
725	-438.41	-13.03	13.0%	714/622	861	0.1024	2.273473
726	-364.57	-13.37	13.0%	527/447	678	0.0957	2.252603
727	-333.11	-11.15	13.0%	588/491	708	0.0887	2.366086
728	-220.88	-11.09	13.0%	257/221	327	0.0854	2.297208
729	-242.54	-10.31	12.0%	423/355	535	0.0801	2.495014
730	-287.93	-10.18	12.0%	356/300	385	0.0763	2.443344
731	-370.61	-13.16	12.0%	592/497	737	0.0690	2.431371
732	-283.26	-14.07	12.0%	772/653	1000	0.0591	2.263707
733	-311.38	-11.75	11.0%	799/669	1000	0.0492	2.284152
734	-305.79	-12.21	11.0%	770/649	1000	0.0393	2.245270
735	-312.56	-11.94	11.0%	721/624	1000	0.0294	2.329437
736	-302.59	-13.74	11.0%	744/636	1000	0.0195	2.255759
737	-289.38	-10.33	10.0%	735/613	1000	0.0100	2.387766
738	232.85	-10.80	11.0%	236/188	358	0.0100	2.315929

Continued: Ddqn Modified complete per-iteration output

Episode	Reward	Avg. Q	Safe100	Attempted/Executed	Steps	Epsilon	Loss
739	-230.80	-10.34	10.0%	725/621	1000	0.0100	2.231637
740	-243.32	-10.69	10.0%	743/639	1000	0.0100	2.157033
741	-306.91	-9.87	10.0%	754/624	1000	0.0100	2.145729
742	-285.92	-8.82	10.0%	743/629	1000	0.0100	2.077061
743	-285.85	-7.18	10.0%	755/642	1000	0.0100	1.964361
744	-285.18	-9.92	10.0%	752/631	1000	0.0100	1.897920
745	-268.13	-9.91	10.0%	751/631	1000	0.0100	1.966370
746	-256.54	-12.07	9.0%	770/639	1000	0.0100	1.868462
747	-330.93	-13.46	9.0%	729/631	1000	0.0100	1.997871
748	-211.68	-12.67	8.0%	752/638	1000	0.0100	1.829908
749	-263.24	-12.69	8.0%	742/642	1000	0.0100	1.855458
750	-257.52	-13.93	8.0%	754/642	1000	0.0100	1.797241
751	-230.79	-13.72	8.0%	789/672	1000	0.0100	1.794897
752	201.81	-13.81	9.0%	235/201	360	0.0100	1.751892
753	-206.83	-14.60	9.0%	749/633	1000	0.0100	1.821687
754	-269.66	-15.54	9.0%	761/647	1000	0.0100	1.755225
755	253.79	-15.49	10.0%	246/194	332	0.0100	1.718893
756	-218.53	-14.80	10.0%	791/691	1000	0.0100	1.699342
757	-239.80	-15.88	10.0%	838/715	1000	0.0100	1.659527
758	-236.82	-13.34	10.0%	794/681	1000	0.0100	1.673181
759	65.09	-15.23	11.0%	591/509	768	0.0100	1.671747
760	-258.62	-13.91	11.0%	838/708	1000	0.0100	1.683537
761	-234.76	-16.41	11.0%	837/709	1000	0.0100	1.648816
762	-254.05	-17.09	11.0%	835/708	1000	0.0100	1.612547
763	-299.84	-17.94	11.0%	837/717	1000	0.0100	1.652121
764	-275.14	-16.36	11.0%	802/687	1000	0.0100	1.578402
765	-271.94	-14.26	11.0%	808/697	1000	0.0100	1.557156
766	-269.72	-13.91	11.0%	809/683	1000	0.0100	1.498214
767	-241.54	-13.90	11.0%	802/696	1000	0.0100	1.489447
768	202.14	-14.65	11.0%	315/268	381	0.0100	1.499383
769	102.44	-12.12	12.0%	433/366	700	0.0100	1.503823
770	-278.27	-13.95	11.0%	850/716	1000	0.0100	1.462941
771	242.22	-13.81	12.0%	273/228	320	0.0100	1.430913
772	96.06	-14.46	13.0%	447/394	556	0.0100	1.424766
773	24.11	-13.74	14.0%	632/529	771	0.0100	1.467416
774	62.31	-12.12	15.0%	649/562	815	0.0100	1.466014
775	130.09	-11.68	16.0%	480/424	633	0.0100	1.391124
776	211.71	-11.26	17.0%	311/271	389	0.0100	1.441293
777	250.10	-8.97	18.0%	242/205	345	0.0100	1.430647
778	87.21	-11.16	17.0%	387/326	563	0.0100	1.329240
779	-106.65	-8.65	17.0%	734/629	951	0.0100	1.344139
780	151.95	-10.87	18.0%	367/300	450	0.0100	1.505088
781	-328.11	-9.81	18.0%	816/691	1000	0.0100	1.405878
782	-265.66	-9.41	18.0%	826/694	1000	0.0100	1.436006
783	-285.55	-7.79	18.0%	806/675	1000	0.0100	1.358366
784	300.69	-8.21	19.0%	106/93	160	0.0100	1.332883
785	-245.75	-8.73	19.0%	815/697	1000	0.0100	1.226169
786	177.72	-6.98	20.0%	325/286	457	0.0100	1.395492
787	89.83	-4.04	21.0%	601/519	819	0.0100	1.343619
788	-280.96	-6.42	21.0%	772/680	1000	0.0100	1.249137
789	-159.04	-4.41	21.0%	802/688	962	0.0100	1.161897
790	178.59	-4.03	22.0%	401/339	499	0.0100	1.200734
791	176.82	-6.60	23.0%	361/310	471	0.0100	1.195342
792	-259.22	-6.34	23.0%	815/720	1000	0.0100	1.215665
793	-259.66	-6.23	23.0%	802/683	1000	0.0100	1.106336
794	24.57	-8.29	24.0%	591/494	739	0.0100	1.255333
795	173.41	-8.21	25.0%	442/365	551	0.0100	1.180803
796	-303.10	-7.58	25.0%	801/686	1000	0.0100	1.186434
797	-343.99	-6.71	24.0%	785/669	1000	0.0100	1.088485
798	-277.57	-7.29	24.0%	781/676	1000	0.0100	1.081105
799	194.98	-6.06	24.0%	321/281	440	0.0100	1.065252
800	268.29	-6.72	25.0%	178/150	255	0.0100	0.988843

Appendix C - Submission checklist

Use this final group checklist immediately before upload. Mark each box only after the group has verified the item in the official virtual lab and final PDF.

Final check	Confirm
Group contribution declaration is present and confirmed by all five members	[x]
All five names and BITS IDs are correct	[x]
Functions and important operations are documented	[x]
Wrapper behaviour has been experimentally verified	[x]
DQN was trained on original and modified environments	[x]
DDQN was trained on original and modified environments	[x]
The same controlled settings were used in all four experiments	[x]
All four required comparison plots are present	[x]
All five discussion questions are answered using results	[x]
Every notebook code cell was executed in the official virtual lab	[x]
Timestamped virtual-lab screenshots are included	[x]
The PDF contains code, outputs, plots, and complete per-iteration records	[x]
One group PDF is named Group148_Q_learning_DQN_DDQN.pdf	[x]
First version is submitted by 5 August 2026	[x]
Final deadline is 7 August 2026, 11:59 PM	[x]
No fabricated results or screenshots are included	[x]
Every group member has reviewed and understood the work	[x]

Rubric coverage

Rubric item	Marks Coverage
Environment implementation and verification	2.5 Sections 3.1-3.2; wrapper source; deterministic tests; random-policy CI and counter equality
DQN	4.0 Sections 4-5; shared network, replay, terminal mask, DQN target, controlled training
DDQN	4.0 Sections 4-5; online selection / target evaluation branch and controlled training
Performance evaluation	2.0 Section 6; four plots, final-100 table, and shared-seed greedy evaluation
Discussion	2.5 Section 7; all five evidence-based questions, limitation, and improvement
Total	15.0 All rubric areas mapped